

MAGNETOM

Function Description System

RF System

Document Version

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As in the previous Harmony/Symphony/Sonata product line, the already successful local coil concept, Integrated Panoramic Array, has been taken to a much more advanced level: Total imaging matrix or Tim. Tim stands for the ultimate in receive coil coverage of the patient using the new and revolutionary Matrix coils. Matrix coils are specially designed coils which have several coil elements -anywhere from 4 to 24 elements depending on the coil - placed in groups, or clusters, in such a way that the individual coil element can be read out and combined to produce three different modes. Depending on the selected mode, parallel acquisition technique (PAT) imaging and classical CP imaging techniques can be performed. Due to the arrangement of the coil element groups, PAT imaging is possible in all directions allowing PAT factors (speed factors) from 2 to 4 in any one direction, depending on the matrix coils involved. The emphasis in the Tim concept is simple: Maximum coverage, maximum signal quality, and maximum speed. This translates into an **improved workflow, greater patient comfort** and **higher revenues**.

Improved workflow because the matrix coil concept allows the user the flexibility to select any type of examination without having to change coils and reposition the patient. The user no longer sees examinations using explicit individual local coils. Instead for any given type of examination, the required number of matrix coils - there are a total of six covering the complete body - are placed on the patient to cover the desired region of interest. **Greater patient comfort** because the matrix coils allow PAT factors up to 4 meaning shorter examination times. Also, the patient does not have to be repositioned for additional exams considered necessary. As a result, overall examination time is further reduced. **Higher revenues** realized through a higher patient throughput.

Complementing this new matrix coil technology is a new intelligent coil control system coupled with a new patient table drive system. The user only requires to position the slices within the region of interest and the patient table will automatically center the ROI in the magnet isocenter. The coil control electronics are capable of detecting the position and orientation of the coils and automatically select the correct coils without any user intervention, providing a true plug-and-play capability.



The RF-system of Symphony Tim Upgrade Systems is very similar to the Avanto design. Please refer to the corresponding sections of this chapter.

Fig. 1: Block Diagram of Avanto RF System

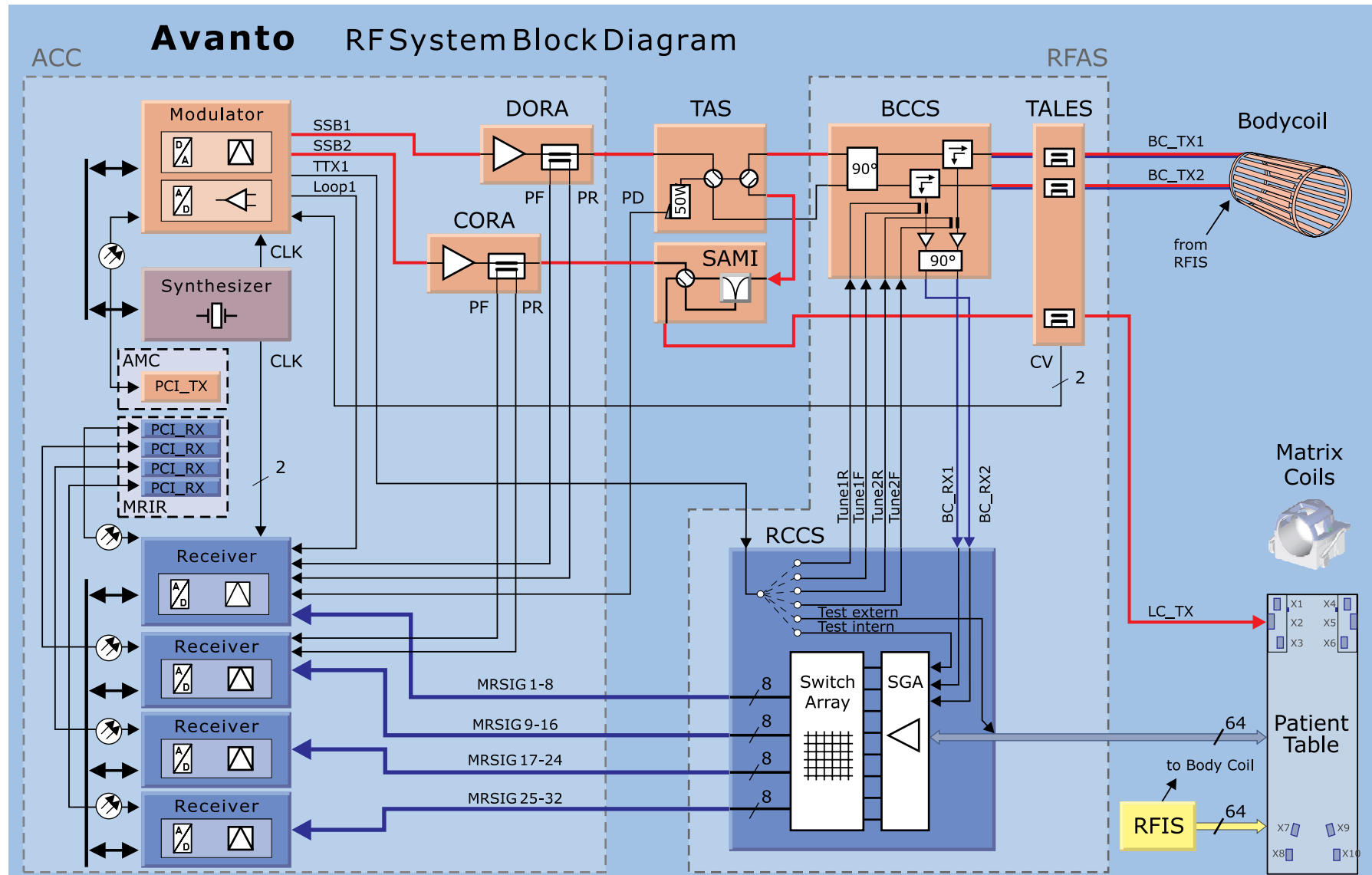
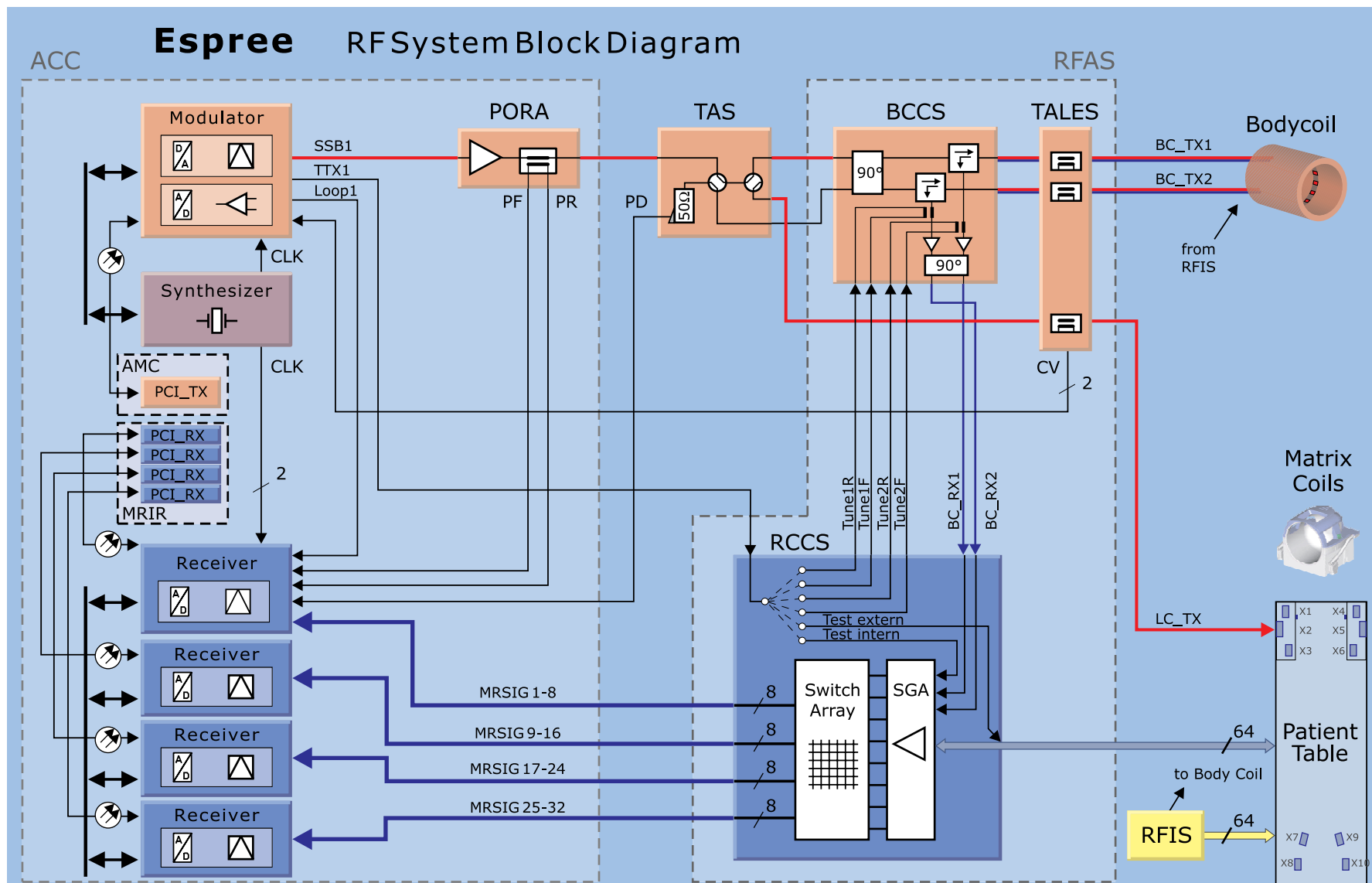


Fig. 2: Block Diagram of Espree RF System



1.1 RF System Components overview

1.1.1 RFSU

The role of the RF signal unit can be best described as a low-to-high and high-to-low frequency converter. The tasks of generating and modulating the RF excitation pulses as well as the demodulation and filtering of the MR signals are performed by digital signal processors (DSPs) and associated circuitry found in the advanced measurement control (AMC) section on the **PCI_TX** and **PCI_RX** boards, respectively. The **Modulator** board is responsible for converting the digitally modulated RF pulses generated on the PCI_TX board to the required system frequency of 63.6 MHz. Conversely, the **Receiver** board is responsible for mixing the received MR signals down to an intermediate frequency of around 1 MHz for quantification (digitalization) by an A/D converter. This 1 MHz digitized IF frequency is then sent to the PCI_RX board(s) located in the Imager where it (they) is (are) digitally filtered, demodulated and decimated to create what is called "image raw data". The raw data is then passed on to the imager for final image reconstruction. The RFSU components are realized as rack-mountable plug-in boards sharing a common backplane for power and control signal distribution. All RF signals between the RFSU components are connected via front-side QLA-type coax cables. The RFSU rack is prepared to accept a maximum of two (2) modulator boards, four (4) receiver boards and one (1) synthesizer board. The RFSU components do not require any adjustments and are maintenance-free. For MAGNETOM Trio another broadband TX Board is used for 3.0T proton imaging (123.2 MHz) and various frequencies for other nuclei.

1.1.2 RFPA

1.1.2.1 DORA

The **DORA** RF power amplifier is identical to the RF amplifier unit used in the 039 products. It is a solid state amplifier with a maximum output power of 15 kW on 50Ω. A solid-state design is in many ways superior to a bulky and complicated tube version: It allows for a very compact unit and includes the benefits of having no moving parts or external power supplies as well as not requiring adjustments or maintenance. Its compactness and high power capabilities, however, require the need of active cooling of its power components. As cooling medium, chilled water is supplied by the cooling system.

1.1.2.2 CORA

For the multi-nucleus spectroscopy option, a solid state power amplifier unit, **CORA**, will be additionally installed.

It is a solid state RF power amplifier with a specified broadband frequency range of 15 MHz to 26 MHz and a peak power of 2.5 kW into 50Ω. The gain of the amplifier is 63 ± 3 dB.

This is the identical unit used in Symphony and Sonata systems for their multi nuclei option as well as in the main RFPA for Concerto systems.

1.1.2.3 PORA

The **PORA** amplifier is of the same design as the DORA in that internally it is actually 2 DORA amplifiers which have been combined for a total output power of 22.5 kW. Although each DORA produces 15kW, power losses of the combiner explain the difference. The PORA has, in contrast to the DORA, a faster CAN bus (CON Open vs. Can proprietary). Functionally, however, the PORA is controlled in the same way as the DORA. New control signals are now available and the internal status of the PORA can be determined via several additional LEDs.

1.1.3 RF Application System

The RFAS is a group of components fulfilling the tasks of interfacing the RF "front-end" to the various RF transmit and receive coils.

The typical **functions** of an RF Application System are:

- **transmit path switch (TAS)** - a relay-actuated mechanical switch required to direct the RF transmit pulse to the desired transmit coil. The Avanto supports a full body transmit coil as well as local transmit coils.
- **90° Hybrid (BCCS)** - provides power and phase splitting of the RF transmit pulse required by a circularly polarized transmit coil - the CP Body Coil.
- **transmit-receive (T/R) switch (BCCS)** - decouples the transmit section from the (very sensitive) receive section whenever using a common transmit and receive coil as in the case of the Body Coil. Transmit-receive capable local coils usually possess a built-in transmit-receive switch.
- **Switch matrix (RCCS)** - is a constellation of row-column lines in a matrix configuration having switches at each of the row-column intersections, called nodes, which can be switched to connect any input attached to the column lines to any of the outputs connected to the row lines. A switch matrix is required by the local coil technology used in Tim.
- **Pre-amplifiers (RCCS)** - the received MR echo signals captured by the receive coils must be properly and adequately amplified for an optimal signal-to-noise (SNR) performance. All local coils have SNR optimized preamplifiers built in to amplify the signals immediately above the noise floor and to levels required to withstand the attenuation of the cables connecting the coils to the RFAS components which can be quite long - several meters. Additional secondary pre-amplifiers, located in the RCCS, may add additional amplification, as necessary. The distance between the RCCS and the receiver circuitry located in the control cabinet is between 8 to 20 meters!

These functional circuits are contained in these devices:

- **TAS_3TM** - Transmit Antenna Selector directs the RF transmit pulses to the Body transmit coil or to a local transmit coil
- **BCCS_63P** - Body Coil Channel Selector. The selector has several functions. These include: Preparing the RF transmit pulse for the circularly polarized Body Coil, providing decoupling (isolation) between the transmit and receive paths via PIN diode I/4 based transmit-receive switches, amplifying the receive signal coming off the Body Coil when it is used for receiving MR signals and providing a recombination of the two 90° phase-shifted MR signals as they come off the circularly polarized Body Coil.

- **RCCS** - Receive Coil Channel Selector has the main task of amplifying the received MR signals off the local coils and routing them to the respective ADC channel in the receiver modules in the RFSU.

1.1.4 Body Coil

The Body Coils for both the Avanto and Espree systems have been designed as no-tune coils. As a result, they no longer require an active impedance matching device (BTB) for patient load dependent power transfer optimization. Instead, the coils have been optimized for heavy loads. The percentage of reflected RF caused by light and medium-load patients can be very high - up to 90% - but the power levels used for these load types are proportionally lower so that the absolute reflected energies at these levels are not large enough to pose problems for the RFPA and the transmit path components.

Despite the absence of a variable impedance matching device there is still the need to perform a calibration of the 90° hybrid in the BCCS. The calibration results are used by the BC tests in the service software and the BC tuning procedure in the tune-up.

As mentioned in the Coil Control section above, detuning of the Body Coil is provided by PIN diodes located between the resonator elements. For dynamic detuning voltage levels of +15V@5A (forward bias) to tune and -30 V (reverse bias) are applied. When using transmit-capable local coils, the Body Coil will be statically detuned by applying a much higher reverse bias voltage of approximately -450V which will assure the PIN diodes will not be brought to conduct (be biased on) by the voltages that can be developed in the Body Coil by the transmit signal of the transmit local coil during the transmit phase.

For the optional spectroscopy feature, an additional component, SAMI, is required to allow for fast switching between the High Power RFPA or Broadband RFPA.

Control for these components is exercised by CAN controllers. In its simplest form, a CAN controller is basically another type of specialized IC which consists of a communication unit and simple one/zero outputs.

1.1.5 Matrix Coils and Local Coils

The new coils present a major innovation in local coil technology. This new technology, called Tim - Total imaging matrix, is supplemented by newly designed multi-element matrix coils. Matrix coils have multiple receive coil elements grouped in clusters which can be combined to create three different modes.

Mode 1 - The first mode is equivalent to a CP-mode and is optimized to obtain maximum SNR at the center of the region of interest.

Modes 2 and 3 - called Dual and Triple mode, map the covered FoV of the coil elements in the direction of their orientation in either two or three regions respectively. The signal derived from these modes contain additional information which can be used to improve signal-to-noise (SNR) in the periphery and/or enable higher PAT factors.

The total signal information contained in all modes is completely equivalent to the information from the original independent coil elements. The matrix mode concept allows a very efficient use of receiving channel resources and the utmost flexibility in parallel imaging acceleration as well as highest signal-to-noise (SNR) and contrast-to-noise (CNR), particularly in the periphery.

All new matrix coil resonator elements employ **circular polarization technology** for optimized homogeneity and built-in preamplifiers for unparalleled S/N performance and are factory-tuned so they do not require patient-dependent impedance matching. The coil casings use a specially formulated plastic with a high static load rating and excellent RF permeability to ensure optimal RF field homogeneity.

1.1.6 Coil Control

- **Dynamic detuning circuits** - are required when separate transmit and receive coils are used. In most cases the Body Coil is used to apply the high power excitation pulses (transmit pulses). The receive-optimized matrix or local coils are used to capture the MR echo signals. In this scenario, the individual coils will need to be dynamically tuned and detuned during the transmit and receive phases.
 - BC transmit = LC detuned (=BC tuned)
 - LC receive = LC tuned (=BC detuned)

This fast-acting switching between tuned and detuned states is achieved with the use of PIN diodes. The diodes are strategically placed in the coils so that by opening or closing the diode the coil can be tuned or detuned. The forward and reverse bias voltages to close and open the diodes are supplied by PIN diode control circuits located in the RFIS unit.

The component responsible for the coil control is the **RFIS**. It has the monumental task of controlling up to 64 separate local coil elements by supplying the dynamic detuning signals and supply voltages for the built-in pre-amplifiers.

1.1.7 RF Safety WatchDog

The task of the RFSWD, also called power absorption limiter (PALI) system, is to monitor the applied RF excitation pulses in order to prevent exceeding safe RF exposure levels which, when exceeded, could cause excessive local and/or global warming of the patient leading to organ damage.

2.1 General

The RFSU is a modular system consisting of three components:

- Modulator board
- Receiver board
- Synthesizer board

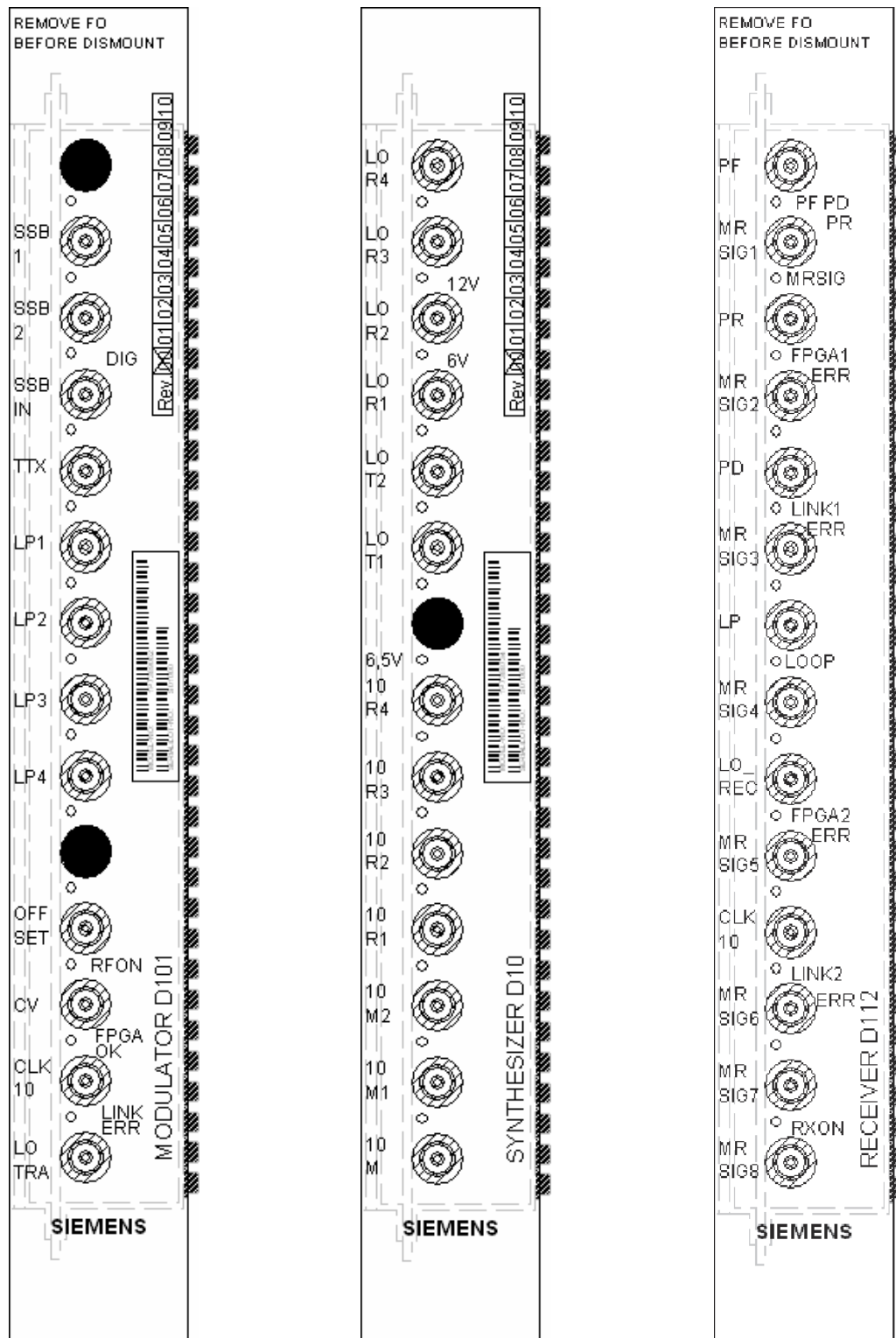
All components are fully digital and realized in the form of PC boards placed in a rack and connected to a common backplane over 96 pole connectors. All RF signal inputs and outputs are available as coax connectors at the front plate of each board.

As was the case in the Harmony / Symphony product line, the role of the RFSU can be best summarized as being low-to-high and high-to-low frequency converters. The RF pulse modulation and MR signal demodulation are performed by DSPs on the PCI_TX and PCI_RX boards respectively. The modulator digital-to-analog converts the digitally modulated RF pulses from the PCI_TX and then mixes the result up to the required system frequency. Conversely, the receiver mixes the MR signal down to an IF and A/D converts this for the PCI_RX which then does the filtering and demodulation digitally.

There are no adjustments required and there are no filters to clean and maintain.

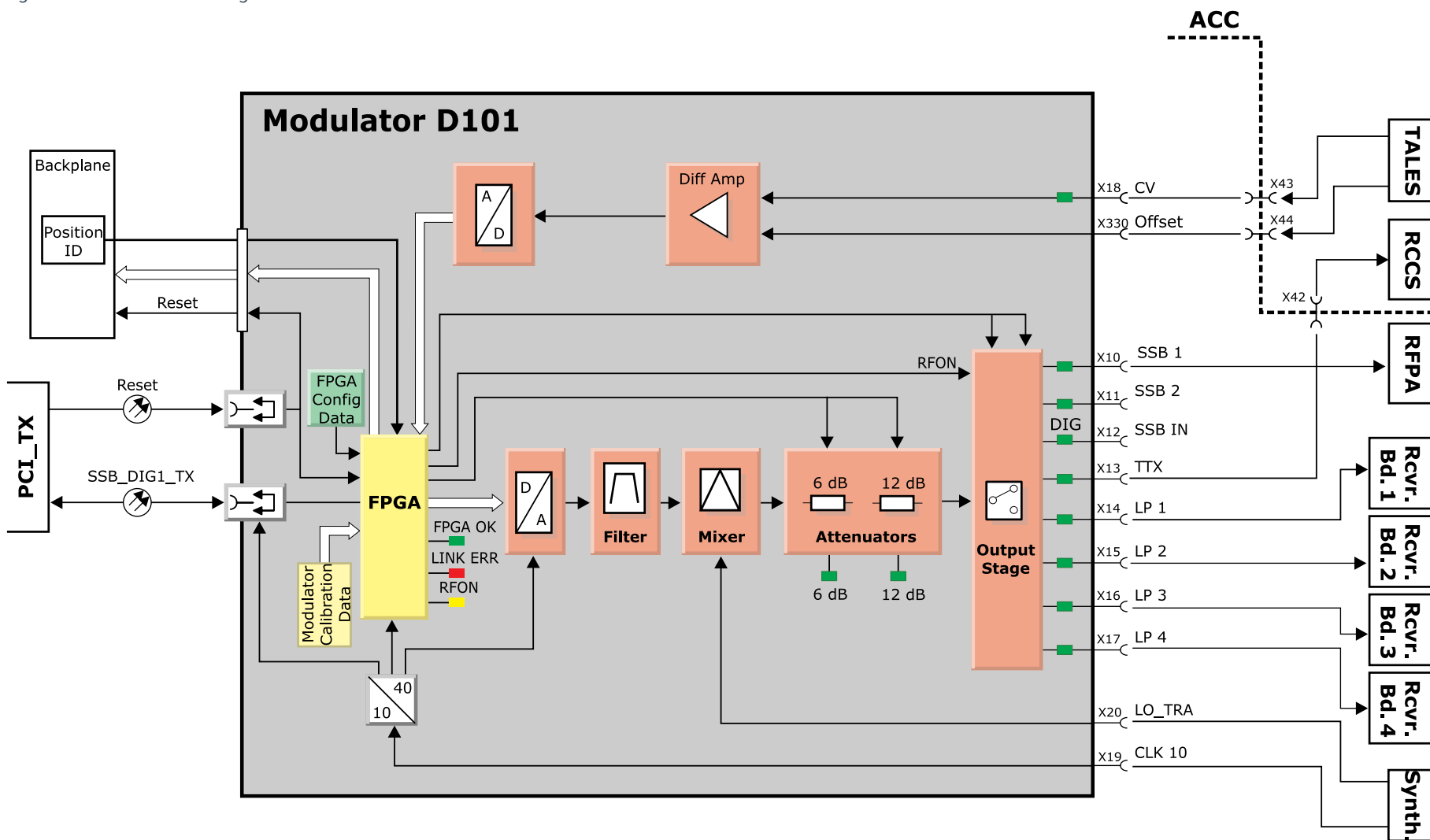
Due to its modular design the RFSU can be expanded to include a second RF modulator and up to 3 additional receiver modules for a total of 32 dedicated receive channels.

Fig. 3: RFSU Front Side Layout



2.2 Modulator

Fig. 4: Modulator Block Diagram



The modulator generates the excitation pulses at the same frequencies from the local oscillator carrier and the digital SSB amplitudes from the digital modulator of the PCI_TX (in the AMC)

2.2.1 Inputs

The digitally modulated single side-band pulse **SSB_DIG1_TX** generated on the PCI_TX board in the AMC is delivered to the modulator via a bi-directional, high-speed (640MBit/s each direction) optical fiber connection located in back of the board.

A 10 MHz clock **CLK10_IN**, produced by the synthesizer, for synchronizing the data transfer and digitalization.

The local oscillator signal **LO_IN** is produced by the synthesizer board D10 and delivered to the modulator via a QLA coax cable at the front side of the board.

The **CV** and **OFFSET** signals is a multiplexed signal coming from the TALEs representing the measured forward and reflected RF transmit voltages applied to the body coil. It is analog-to-digital converted with 16 bits at a 500 kHz sampling rate and sent to the AMC over the SSB_DIG1_TX fiber optic line. These values are used by the RF Safety Watchdog (RFSWD) software for calculating the SAR.

Tab. 1 Modulator inputs

Signal	Description	Level
CLK_10	10 MHz synchronization clock	$3 \pm 0,5$ dBm
LO_TRA	Local oscillator of 63.6 MHz	$-1 \pm 0,1$ dBm
CV	Forward and reflected RF transmit voltages	0 to 700 mV typical, -100 to -900 mV max. 1 M Ω input impedance
OFFSET	Negative input for elimination of offsets between RFAS-GND and RFSU-GND	0mV typical, -100mV to +100mV maximum

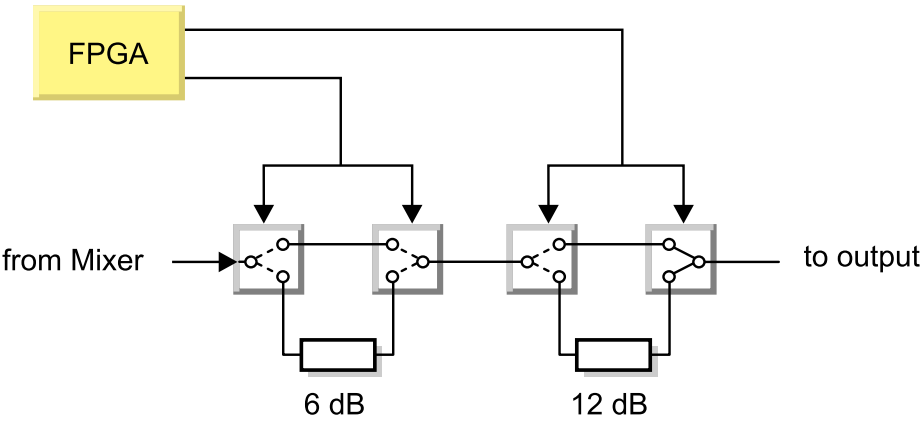
2.2.2 Function

A de-serializer converts the serial data stream into parallel data blocks which are then sent to the DAC for conversion.

The DAC output is a SSB IF (intermediate frequency) signal of around 1 MHz. This is mixed up to the system frequency of 63.6 ± 250 kHz with local oscillator signal coming from the synthesizer board D10. The signal is filtered for a frequency spectrum free of spurious and inter-modulation harmonics.

The amplitude of the resultant RF signal is adapted to the required output level through a two-stage attenuator matrix consisting of a 6 dB and a 12 dB attenuator providing an overall attenuation of 0, 6, 12 and 18 dB respectively.

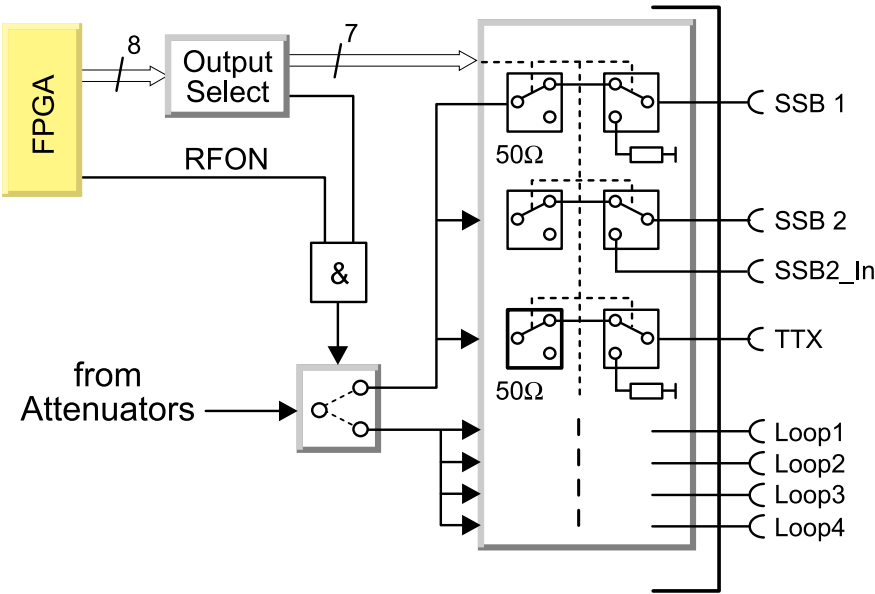
Fig. 5: Modulator Attenuators



2.2.3 Outputs

The final **SSB** signal is applied over a switch array to one of seven outputs. The data for switch selection and attenuator selection as well as the frequency selection for the synthesizer is also sent to the modulator over the high-speed fiber optic line.

Fig. 6: Modulator Outputs



Tab. 2 Modulator Outputs

Output	Function	Level
SSB_1	Main output for normal proton imaging	3 dBm max
SSB_2	Secondary output used for NOE option	3 dBm max
TTX	Used for tuning and service testing.	3 dBm max
Loop 1	Used for testing receiver 1	16 dBm max
Loop 2	Used for testing receiver 2	16 dBm max

Output	Function	Level
Loop 3	Used for testing receiver 3	16 dBm max
Loop 4	Used for testing receiver 4	16 dBm max

2.2.4 LEDs

For service there are several LEDs mounted on the front plate displaying the settings of the attenuators (ATT), the output selection (OUTSEL), RF transmission has been enabled (RFON) and the FPGA status (FPGA_OK and LINKERR).

Tab. 3 Modulator LEDs

LED	LED 6db	LED 12db
0	ON	ON
6	Off	ON
12	ON	Off
18	Off	Off

2.2.5 Specifications

Tab. 4 Modulator Specifications

Value	Specifications
Frequency response (in band)	± 0.2 dB uncorrected ± 0.1 dB corrected
Center Frequency Range	250 kHz
Bandwidth	± 250 kHz
Transmit amplitude resolution	16 bit
Variation 1 within band of 125 kHz	< 2.0 dB (peak-peak)
Variation 2 within band of 250 kHz	< 3.0 dB (peak-peak)
Reproducibility amplitude <1s	< 0.05 dB (peak-peak)
Reproducibility phase <1s	< 0.5 deg (peak-peak)
Reproducibility amplitude >1s	< 0.3 dB (peak-peak)
Reproducibility phase >1s	< 3.0 deg (peak-peak)
Linearity	2% over 40dB dynamic

2.3.1 Inputs

Operating voltages of +6.5 and 15 V.

Control inputs for determining the frequency and phase and selection of DDS 1 or 2 outputs. The control signals are originating on the modulator and sent to the synthesizer over the backplane connection.

2.3.2 Function

The synthesizer generates from an over-stabilized reference oscillator the highly-stable local frequencies. The local oscillator signals are generated by two DDS (Direct Digital Synthesizer). Both DDSs are fed with a common 160 MHz clock also created from the 10MHz reference oscillator via a frequency multiplier. The DDSs have a frequency bandwidth of $300 \pm \text{kHz}$.

2.3.3 Outputs

The output signals of either DDSs can be coupled to the six LO output connectors via four fast-acting switches. The control signals of the switches are also originating from the modulator.

All outputs are always active when power is applied.

Tab. 5 Synthesizer outputs

Signal	Description	Level
TX1	Local oscillator for modulator 1	$-1 \pm 0.1 \text{ dBm on } 50\Omega$
TX2	Local oscillator for modulator 2 (option)	$-1 \pm 0.1 \text{ dBm on } 50\Omega$
RX1	Local oscillator for receiver 1	$-1 \pm 0.1 \text{ dBm on } 50\Omega$
RX2	Local oscillator for receiver 2 (option)	$-1 \pm 0.1 \text{ dBm on } 50\Omega$
RX3	Local oscillator for receiver 3 (option)	$-1 \pm 0.1 \text{ dBm on } 50\Omega$
RX4	Local oscillator for receiver 4 (option)	$-1 \pm 0.1 \text{ dBm on } 50\Omega$
CLK10_M1	10 MHz clock for modulator 1	$3 \pm 0.5 \text{ dBm on } 50\Omega$
CLK10_M2	10 MHz clock for modulator 2	$3 \pm 0.5 \text{ dBm on } 50\Omega$
CLK10_R1	10 MHz clock for receiver 1	$3 \pm 0.5 \text{ dBm on } 50\Omega$
CLK10_R2	10 MHz clock for receiver 2	$3 \pm 0.5 \text{ dBm on } 50\Omega$
CLK10_R3	10 MHz clock for receiver 3	$3 \pm 0.5 \text{ dBm on } 50\Omega$
CLK10_R4	10 MHz clock for receiver 4	$3 \pm 0.5 \text{ dBm on } 50\Omega$

2.3.4 LEDs

Tab. 6 Synthesizer LEDs

LED	Description
6.5 V	ON indicates 6.5 V is ok
12 V	ON indicates the 15 V is ok
LOCK	???

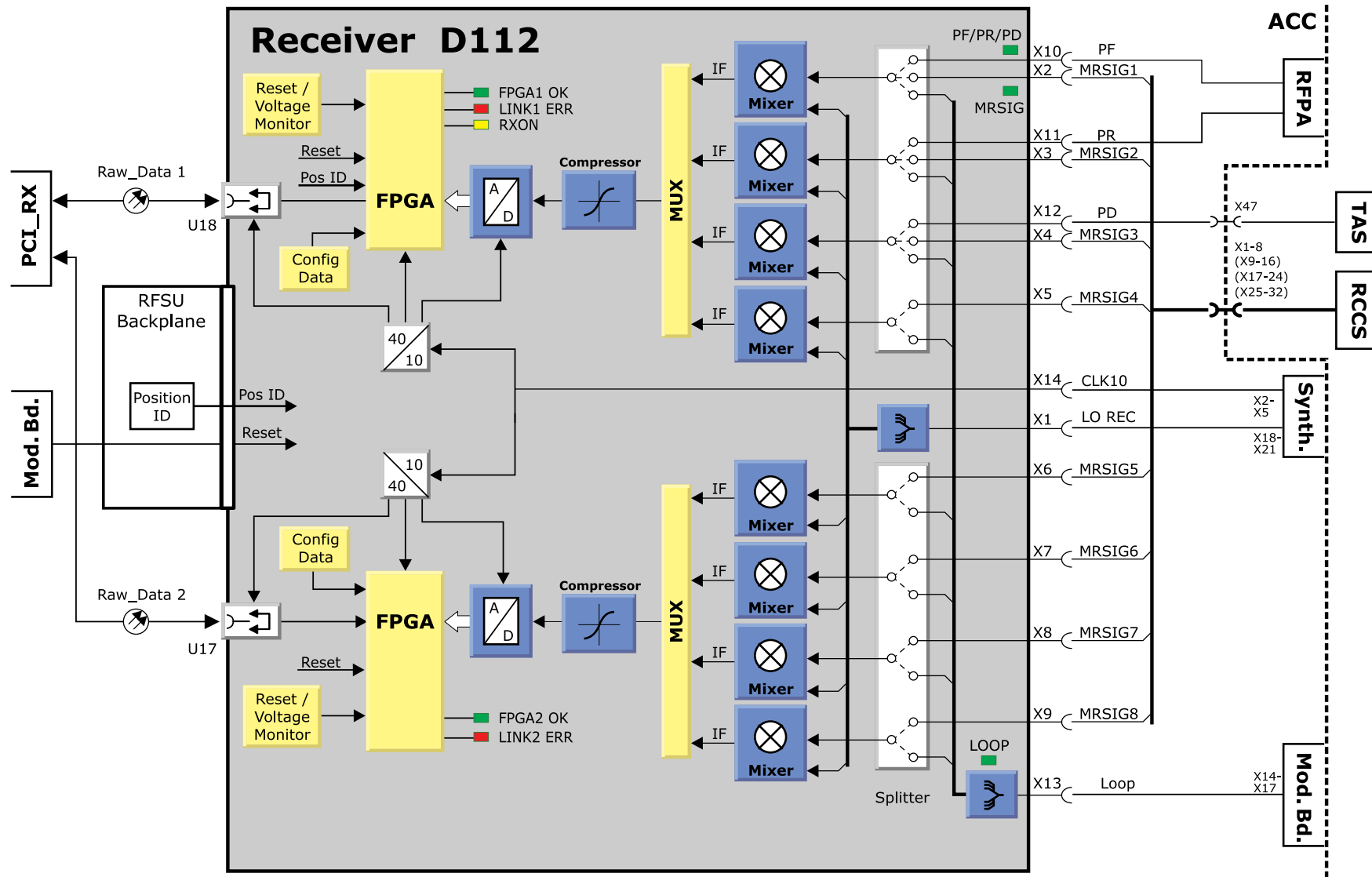
2.3.5 Specifications

Tab. 7 Synthesizer specifications

Value	Specifications
Frequency Range	10 to 120 MHz
Frequency bandwidth	300 kHz
Frequency instability	$4 \cdot 10^{-10}$ pp max over 5min
Amplitude instability	0.02dB max
Frequency switch-over time	1ms max
DDS switch-over time	10 μ s max

2.4 Receiver

Fig. 8: Receiver Block Diagram



The receiver creates a digital IF signal from the MR echo signals coming from the RCCS which will be digitally demodulated and filtered by the PCI_RX in the AMC.

2.4.1 Inputs

Tab. 8 Receiver inputs

Signal	Description	Level
MRSIG 1-8	MR echo from receive coils	3 dBm max
PF (Mod 1)	RFPA forward power	3 dBm max
PR (Mod 1)	RFPA reflected power	3 dBm max
PD	Sampled power level at the 50 ohm dummy load in the TAS	3 dBm max
Loop	Loop testing of RFSU components	16 dBm max
CLK 10	10 MHz clock used for internal synchronization	3 ± 0.5 dBm on 50Ω
LO REC (RX1-4)	Local oscillator for mixer	-1 ± 0.1 dBm on 50Ω

2.4.2 Function

The receiver board functions as a frequency converter and Analog-to Digital converter. The MR signals received by the receive coils are at the nominal system frequency of 63.6 MHz. The receiver board reduces this frequency to an IF of approximately 1 MHz. This is then digitized with a 10 MHz sample rate and sent to the PCI_RX boards of the MRIR over a bi-directional, high-speed optical fiber link.

2.4.3 Outputs

The digitized signals are sent via a high-speed fiber optic line to the PCI_RX board for demodulation and filtering.

2.4.4 LEDs

Tab. 9 Receiver outputs

LED	Description
PF/PR/PD	Signifies that these three inputs are being input.
MRSIG	Signifies that the MR signal inputs are being input
FPGA 1,2	Lights green when FPGA is successfully loaded
LINK 1,2 error	Lights red when an error occurs in the fiber optic output section

LED	Description
Loop	Signifies the LOOP inputs are activated. Active during service testing.
RX ON	Indicates the system is in the receive mode (i.e. NOT in transmit mode)

2.4.5 Specifications

The code of the board positions is read in over the optical interface.

Tab. 10 Receiver specifications

Value	Specifications
Center frequency	63.6 MHz
Frequency range	150 kHz
0.1dB bandwidth	250 kHz
Max input level	± 1.2 dBm
Stability (5 min.)	≤ 0.01 dB
Receiver resolution	18 bit effective @10 MHz sample rate
Noise figure	12 dB

[Click for an overview description of the RFPA](#)

Fig. 9: Avanto: DORA RFPA Front Panel



3.1 Inputs

Tab. 11 RFPA DORA inputs

Signal	Description
AC_IN	Three-phase AC voltage input (93V phase to neutral). An internal power supply generates all required voltages.
RF_IN	Input of RF signal to be amplified. The nominal input level for the full output power level of 15kW is 0dBm.
RFPA_UNB	To reduce noise and loss of the MR echo during reception, the amplifier is blanked during the receive cycle. This is accomplished with the fiber optic signal RFPA_UNB. When the signal is active (light on), the amplifier is activated.
RFPA_ON	An enable signal (light on) is required to bring the amplifier to operating condition. The purpose of this signal is to disable the amplifier, a requirement for SAR monitoring. (see also RFPA CORA description)
CAN1, 2	Fiber optic CAN bus communication interface.

3.2 Function

3.2.1 Power up

To bring the amplifier up to operating condition, the **RFPA_ON** signal must be present and a command must be sent to the CAN unit to enable the internal enable signals. After a successful power up, the **Ready** LED lights to signify that the amplifier is operational.

3.2.2 Monitoring

The RFPA has several internal circuits monitoring a number of vital functions: overvoltage and overcurrent of power supply, peak output power levels or reflected power is too high (RF Power), unblank duty-cycle, and power stage temperature. In the event of an internal failure or an operational limit of the amplifier has been exceeded, the amplifier provides a status report via the CAN bus.

3.3 Outputs

Tab. 12 RFPA DORA outputs

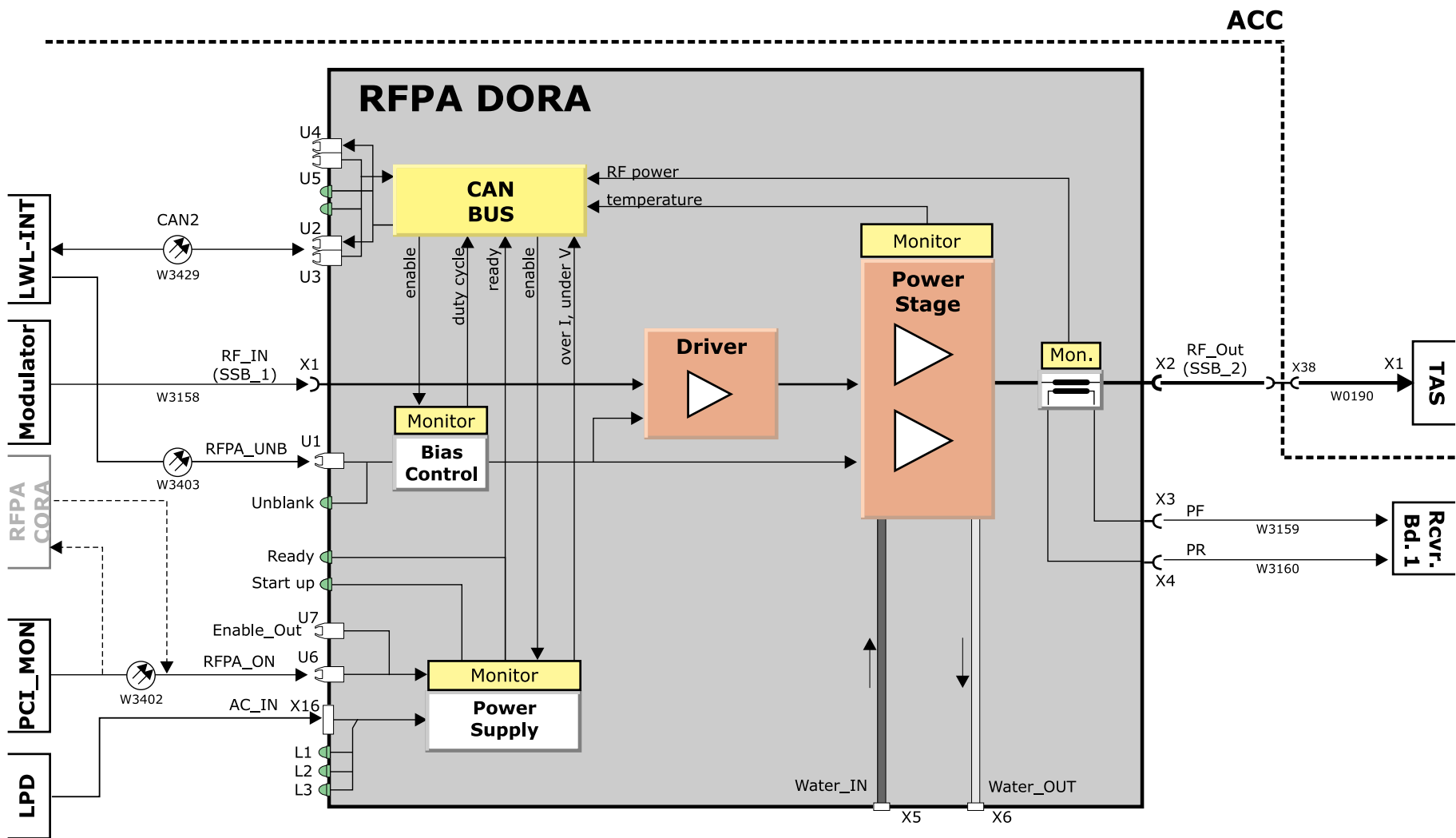
Signal	Description
PF, PR	Directional coupler (DICO) outputs the forward and reflected power levels of the RFPA output.
RF_OUT	This is hopefully where the amplified RF comes out.

3.4 LEDs

Tab. 13 RFPA DORA LEDs

LED	Description
UNBLK	Lights when a gating pulse is applied.
Start-up	RFPA put into power-up mode (command via CAN).
READY	Amplifier power-up completed and ready.
CAN TX	Lights when activity on CAN bus.
CAN RX	Lights when activity on CAN bus.
L1-L3	AC power supplied to unit.

Fig. 10: Block Diagram of DORA RF Power Amplifier



Click for an overview description of the RFPA. The functional description is the same as for DORA. The maximum output power is greater at **22.5 kW**.

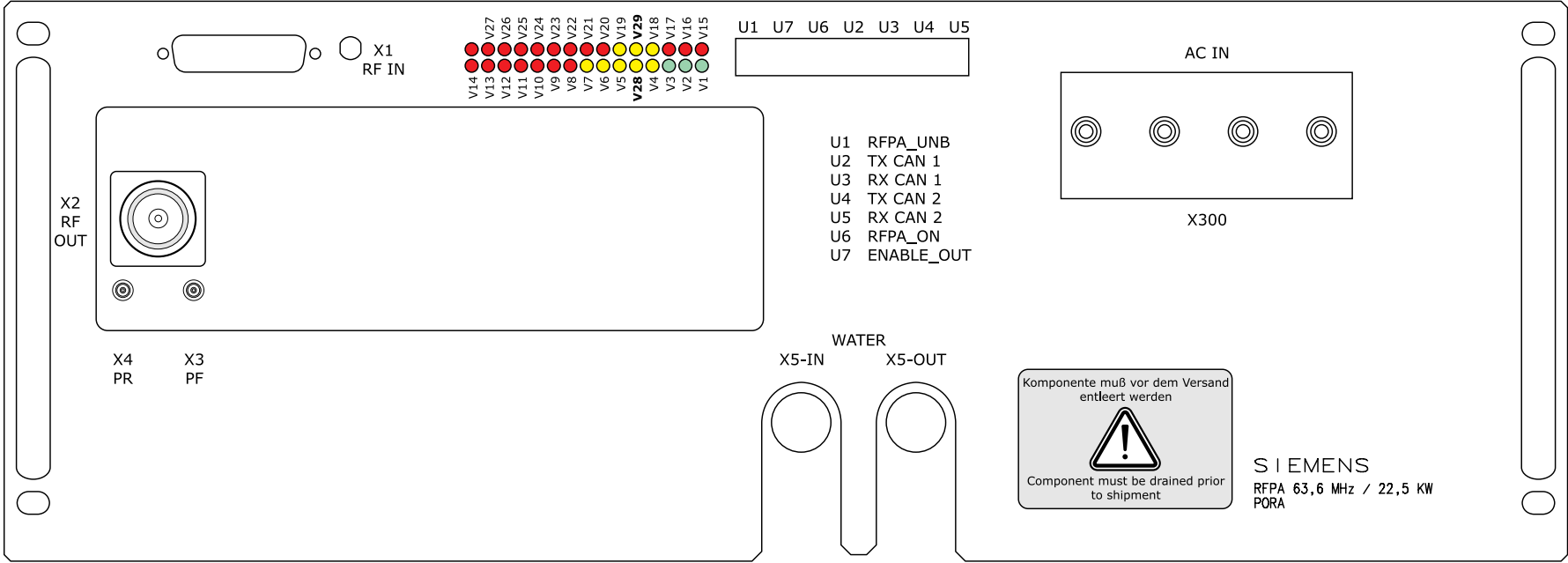
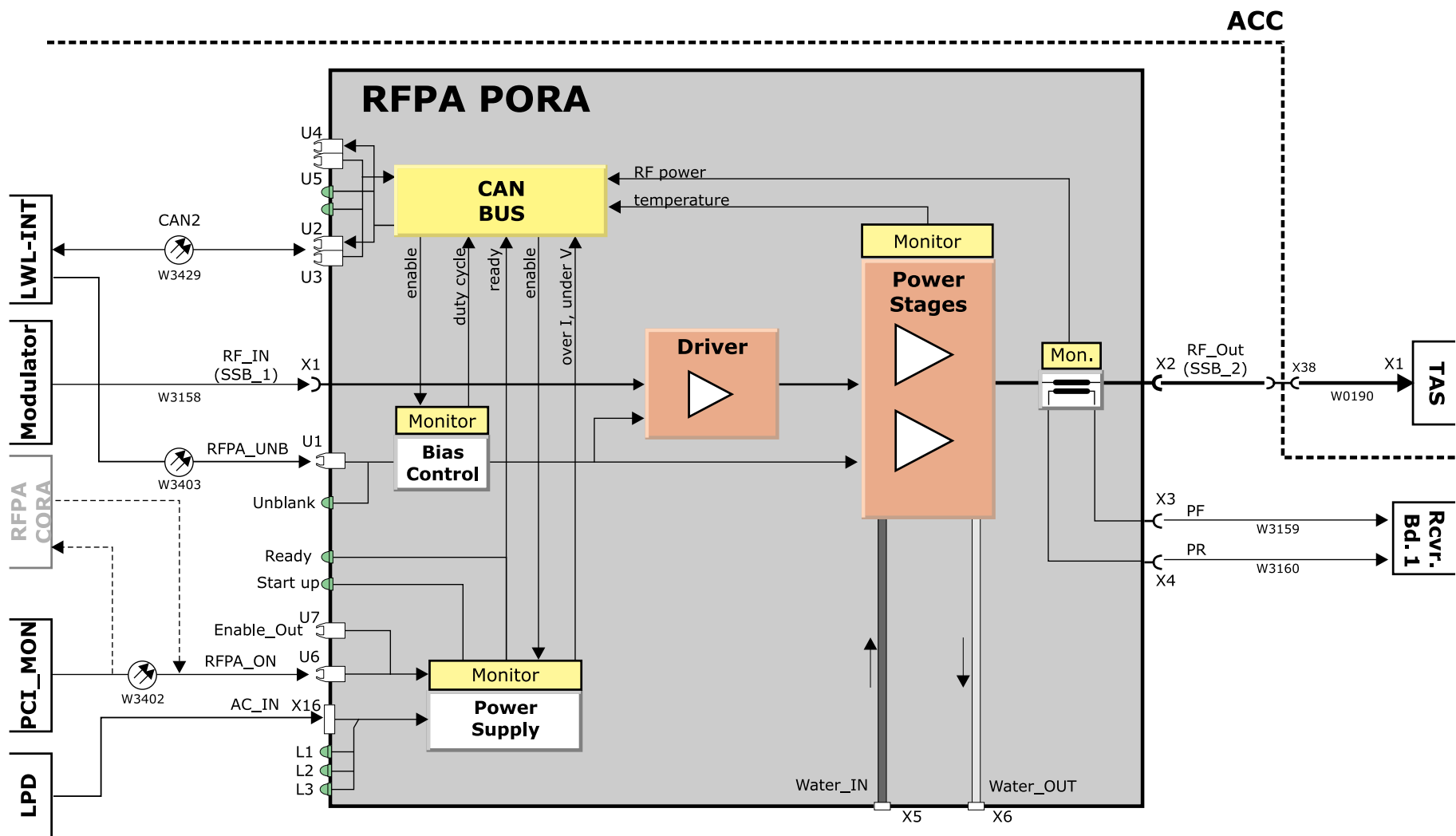
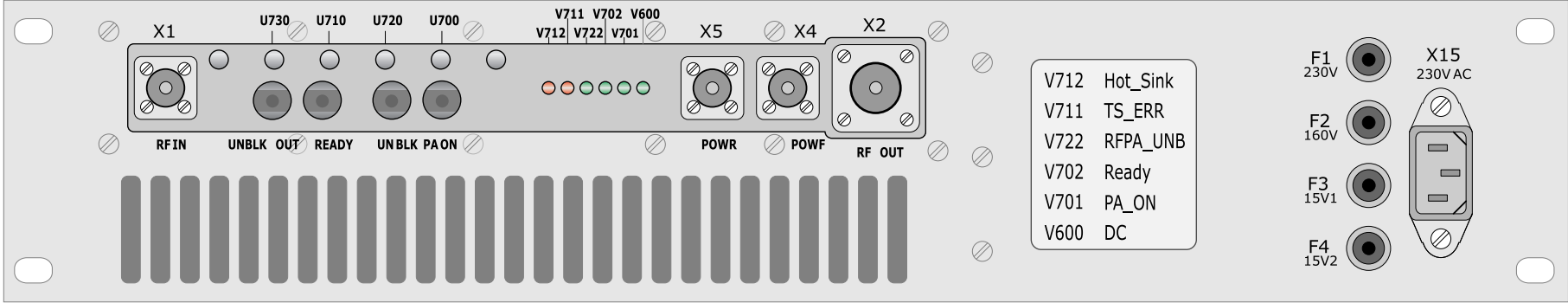


Fig. 12: PORA RF Power Amplifier Block Diagram



[Click for an overview description of the RFPA](#)

Fig. 13: RFPA CORA



5.1 Inputs

Tab. 14 RFPA CORA inputs

Signal	Description
AC_IN	The two-phase AC primary voltage of 230V. An internal power supply generates all required voltages.
RF_IN	Input of RF signal to be amplified. The nominal input level for the full output power level is 0dBm.
RFPA_UNB	To reduce noise and loss of the MR echo during reception, the amplifier is blanked during the receive cycle. This is accomplished with the fiber optic signal RFPA_UNB. When the signal is active (light on), the amplifier is activated.
PA_ON	An enable signal (light on) is required to bring the amplifier to operating condition. The purpose of this signal is to disable the amplifier, a requirement for the SAR monitoring.

5.2 Function

5.2.1 Power up

To bring the amplifier up to operation the **PA_ON** signal must be applied. When the amplifier is ready, it will set the **Ready** signal (light on). This signal is routed to the RFPA_On input of the DORA amplifier. This will tell the AMC via the CAN Bus that CORA is okay and will cause the AMC to enable the DORA amplifier via the CAN bus; the start up - LED in DORA lights up. Finally the start up - LED switches off and the ready LED in the DORA lights up. Both RFPAs are ready now.

5.2.2 Monitoring

The CORA has integrated monitoring circuitry for various internal functions. If any of these monitored functions fails, the **Ready** condition is extinguished (LED goes out) and fiber optic signal to DORA RFPA is removed (light out).

Two LEDs represent temperature-related errors - **TS-ERR** and **HOT-Sink**.

In case of an error the Ready fiber optic signal to the RFPA_ON input of the DORA RFPA will be set to low (light out), causing the DORA RFPA to shut down. This will be reported back to the MR system control unit (AMC) by the DORA's CAN unit. Currently, the AMC will react by setting the PA_ON signal to CORA low and hence disable CORA and DORA.

5.3 Outputs

Tab. 15 RFPA CORA Outputs

Signal	Description
PF, PR	Directional coupler (DICO) outputs of the forward and reflected power levels of the RFPA output
RF_OUT	This is hopefully where the amplified RF comes out

5.4 LEDs

Tab. 16 RFPA CORA LEDs

LED	Description
DC	AC power supplied to unit
PA_ON	Remote power on signal present
READY	Amplifier start-up completed, system ready
UNBLK	Lights when a gating pulse is applied to the amplifier
TS-ERR	Transistor junction temperature too high (non-latched)
HOT-TSINK	Heat sink temperature too high (non-latched)

6.1 General

Click for an overview description of the RFAS

The RFAS is that group of components needed to apply the RF transmit and receive signals between the coils and the RF front-end which consists of the modulator and receiver boards.

The RFAS consists of the components

- TAS_3T, SAMI_3T
- BCCS
- RCCS
- TALES

Fig. 14: RFAS Parts Location (1)

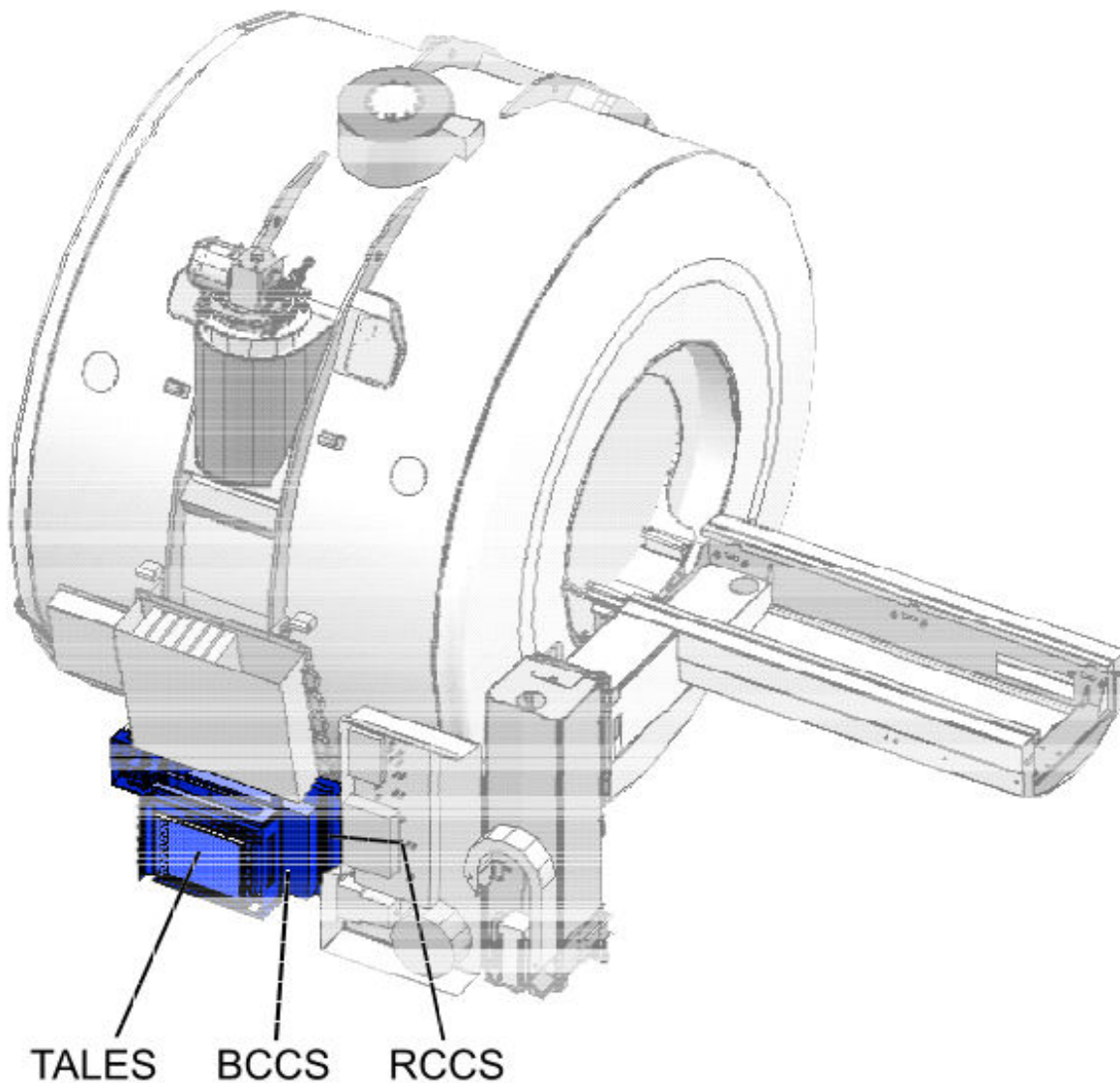
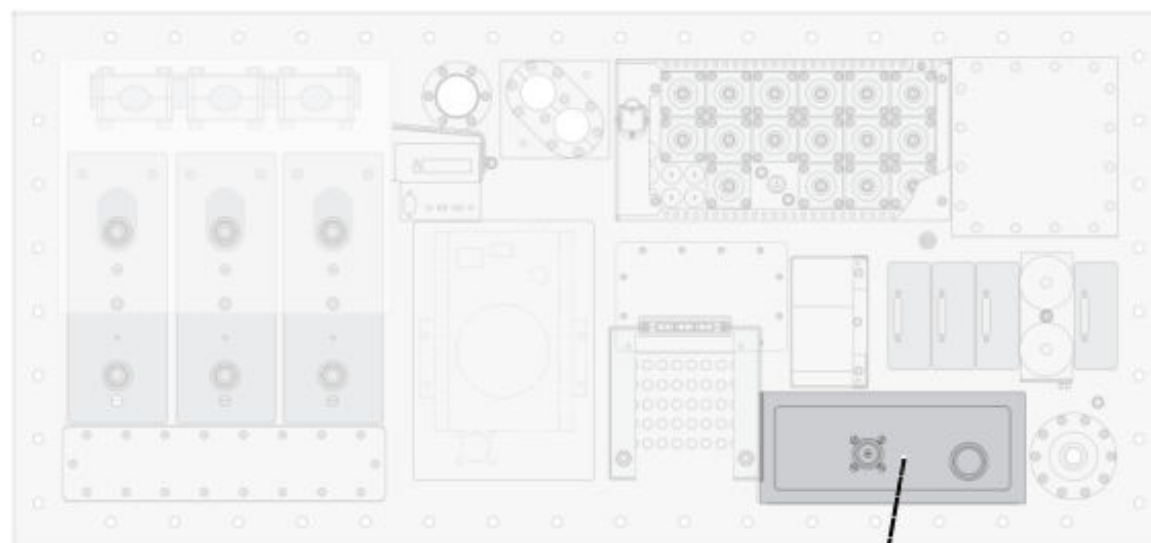


Fig. 15: RFAS Parts Layout (2)



Filter Plate

TAS

6.2 TAS_3TM

6.2.1 Mechanical switching

The Transmit Antenna Selector (TAS) version 3TM (**3T** = for use on systems with a max RF transmit power of 35 kW; **M** = measure output of the dummy load voltage) supplies mechanical switching of the RF transmit signal to two output paths.

6.2.2 50Ω Termination

In addition it contains a 35kW (PEP) 50Ω dummy load used for termination of the 90° hybrid of the BCCS and the RFPA for running service test and calibration loops.

6.2.3 Relay control

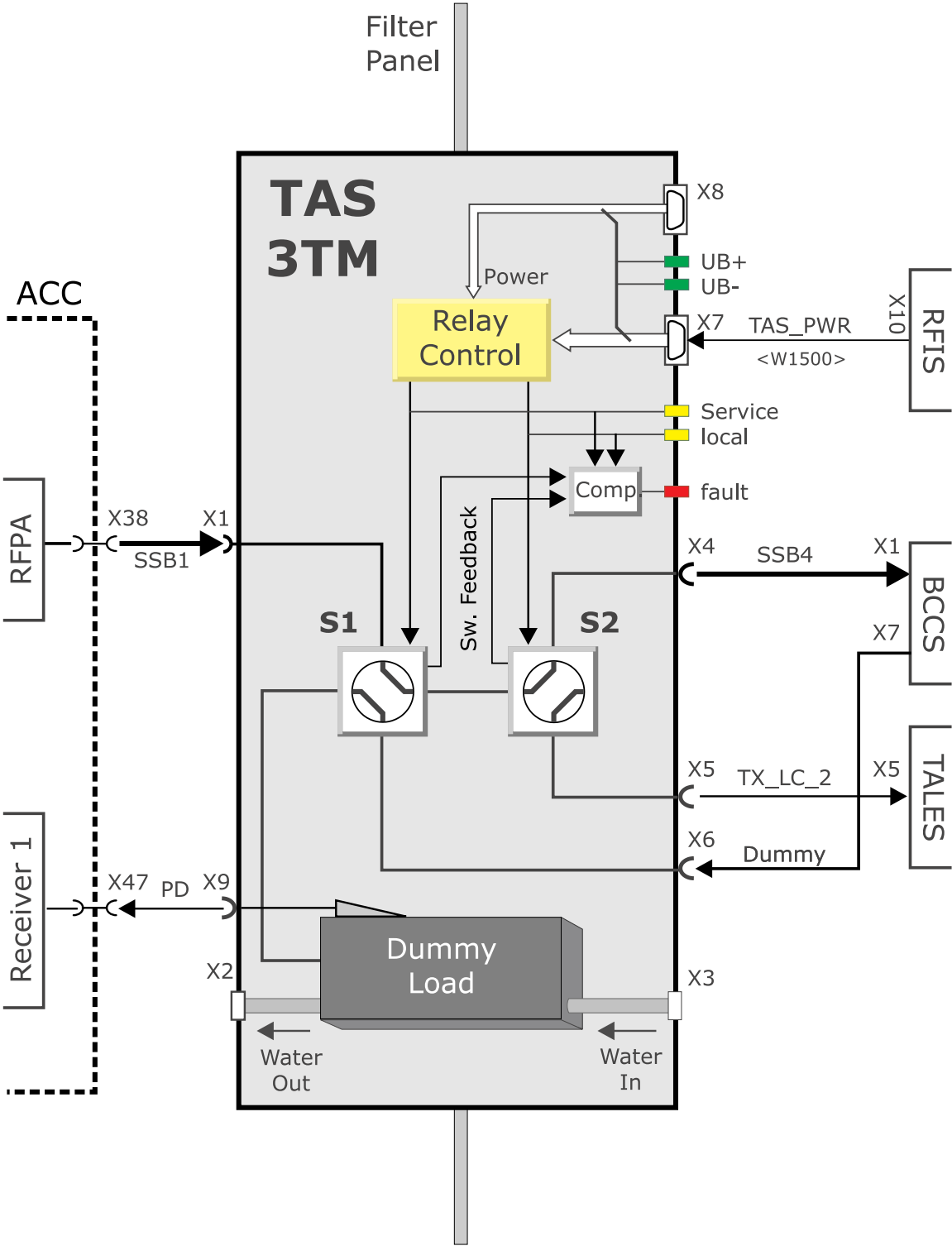
The switches are actuated with DC relays operating at 15 volts. The supply voltage of 15 volts and the control signals for S1 and S2 are supplied by the RFIS.

6.2.4 LEDs

Tab. 17 TAS_3TM LEDs

LED	Status	Description
UB+	ON	+15 V OK
UB-	ON	-15V OK
Service	ON OFF	S1 set RFPA to dummy Imaging (RFPA to BCCS or TX local coil), depending on S2
Local	ON OFF	TX to local coil (via TALES) TX to body coil (via BCCS)
Fault	ON	Discrepancy between relay position and relay control signal

Fig. 16: TAS_3TM Block Diagram



6.3 SAMI_3T

The differences are:

- No S1 switch and hence no relay and no control LEDs
- No control signals and no +15V are needed, hence there are no connectors X4 and X5.

S1 is not necessary, because the CORA broadband amplifiers transmits frequencies not higher than 26MHz. This low frequency will always be routed to the local transmit coil via the frequency switch.

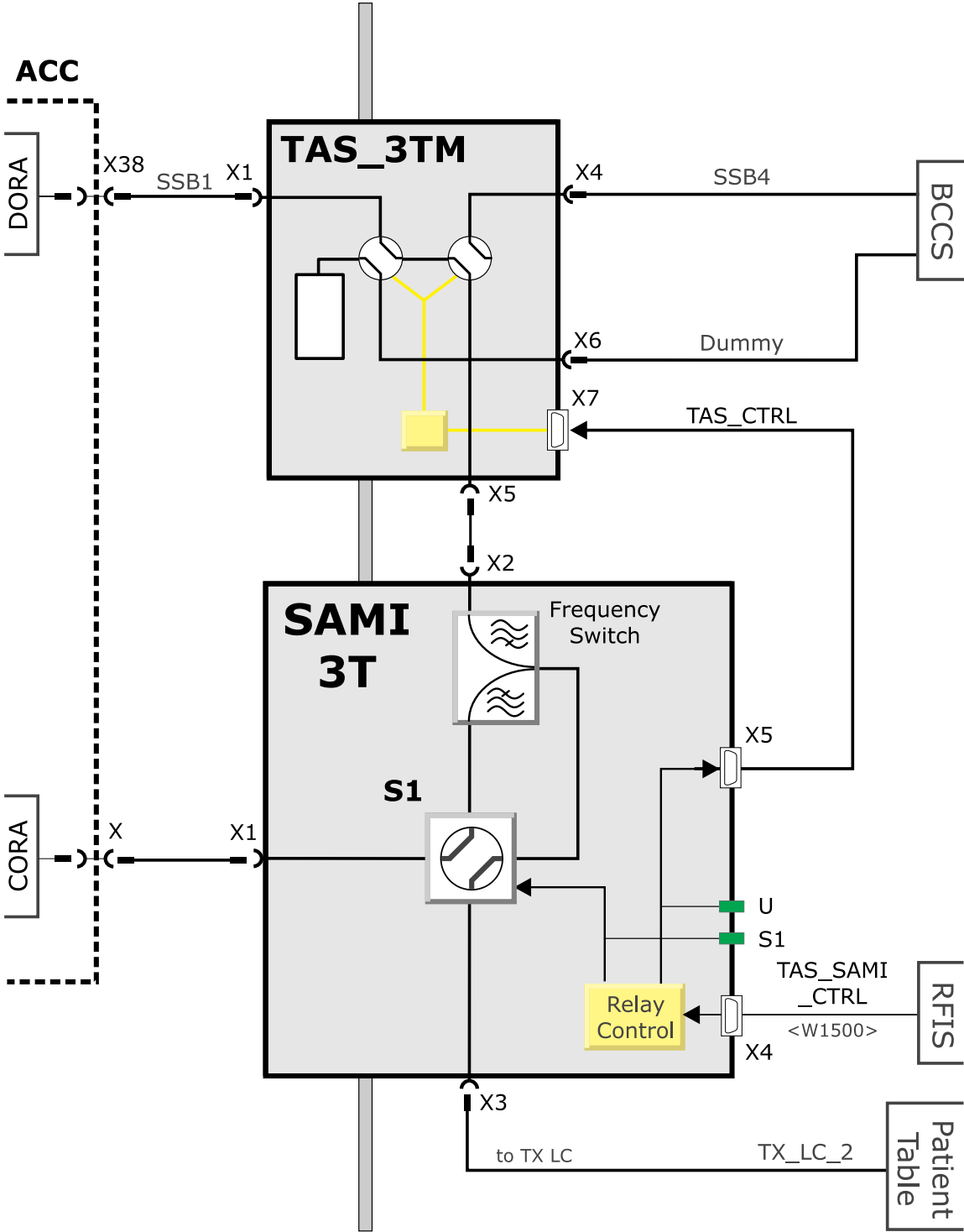
6.3.1 Function

6.3.1.1 Decoupling Experiment

During the reception of the x-nucleus spectrum, a RF pulse at the proton frequency is transmitted which acts to decouple surrounding hydrogen protons. The result is a more distinct line spectra and a better signal-to-noise ratio. S1 is in Combine position.

The **Decoupling** works as follows: A local Tx/Rx dual resonance coil is used. The broadband RFPA generates x-nucleus magnetization and during receiving the x-nucleus signal the 35kW RFPA applies a proton frequency RF pulse to the local coil in order to decouple the spectra and hence to get a better signal to noise ratio.

Fig. 17: SAMI_3T Block Diagram



6.4 BCCS_63_P

The Body Coil Channel Selector (BCCS) provides the necessary circuits to interface the RFPA to a circular polarized body coil. P = power: prepared for up to 35kW transmit power.

6.4.1 TX_Hybrid

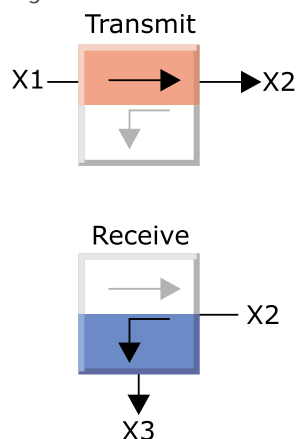
The TX_Hybrid effects a 90° phase shifting and power splitting of the transmit signal required by the CP Body Coil. To achieve this with the highest efficiency, the TX_Hybrid requires 50 ohms on all ports. The connection at port at X7 is terminated by the dummy load in the TAS.

The output characteristics of the Hybrid will be calibrated during system tune-up.

6.4.2 Transmit / receive switch

The T/R switch is required when imaging with the Body Coil. It is actively switched to provide a linear feed-through characteristic.

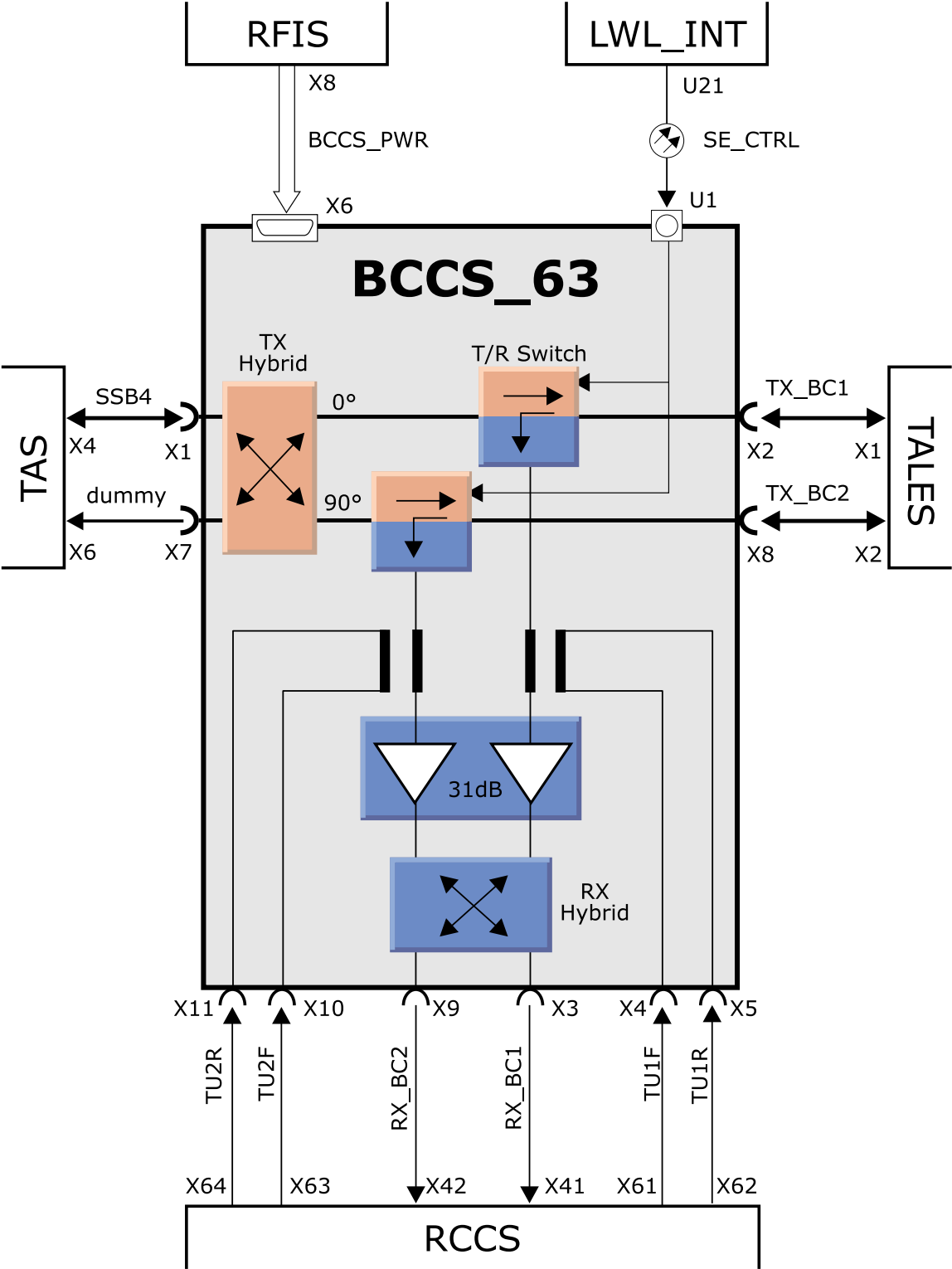
Fig. 18:



When transmitting, **SE_CTRL** is active closing two diode switches. The first switch allows the RF signal to pass between X1 and X2 unimpeded. The path between X1 and X3 however is electrically isolated with a very high impedance. It therefore prevents the signal from passing through to X3. Even if the **SE_CTRL** signal is missing, the switches will close as soon as the signal exceeds 0.7 V. The switch action just described will still work, although in non-linear mode.

When receiving, the **SE_CTRL** is inactive (0V) and X2 is connected to X3. Path X1 to X2 is electrically isolated as long as the MR signal does not exceed 0.7 V (it never will).

Fig. 19: BCCS Block Diagram



6.4.3 Directional Coupler

The directional coupler provides a port for tuning. A tune signal is applied alternatively to both the forward and reflected side of the directional coupler. The ratio of the amount of tune signal being coupled in the forward and reflected directions between the body coil and the amplifier is influenced by the impedance of the body coil.

6.4.4 Pre-amplifier

The narrowband, low noise preamp has a fixed gain of 31dB.

6.4.5 RX_Hybrid

The MR signal received from the CP Body Coil, after being amplified, is recombined by the RX_Hybrid. The output is taken from X3. It is possible to feed the uncombined MR-signals from both coil systems, but is not supported by software.

6.5 TALES

The Transmit Antenna Level Sensor (TALES) is a precision RMS RF voltmeter for measuring forward and reflected RF levels into the transmit coils (Body Coil or TX-local coil) for use in SAR calculations.

6.5.1 Demodulators

The RF signal picked up over the directional couplers is demodulated and filtered producing a proportional dc voltage.

6.5.2 MUX

During transmission, the AMC selects each of the six values via a MUX with the select signals **MH_SEL**. The signal **CV** is output to the receiver board where it is digitized and read in by the PCI_MON in the AMC for calculating the SAR values.

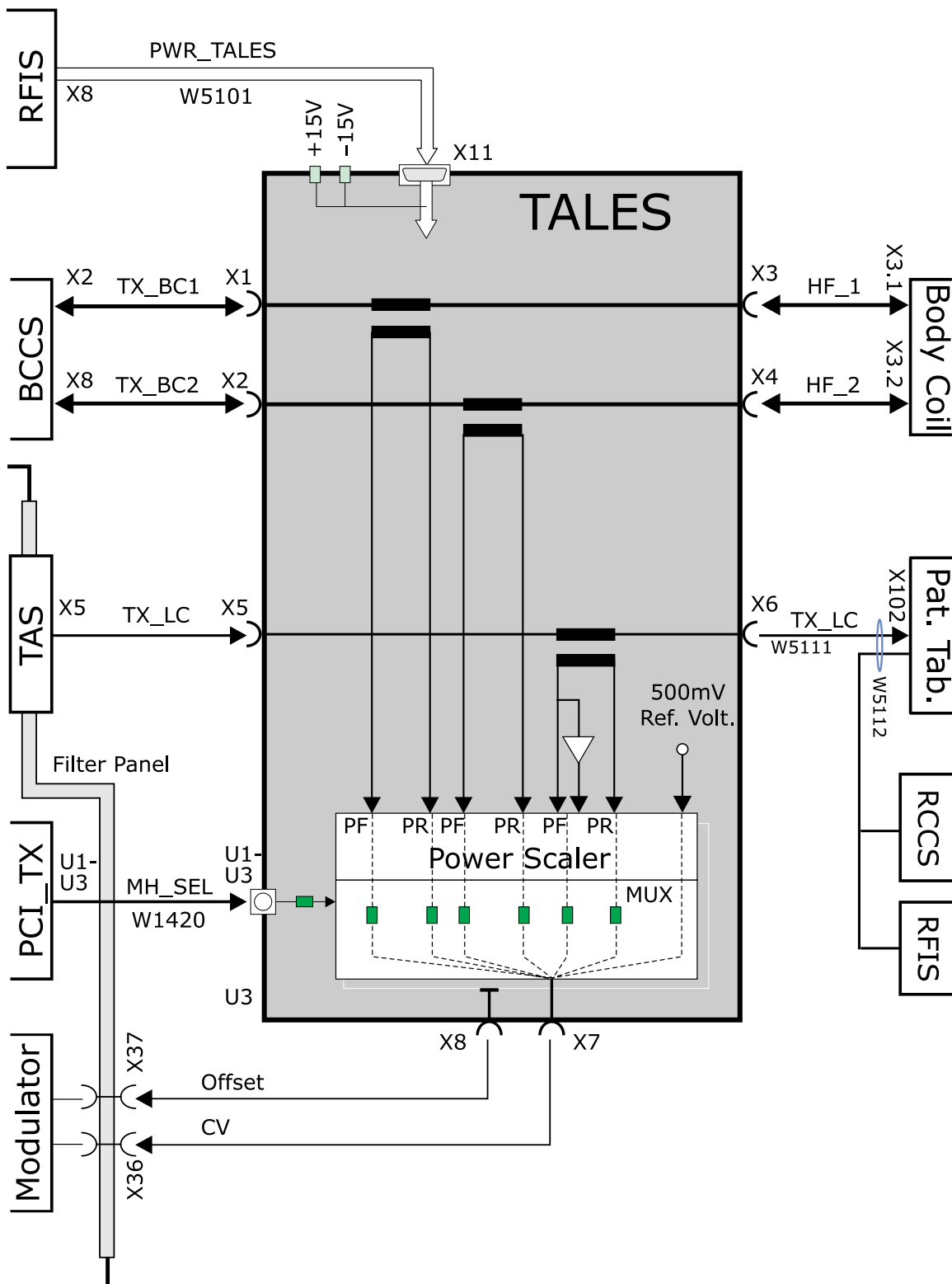
6.5.3 LEDs

The LEDs display the selected signal being measured by the AMC.

6.5.4 Reference voltage

In addition to the RF signal being measured, the TALES outputs a control dc reference voltage of 500mV. The reference is used to verify the accuracy of the signal coming from the TALES. The TALES must be replaced every two years as part of the preventive maintenance. The old one will be sent back to Erlangen for re-calibration.

Fig. 20: TALEs Block Diagram



6.6 RCCS

The Receive Coil Channel Selector (RCCS) is a 64 by 32 switch matrix with pre-amplification supporting 76 seamlessly integrated coil elements.

6.6.1 Switch Matrix

6.6.1.1 Local coil inputs

The Switch Matrix is constructed as a 64 by 32 array over which any one of the 64 local coil (LC) input channels can be switched to any of the 32 outputs channels. Due to the new Matrix Mode in contrast to the LCCS (Harmony/Symphony systems), it is no longer necessary, or possible, to combine two LC inputs to one output. Each input will be connected to one output only.

6.6.1.2 Body Coil inputs

There are 8 Body Coil (BC) inputs, only 2 of which are being currently used. The 8 BC inputs are not switched through the 64 x 32 matrix but over a set of 8 additional switches directly to the first 8 outputs, i.e. BC1 to output 1, BC2 to output 2, etc. As a consequence, each switched BC input makes the corresponding output unavailable for LC inputs. All unused LC and BC inputs are terminated with 50 ohm.

6.6.2 SGA gain switching

Each LC/BC input channel includes a switched gain amplifier (SGA) whose gain can be programmed to be either fixed low or high or dynamically switched during the sequence. The **RFAS_Low_Gain** signal is used for dynamic gain switching.

6.6.3 Tuning-multiplexer

The tuning signal **TTX** supplied by the Modulator board in the RFSU can be diverted via a multiplexer to one of 4 output paths all of which are going to the BCCS and are used for the Body Coil tune check procedure.

6.6.4 1-72 Multiplexer test path

The **TTX** signal can also be switched to any of the 72 input paths for testing purposes. The signal can be fed via to any of the amplifiers and switch matrix paths via the Tuning MUX and the 1-72 multiplexer to any of the amplifier and switch matrix paths. In this way, all RCCS amplifiers and matrix nodes can be tested.

6.6.5 LC Voltage supply & monitoring

The RCCS supplies the local coils with power for their preamplifiers via a 10.5 V supply voltage. The voltage is fed to the LC over the cables connecting the LC connectors and the RCCS inputs. The circuit responsible for generating and regulating the 10.5 V from the 15 V supply is also able to switch the 10.5 V on and off. Each of the 64 supply channels is monitored and has a corresponding LED for indicating a shorted supply. If the LED lights, it indicates there is a short somewhere between the RCCS output pin and the Matrix or local coil.

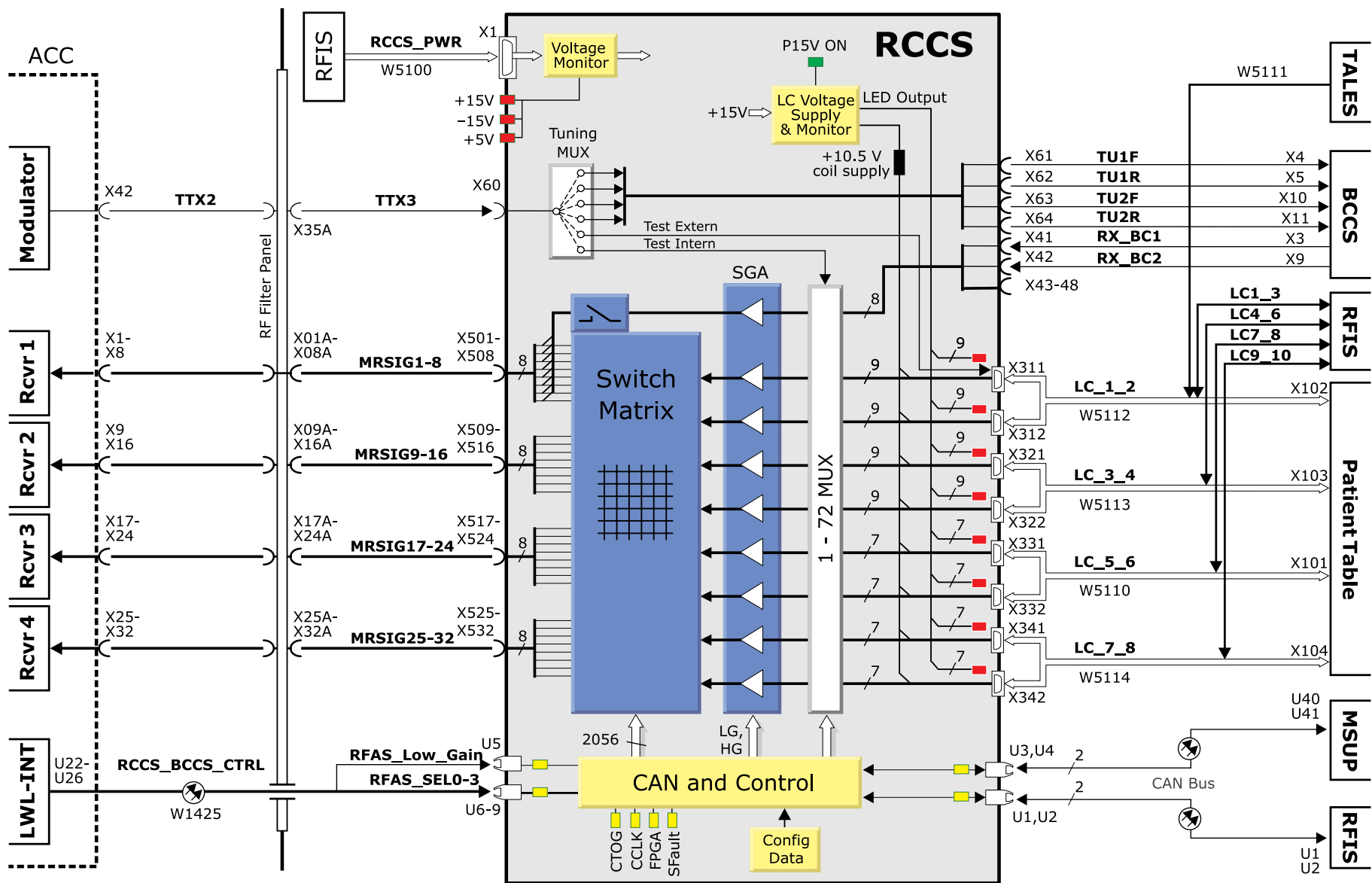
6.6.6 Settings

The setting of the Matrix for input to output switching, the setting of the Tuning Mux, the setting of the static or dynamic gain mode and high or low gain, and the setting of the Test Path Mux are made by the CAN and the control block. Up to 16 different settings can be stored in the CAN and control section and selected dynamically via the setting group address signals **RFAS_SEL 0-3**.

The settings are stored at the MPCU as well. If a particular setting has been previously loaded into the RCCS, the MPCU does not have to retransmit it to the RCCS.

For whole-body imaging, up to 13 of the 16 settings will be used for that purpose.

Fig. 21: Block Diagram of Receive Coil Channel Selector



6.6.7 LEDs

As a rule of thumb, the RCCS is in normal fault-free operation mode if no red LED is on but the one and only green LED is on.

Tab. 18 RCCS LED description

LED	Description
Green LEDs	
P15V ON	On as long as the +15V are supplied to the RCCS.
Yellow LEDs	
All yellow LEDs are status LEDs, i.e. they define the status of a certain signal. They can be used for troubleshooting but their status is not connected a priori to a failure mode.	
FPGA	FPGA ready signal - turns on shortly after turning on the DC supply for the RCCS. Indicates that all FPGAs are ready for operation.
CCLK	Control clock - is on if the internal RCCS clock is active. Is off when the RCCS CAN bus is in sleep mode.
CTOG	Control toggle - each status change of this LED indicates a data input into the RCCS FPGA
SFAULT	Sum Fault - This LED turns off whenever a sum voltage fault occurs i.e. whenever -15 V, +5 V, +15 V or +10.5 V are not OK.
CAN1, CAN2	Shows the status of the transmitting diode of the optical transmitter for the CAN-Bus interface.
GAIN	Shows the status of the RFAS_Low_Gain Control Signal. Light at this optical input turns all those SGA simultaneously to low gain that are programmed for dynamic gain setting.
Red LEDs	
P5V, P15V, N15V	ON if the corresponding voltage is too low.
10.5 V LC supply voltage	Each of the 64 LC inputs has an associated red LED labelled with the input number. If the 10.5 V LC supply voltage is overloaded or shorted, the monitor switches off the voltage for that channel and the corresponding red LED is enabled. The fault status is latched in order to catch sporadic or intermittent faults. To recover from this condition, the RCCS must be reset by rebooting the MR scanner.

6.6.8 Specifications

Tab. 19 RCCS specifications

Value	Specifications
frequency range	15 to 125 MHz

Value	Specifications
noise figure:	
low gain	max 10 dB
high gain	max 6dB

7.1 Overview

The Body Coil (BC) generates the B1 field for excitation of the proton nuclei in the MR-experiment.

- **Transmit-only mode** - whereby the body coil transmits the excitation pulse and one or more local coils receive the MR echoes.
- **Transmit/Receive mode** - the body coil is additionally used for receiving the MR echo signal.

The primary use of the Body Coil is transmitting the excitation pulses and localizing imaging of scouts or topology images used for slice positioning.

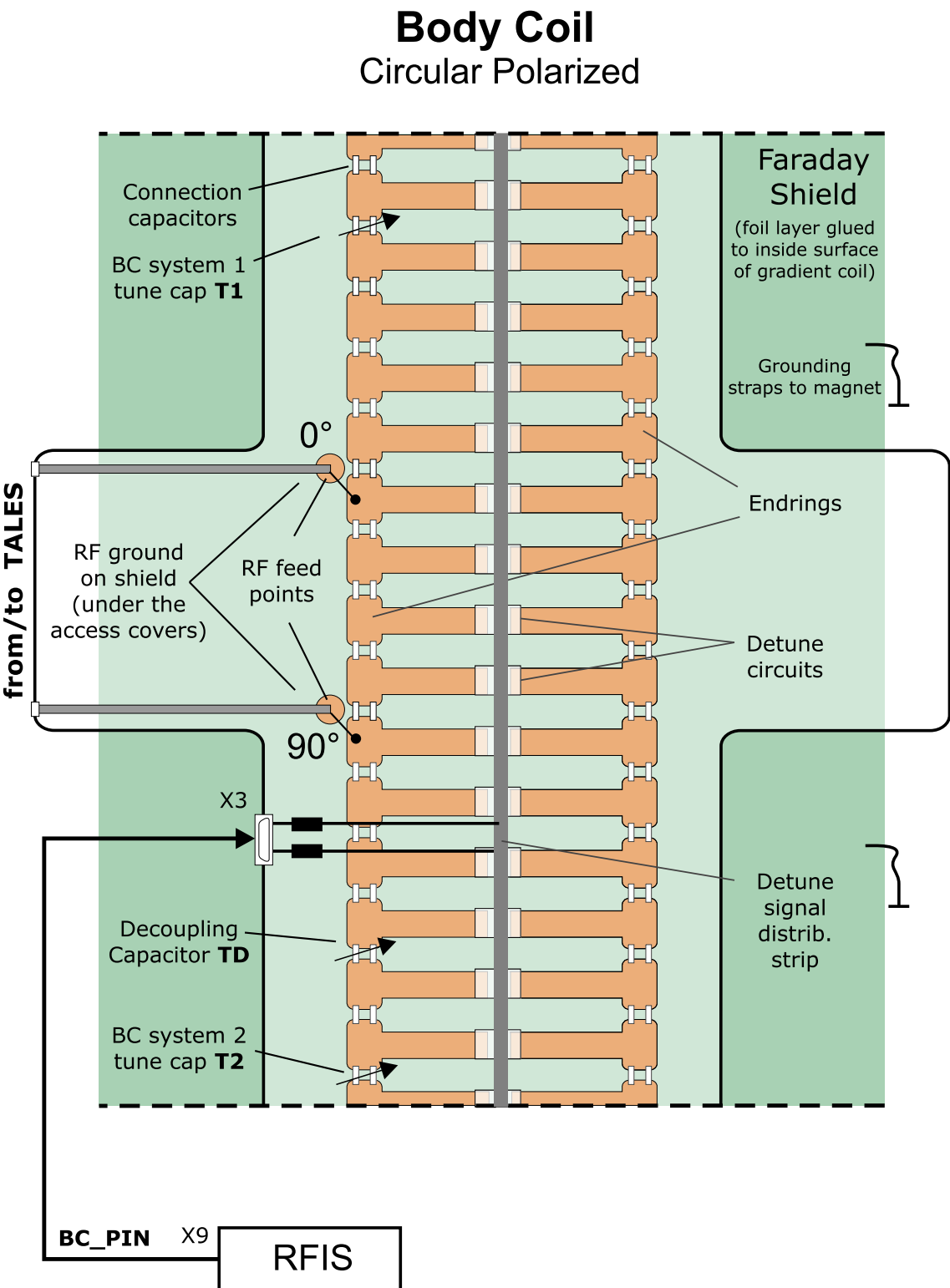
The Avanto Body Coil is a circularly polarized (CP) coil and based on the same high-pass birdcage design as used for the Symphony BC2. A principal advantage of the birdcage design is that the RF field amplitude drops very quickly outside the 500mm imaging volume thereby greatly reducing field ambiguity artifacts (aka third arm artifact) and also distributes the RF field more uniformly over the length of the coil.

In order to prevent the development of eddy currents on the conductive surfaces of the Body Coil, all large-area conductive structures will be lacerated to suppress eddy currents.



The Body Coil for Symphony Tim Upgrade systems is mechanically different from the Avanto coil due to the longer patient bore.

Fig. 22: Block Diagram of Avanto Body Coil



7.2 Function

When the incoming RF signal is at the resonant frequency of the coil, the current flow through the coil resonator elements is distributed through the coil elements via the end rings depending on the signal phase. Being a CP coil, decoupling between resonator systems is an important issue. Strong coupling between the two resonator systems leads to high transmitter reference values and hence reduces the performance of the RF-transmit path. A procedure for checking and, if necessary, optimizing the decoupling is located in the Service Software platform under Tune Up / BC Tuning.

The body coil is a resonant device transducing the current/voltage signals from the RF power amplifier to an electro-magnetic field. The magnetic component is designated as the B1 field.

When using local coils, a detune circuit brings the body coil to off-resonance to prevent coupling with the local coils and thus absorbing any MR echo signal. For receive-only local coils, the detuning is achieved dynamically (fast switching times). When using local coils capable of transmitting, the body coil will be statically detuned. PIN diodes are the devices used to accomplish the detuning. A control voltage of -30 V is applied for dynamic detuning, and a control voltage of approximately -440 V is used for static detuning. The detuning circuitry is integrated into the Body Coil.

The RF shield (Faraday shield) is laminated on the inside of the Gradient Coil and connected via the magnet shell to ground. The RF shield reduces B1 field losses into the gradient coil as well as serves to prevent possible disturbances.

7.2.1 Fixed Impedance Matching

In contrast to previous body coils, the Avanto body coil has been designed as a no-tune coil, that is, patient-dependent impedance matching is no longer necessary. The impedance matching is set to a fixed value optimized for heavy loads where the power requirement is greatest.

As a consequence, a very high reflection will occur in light load situations. These reflections will be seen by the TALEs and BCCS. In the BCCS, most of this reflection will be directed over the hybrid to the dummy load located in the TAS. Assuming a properly functioning BCCS, the RFPA should only see a maximum of 20% reflection in the worst case situation.

The maximum frequency shift as a result of a varying loading of the coil will be no greater than 200 kHz.

7.2.2 Static and Dynamic Detuning

PIN diodes, located on the detune circuits, are used to both dynamically and statically detune the Body Coil. The detune voltages are described in the RFIS section.

7.3 Specifications

Tab. 20 Body Coil Specifications

Value	Specifications
Center frequency	63.60 MHz
Frequency tolerance	100 kHz
Bandwidth	250 kHz
Variation 1 within band of 125 kHz	< 2.0 dB (peak-peak)
Variation 2 within band of 250 kHz	< 3.0 dB (peak-peak)
Asymmetry1 (+125 kHz versus -125 kHz)	< 1.0 dB
Asymmetry 2 (+250 kHz versus -250 kHz)	< 2.0 dB

8.1 Overview

The Body Coil (BC) is a resonant device transducing the current/voltage signals from the RF power amplifier to an electro-magnetic field. The magnetic component, designated as the B1 field, is what interacts with the proton nuclei in the MR experiment.

- **Transmit-only mode** - whereby the body coil transmits the excitation pulse and one or more local coils receive the MR echoes.
- **Transmit/Receive mode** - the body coil is additionally used for receiving the MR echo signal.

The primary use of the Body Coil is for transmitting excitation pulses and localizing imaging of scouts or topology images used for slice positioning. The Espree Body Coil is a new circularly polarized (CP) coil which has been integrated into the gradient coil. The integration of the Body Coil into the gradient coil saves much space and allows for a much larger bore opening. To prevent eddy currents from developing on the conductive surfaces of the Body Coil, all large-area conductive structures will be lacerated to suppress eddy currents.

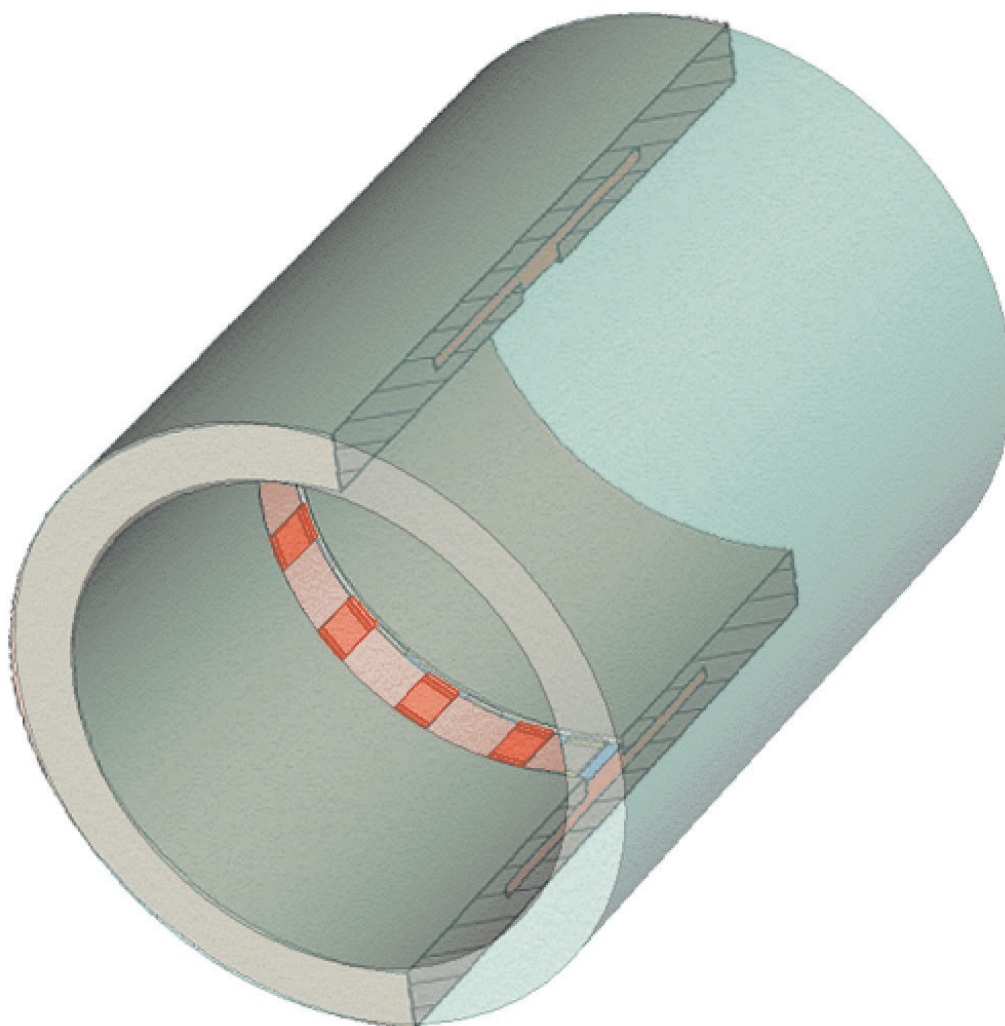
8.2 Function

When the incoming RF signal is at the resonant frequency of the coil, the current flow through the coil resonator elements is distributed through the coil elements via the end rings depending on the signal phase.

When using local coils, a detune circuit brings the body coil to off-resonance to prevent coupling with the local coils and thus absorbing any MR echo signal. For receive-only local coils, the detuning is achieved dynamically (fast switching times). When using local coils capable of transmitting, the body coil will be statically detuned. PIN diodes are the devices used to accomplish the detuning. A control voltage of -30 V is applied for dynamic detuning, and a control voltage of approximately -440 V is used for static detuning. The detuning circuitry is integrated into the Body Coil.

The RF shield (Faraday shield) is laminated on the inside of the gradient coil and connected via the magnet shell to ground. The RF shield reduces B1 field losses into the gradient coil as well as serves to prevent possible disturbances.

Fig. 23: Block Diagram of Espree Body Coil



8.2.1 Fixed Impedance Matching

The Espree body coil has been designed as a no-tune coil, that is, patient-dependent impedance matching is no longer necessary. The impedance matching is set to a fixed value optimized for heavy loads where power requirement is greatest. As a consequence, light loads will result in high reflections, however, since the power levels for light loads are lower, the RF power amplifier has much reserve to compensate. In the BCCS, most of this reflection will be directed via the hybrid to the dummy load located in the TAS. Assuming a properly functioning BCCS, the RFPA should only see a maximum of 20% reflection in worst case situations.

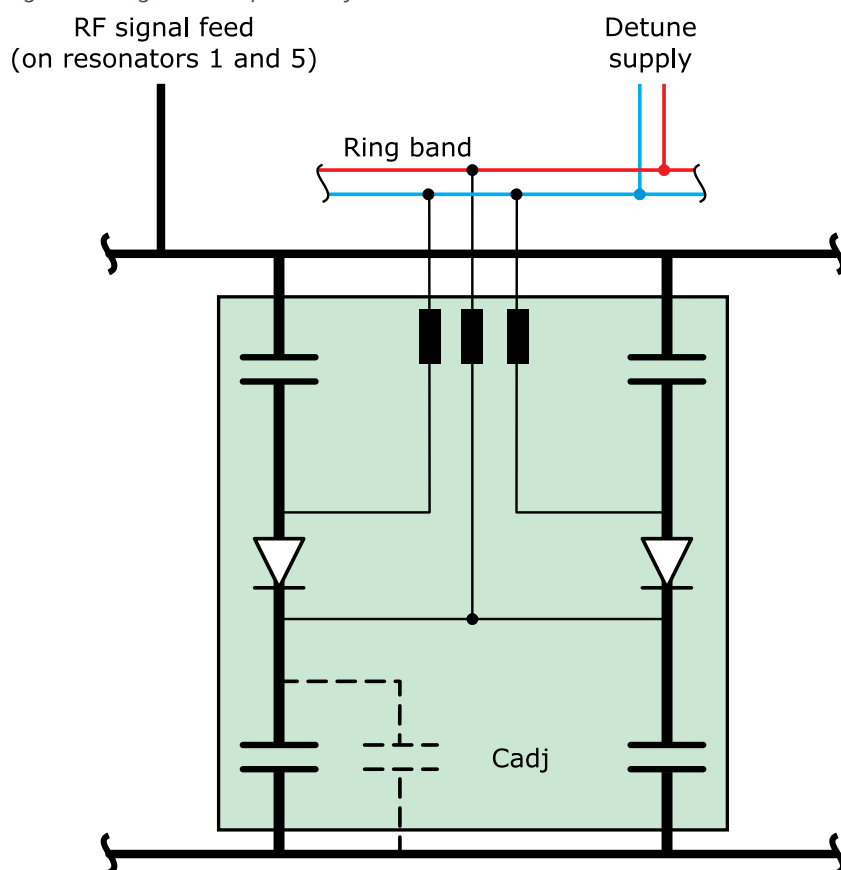
The maximum frequency shift as a result of a varying loading of the coil will be no greater than 200 kHz.

8.2.2 Static and Dynamic Detuning

PIN diodes, located on the resonator boards, are used to both dynamically and statically detune the Body Coil. The detune voltages are described in the RFIS section. The detune signals are fed to the 16 resonator boards via a ring band. When the diodes are closed (positive detune signal), the resonators are tuned. Opening the diodes with a reverse voltage opens the diodes and the resonators are detuned.

The resonator boards are moulded within the gradient coil and cannot be removed or replaced.

Fig. 24: Diagram of Espree Body Coil Resonator



8.3 Specifications

Tab. 21 Body Coil Specifications

Value	Specifications
Center frequency	63.60 MHz
Frequency tolerance	100 kHz
Bandwidth	250 kHz
Variation 1 within band of 125 kHz	< 2.0 dB (peak-peak)
Variation 2 within band of 250 kHz	< 3.0 dB (peak-peak)
Asymmetry1 (+125 kHz versus -125 kHz)	< 1.0 dB
Asymmetry 2 (+250 kHz versus -250 kHz)	< 2.0 dB

The RF Infrastructure (RFIS) contains the electronics providing:

- control signals for the dynamic and static detuning PIN diodes in the both the matrix and local coils and the Body Coil
- the detection and recognition of the connected matrix and local coils

The RFIS is built in a modular fashion consisting of a main motherboard and functional plug-in PCBs.

Fig. 25: RFIS Location (Avanto, Espree systems)

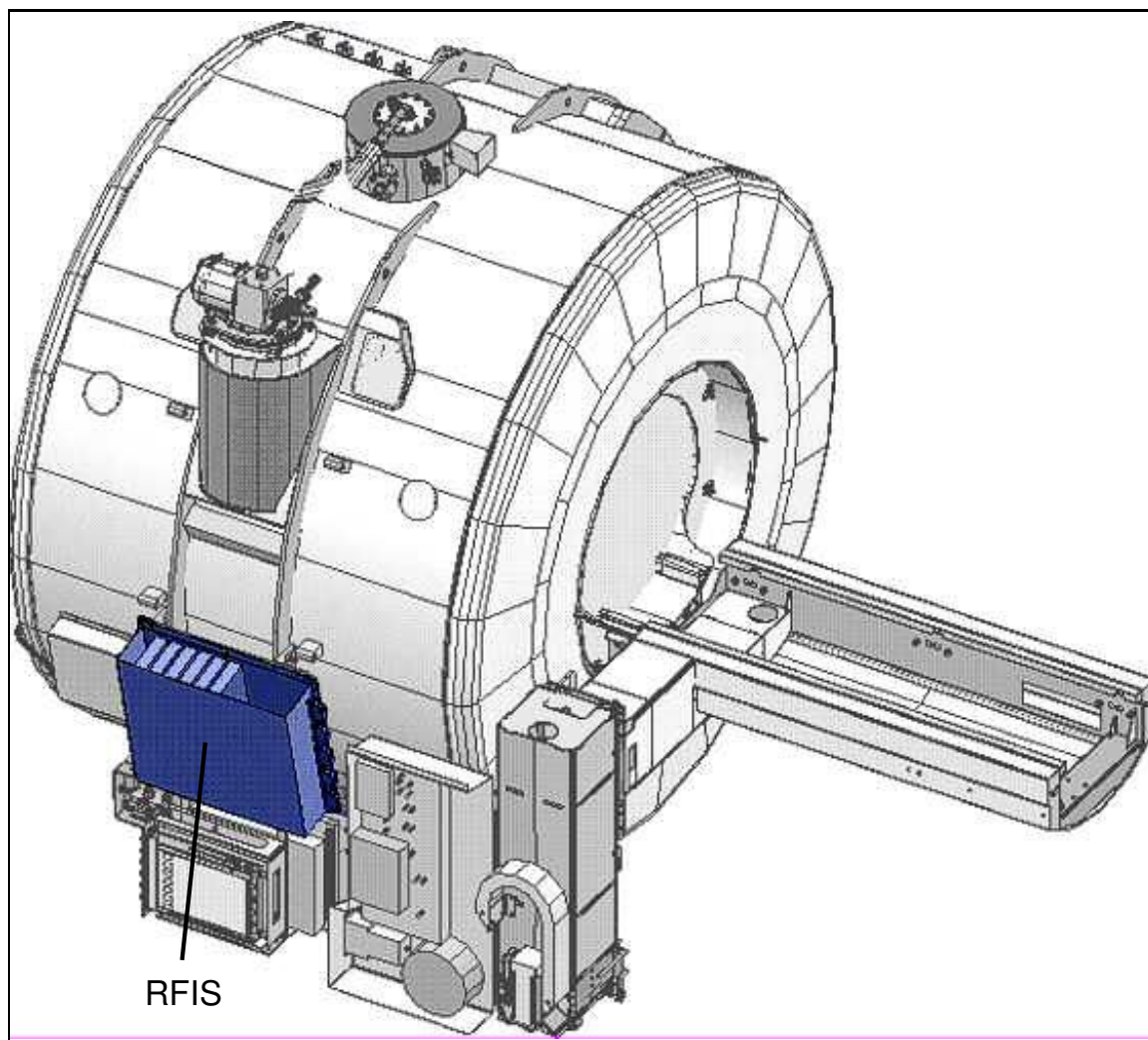
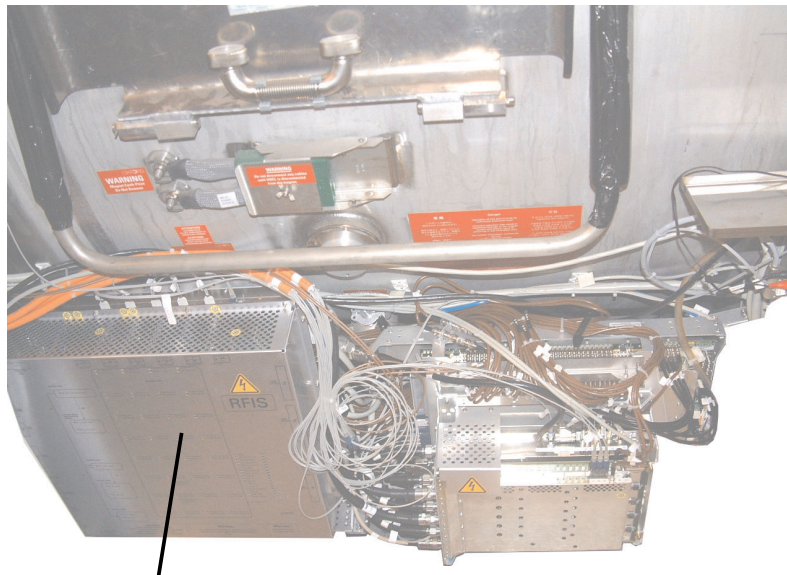
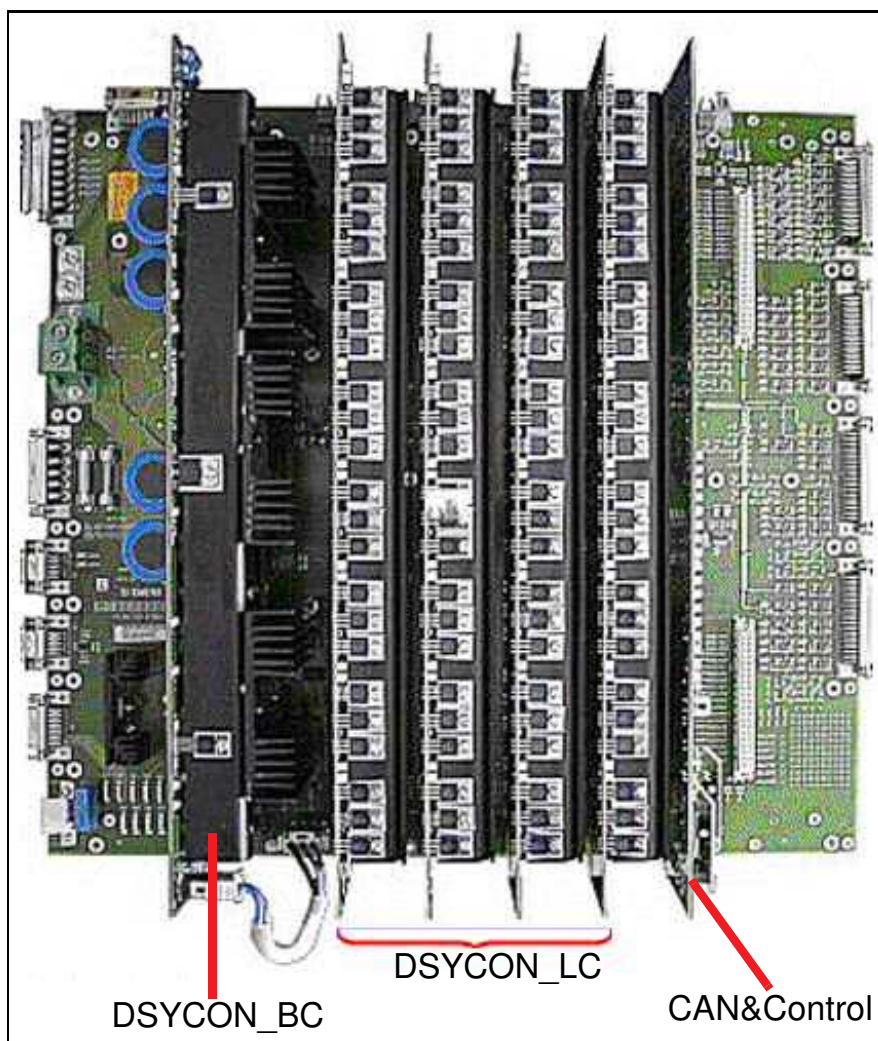


Fig. 26: RFIS Location (Symphony Tim Upgrade Systems)



RFIS

Fig. 27: RFIS Assembly

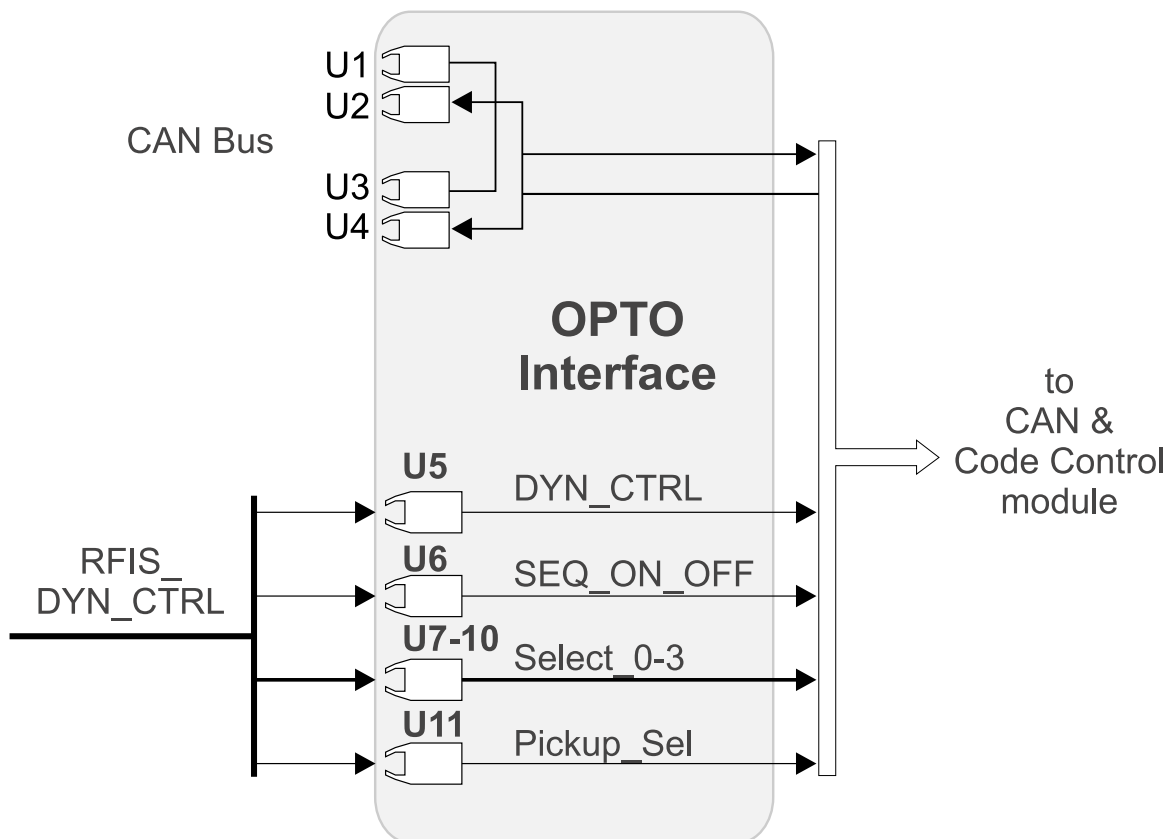


9.1 Motherboard

The motherboard provides a common interface to the periphery and the functional plug-in PCBs.

9.1.1 OPTO interface

Fig. 28: RFIS OPTO Interface Diagram



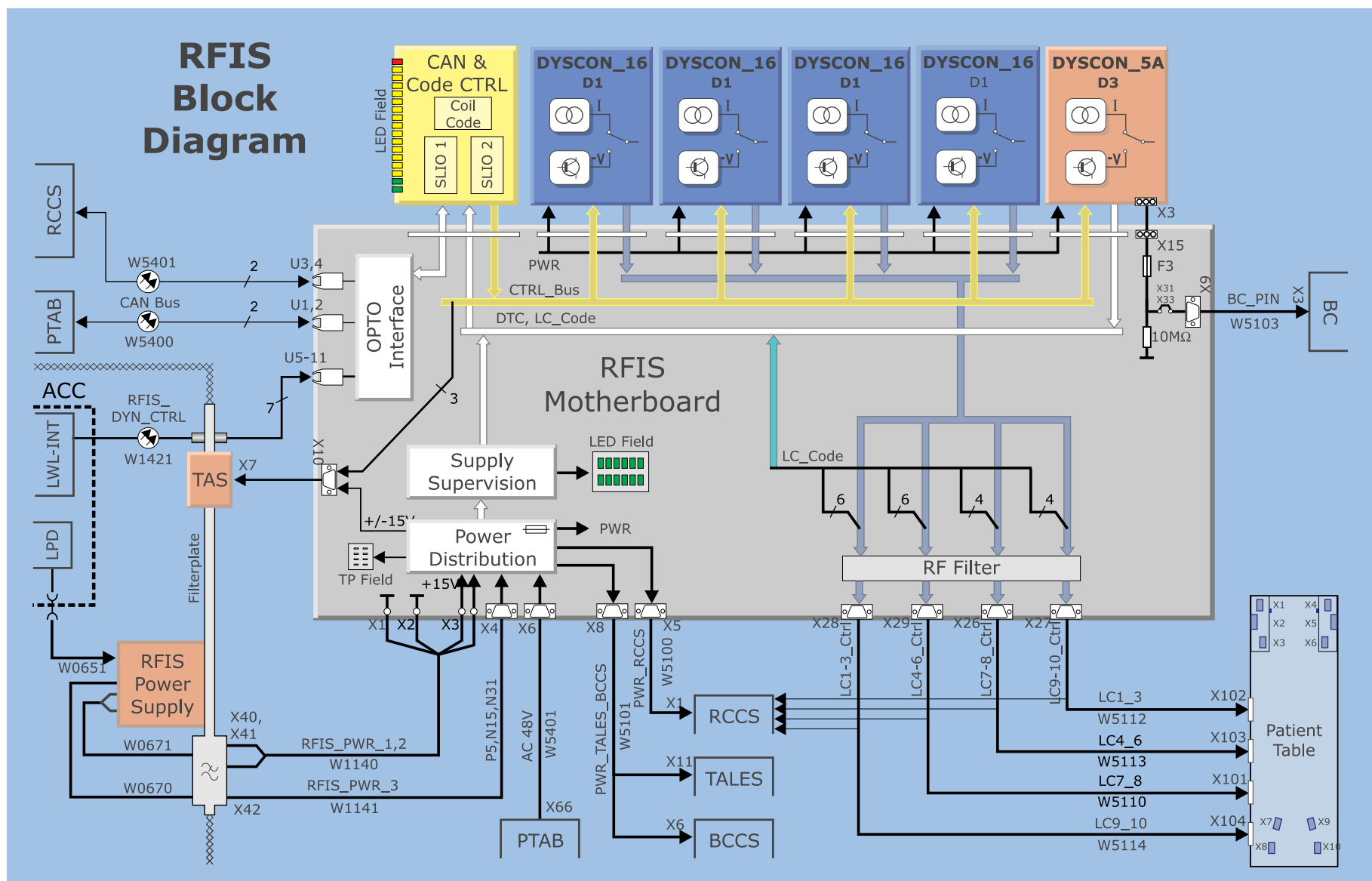
The OPTO interface converts incoming optical signals into electrical and vice versa. The following are the FO signals coming in and going out from the RFIS:

- DYN_CTRL - indicates the transmit or receive mode
- Seq_On_Off - signal for deactivating all PIN diode signals when no sequence is running (to prevent temp increases)
- Select_0-3 - select signals for the selection of the desired mode settings (1 of 16)

9.1.2 LEDs

Each of the voltage monitors have an associated LED. When the LED is on, it indicates that voltage is present within the monitored tolerances.

Fig. 29: RFIS Block Diagram



9.1.3 Power Distribution

This block is responsible for distributing the various operating voltages to the RF components within the RF room.

Tab. 22 RFAS voltages

Monitored voltage name	Remarks
DIG_P5	Extracted from the P15V supply via a 3-pin regulator
ANA_P15V	Supplied by the RFIS Power Supply
ANA_N15A	Supplied by the RFIS Power Supply
ANA_N31V	Supplied by the RFIS power supply
AC48V	Supplied by the patient table electronics
ANA_P31V	Reserved for future use
ANA_NRES	Reserved for future use
RFAS_SUPPLY	A composite signal consisting of the outputs of the F1 fused voltage, F2 fused voltage and the main P5V supply monitors

9.1.4 Jumpers

Tab. 23 RFIS jumpering

Jumper	Pins	Description
x51	2-3	AC48 monitoring (enabled)
x52	1-2	ANA_P31V monitoring (not enabled)
x53	1-2	Reserved voltage monitor (not enabled)

9.1.5 Supply supervision

The supply voltages for all the units being supplied by the RFIS are monitored here. The status of the voltages is displayed via LEDs. In the event of a voltage failure a **N_Supply_Int** interrupt is generated and passed on to the AMC via the CAN and Code Control plug-in module.

Tab. 24 RFIS Supply supervision

Voltage	Use	Monitored threshold
P5V	RCCS	4.25 V $\pm$ 5 %
P15V	RFIS and all RF components	13.46 V $\pm$ 5 %
P15V TXComp	TAS, SAMI, TALES, BCCS, Pickup MUX	13.46 V $\pm$ 5 %
P15V RXComp	RCCS	13.46 V $\pm$ 5 %

Voltage	Use	Monitored threshold
N15V	RFIS and all RF components	- 13.46 V $\pm$ 5 %
N31V	RFIS	- 27.48 V $\pm$ 5 %
AC48V	RFIS	37 VAC $\pm$ 3 V

9.1.6 LEDs

Each of the voltage monitors have an associated LED. When the LED is on, it indicates the voltage is present and within the monitored tolerances.

The diagram illustrates the power distribution and supply supervision logic for the RFIS Motherboard. It is divided into three main functional blocks: Supply Distribution, Supervision RFAS Supplies, and Supervision RFIS Supplies.

Supply Distribution: This block shows the input power sources (PTAB and RFIS P.S.) and their distribution to various components. The PTAB provides P15V to a Regulator (Reg) and AC48V. The RFIS P.S. provides P15V, AC48V, and various RFAS supplies (RFAS_P5V, RFAS_TXCOMP_P15V, RFAS_RXCOMP_P15V) through connectors X1, X2, X3, X4, X5, X6, X7, and X8. The P15V is regulated to DIG_P5V (only for RFIS) and distributed to other components.

Supervision RFAS Supplies: This block shows the supervision logic for the RFAS supplies. It includes RFAS_TXCOMP_P15V, RFAS_RXCOMP_P15V, and RFAS_P5V. These supplies are monitored by LEDs (indicated by green squares) and connected to an AND gate (&). The output of this gate is RFAS_Supply, which is connected to the Latch and the N_Supply_Intr input of the CAN & Code Control block.

Supervision RFIS Supplies: This block shows the supervision logic for the RFIS supplies. It includes DIG_P5V, ANA_NRES, ANA_P15V, ANA_N15V, ANA_N31V, ANA_P31V, and AC48V. These supplies are monitored by LEDs (indicated by green squares) and connected to an AND gate (&). The output of this gate is connected to the Latch and the N_Supply_Intr input of the CAN & Code Control block.

Supply Supervision RFIS Motherboard: This block shows the overall supervision logic. It includes a Latch, a DTC (Data Transfer Control) block, and a CAN & Code Control block. The Latch is connected to the RFAS_Supply and the output of the AND gate from the Supervision RFIS Supplies. The DTC block is connected to the Latch and the N_EN_Supply_Err input of the CAN & Code Control block. The CAN & Code Control block is connected to the N_Supply_Intr input of the Latch and the Volt_OK output of the Latch.

LEDs indicate LOW voltages. Upper limits are not monitored.

9.2 DYSCON_5A module

This module serves to control the detuning PIN diodes in the Body Coil for dynamic and static detuning.

9.2.1 Dynamic detuning

The switch and monitor block provides output switching of the dynamic detuning supply levels of +12V@5 A or -31V for tuning and detuning the Body Coil respectively via the **BC_PIN_CTRL** signal.

9.2.2 Static detuning

When using transmit- local coils the Body Coil will be statically detuned with a high voltage of approximately -450V being applied to the Body Coil PIN diodes for the duration of the use of the transmit local coil. The static detuning voltage is enabled with the **BC_Dyn_On_Static_Off**. See LED table for logic levels.



WARNING

There are life-threatening voltages present on the DYSCON_BC board. Exercise caution when servicing

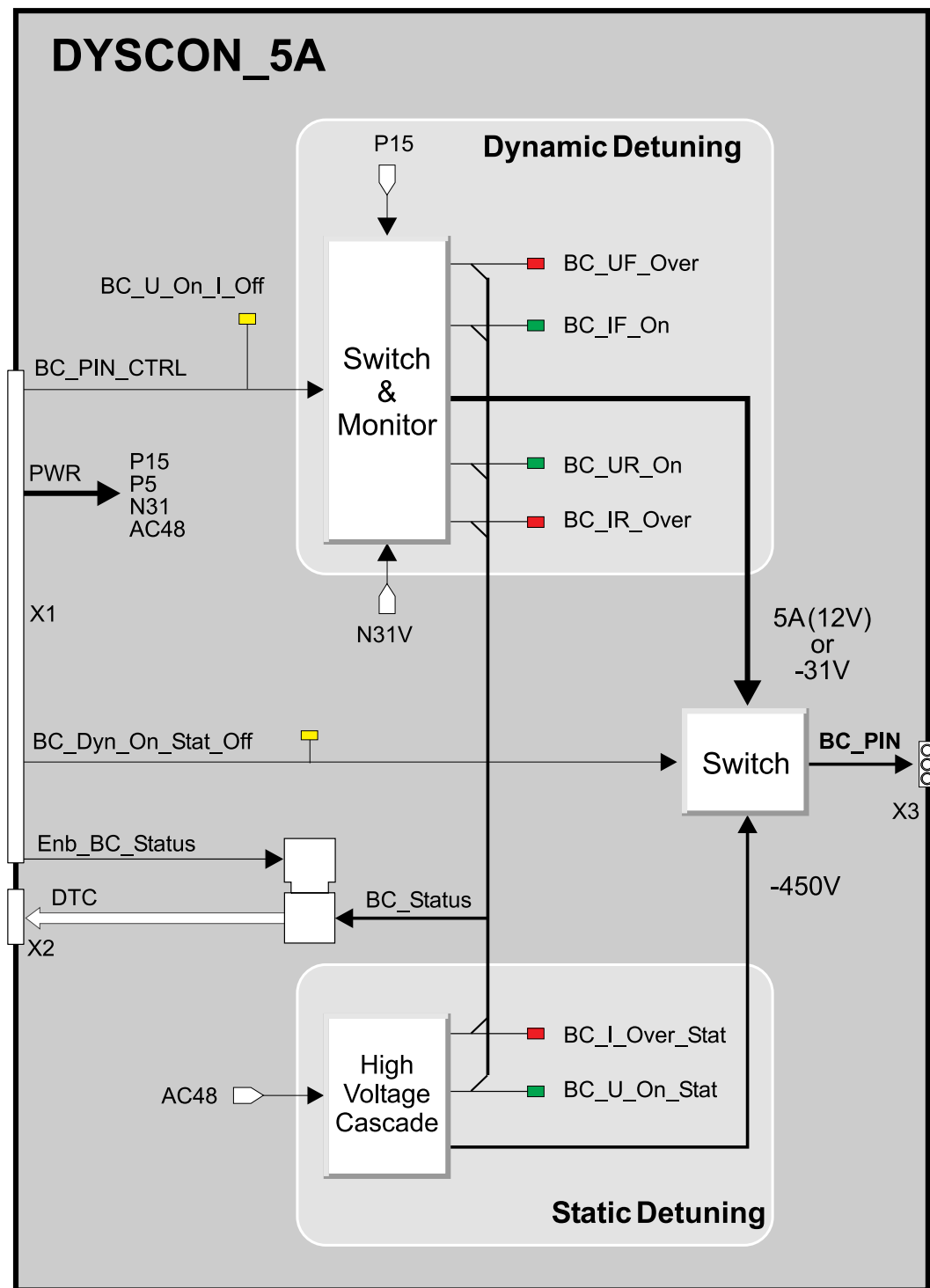
»

9.2.3 Monitoring

The dynamic and static detune signal levels are monitored for proper levels. See LED description for explanations.

The CAN module reads in the status of the monitoring by applying the **En_BC_Status** signal. This is done at every system boot, scanner reboot, when service tests are run and prior to sequence start.

Fig. 31: DYSCON_BC Block Diagram



9.2.4 LEDs

Tab. 25 Monitor LED statuses

LED	ON	OFF
BC_U_On_I_Off	-31V applied to PIN diodes, de-tuning the coil	+15V@5A is applied to the PIN diodes, tuning the coil
BC_Dyn_On_Stat_Off	BC dynamically detuned	BC statically detuned (-450 V applied)
BC_UF_OVR	Dynamic current source at max voltage (e.g. a break in the cable)	Voltage level of current source is ok
BC_IF_ON	Dynamic current output level is ok	Dynamic current output level is too low
BC_UR_ON	Reverse voltage ok	Reverse voltage too low
BC_IR_OVR	Too much current in reverse mode	Current level in reverse mode ok
BC_I_OVR_STAT	Current at static reverse voltage too high	current level in reverse mode ok
BC_U_ON_STAT	Reverse voltage level ok	Reverse voltage too low

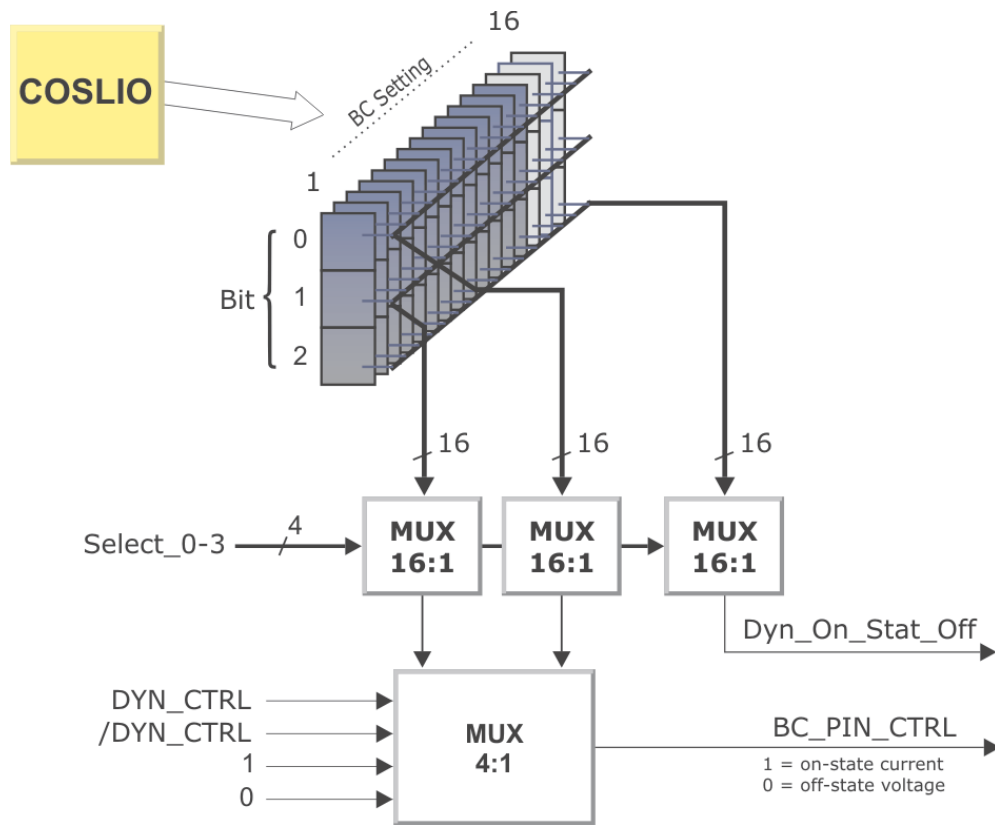
9.2.5 BC settings

The BC_PIN_CTRL and Dyn_On_Stat_Off signals are generated in the FPGA on the CAN module according to the circuitry shown in Figure 38. The circuit contains a mode table consisting of a 3 x 16 matrix. The COSLIO receives the values from NUMARIS and writes these into the mode table. Each bit-layer of the table is input to a 16:1 MUX. The dynamic control signals **Select 0-3** are used to address these 16:1 MUXs. Mode table bits 0 and 1 in turn address a 4:1 MUX applying to the output signal **BC_PIN_CTRL** one of the four signals: DYN_CTRL, DYN_CTRL inverted, 1, or 0.

The bit 2 layer is output via the third 16:1 MUX as **Dyn_On_Stat_Off**.

The 16 settings correspond to positions on the patient table, of which there are only 13, therefore the last 3 settings (designated by gray boxes in the graphic) are not required. The concept of this variable setting is to allow to control the Body Coil dynamically for Move During Scan examinations.

Fig. 32: BC Mode Table



9.3 DYSCON_16 plug-in module

Each module contains 16 PIN diode switching circuits for the dynamic detuning of local coil elements. The signals can be programmed to be static during the sequence (i.e. the Body Coil is being used as receive coil) or to be dynamic, i.e. the Body Coil is used only for transmission and the local coils are used for reception. The signal programming is achieved on the CAN & Code Control module. The dynamic control signal

The RFIS motherboard can accept up to 4 DYSCON_16 modules to control a total of 64 coil elements.

9.3.1 LEDs

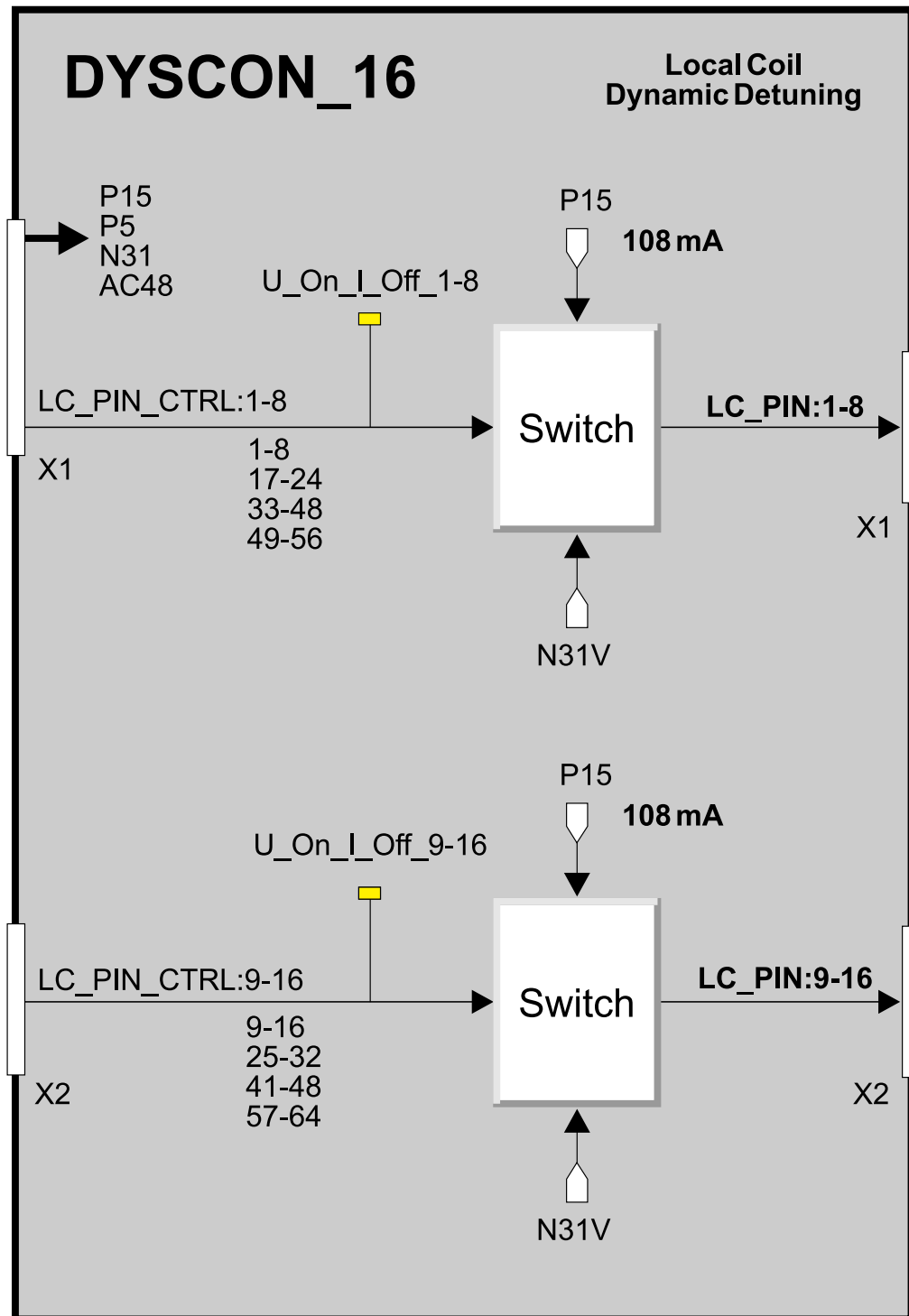
Tab. 26 DXSCON_16 LEDs

LED	Output	Coil
On	Voltage	Tuned
Off	Current	Detuned

9.3.2 Monitoring

There is no monitoring circuitry for this module. The module can be tested with the service tests and the specially designed service plug. See Troubleshooting Guide.

Fig. 33: DYSCON_LC Block Diagram



9.3.3 LC settings

The LC_PIN_CTRL signals are generated in the FPGA on the CAN Module according to the circuitry shown in (→ Fig. 35 Page 81). The circuit contains a mode table consisting of a 2 x 16 x 64 bit matrix. The COSLIO receives the values from NUMARIS and writes these into the mode table. Each bit-layer of the table is input to a 16:1 MUX. The dynamic control signals **Select 0-3** are used to address these 16:1 MUXs. Mode table bits 0 and 1 in turn address a 4:1 MUX applying to the output signal **LC_PIN_CTRL** one of the four signals: DYN_CTRL, DYN_CTRL inverted, 1, or 0.

The 16 settings correspond to positions on the patient table, of which there are only 13. For this reason, the last 3 LC settings (designated by gray boxes in the graphic) are not required. The concept of this variable setting is to allow control the matrix coils dynamically for Move During Scan examinations.

Fig. 34: Table Zones

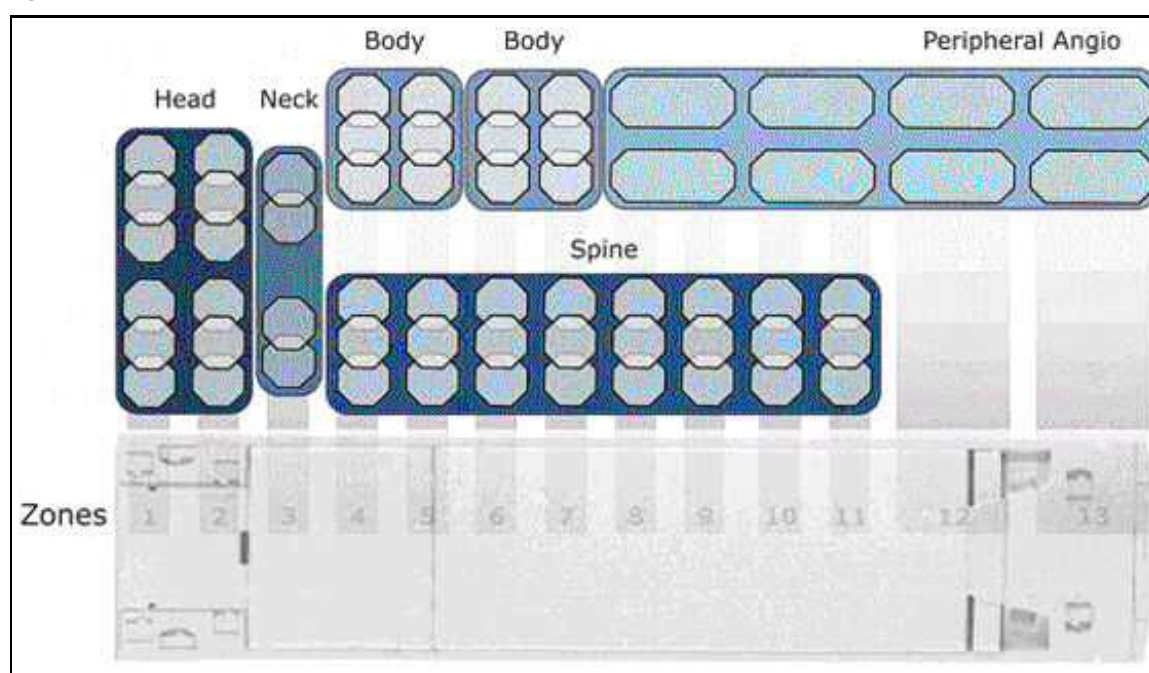
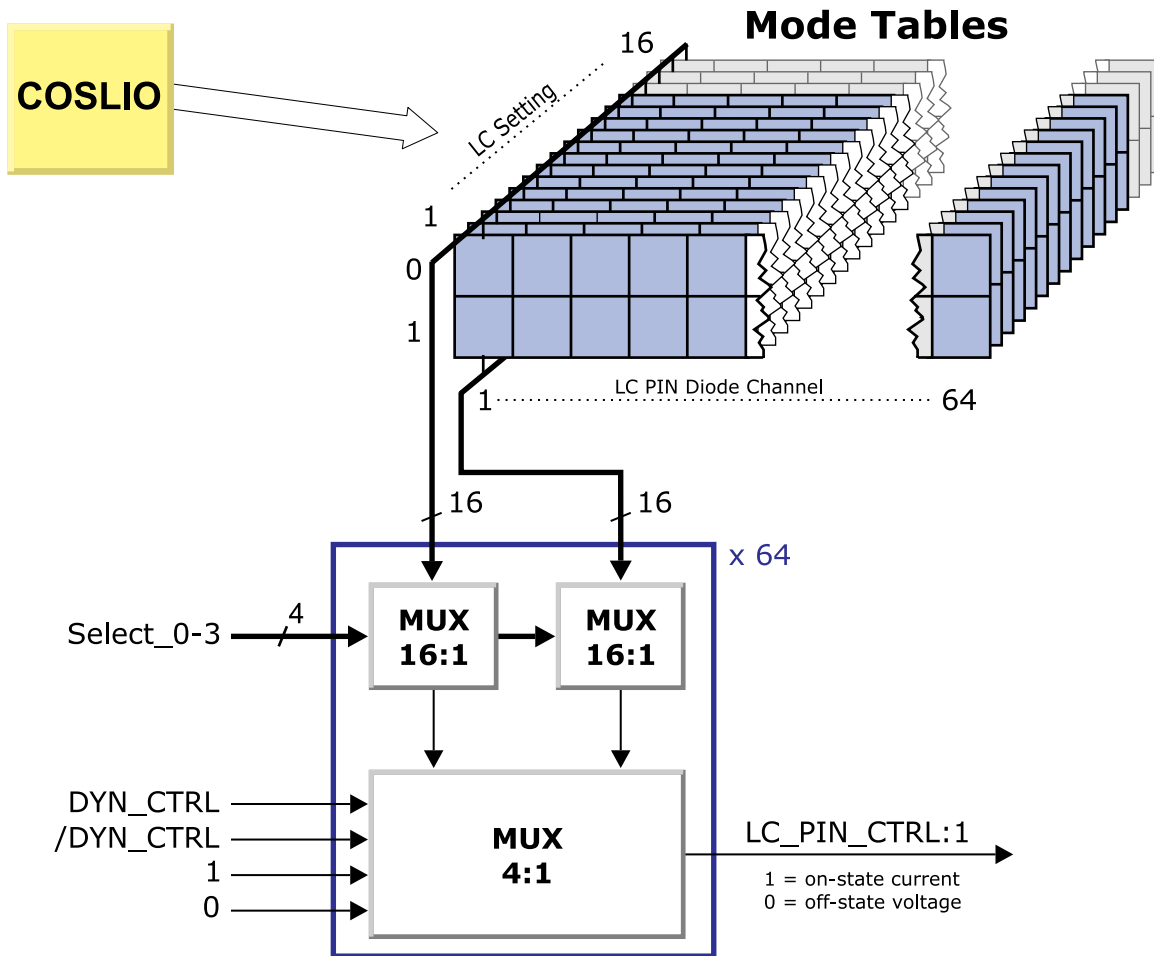


Fig. 35: LC Settings Memory

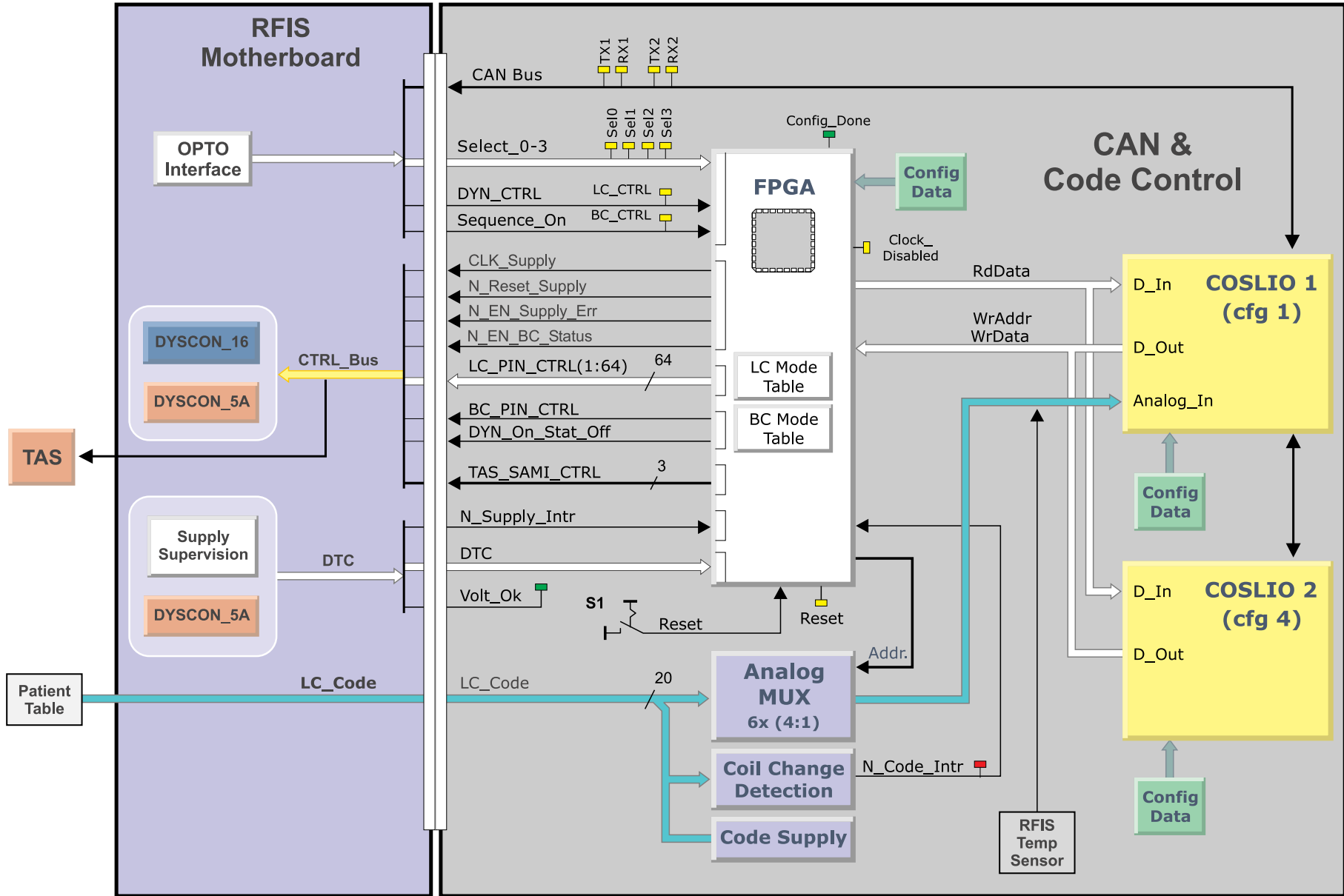


9.4 CAN & Code CTRL module

This module fulfills the following functions:

- Provides a CAN communications channel to system control for receiving control data and sending feedback of the monitoring status
- Generate the PIN diode control and timing signals
- Detect the connected local coils via their coil codes and report these to the System Control
- Detect coil changes during a measurement and report this to the System Control
- Report supply status and RFIS temperature
- Handling of sleep mode
- Detect and report supply errors occurring during a measurement

Fig. 36: Block Diagram of RFIS CAN & Code Control Module



9.4.1 Coil Code recognition

The coil section has two functions:

- Coil code detection
- Coil code change recognition

9.4.2 Coil Code detection

To recognize the matrix or local coils which are connected, a coil code recognition circuit has been realized using resistive coding. A 10V source with an internal impedance of 1.5 k Ω is terminated with resistors inside the coil connectors. These coding resistors have 13 defined values and since each coil connector is provided with two code lines per coil code, 2 to the power of 14 coil code combinations are possible. A list of the Matrix Coil codes can be found in the corresponding table below.

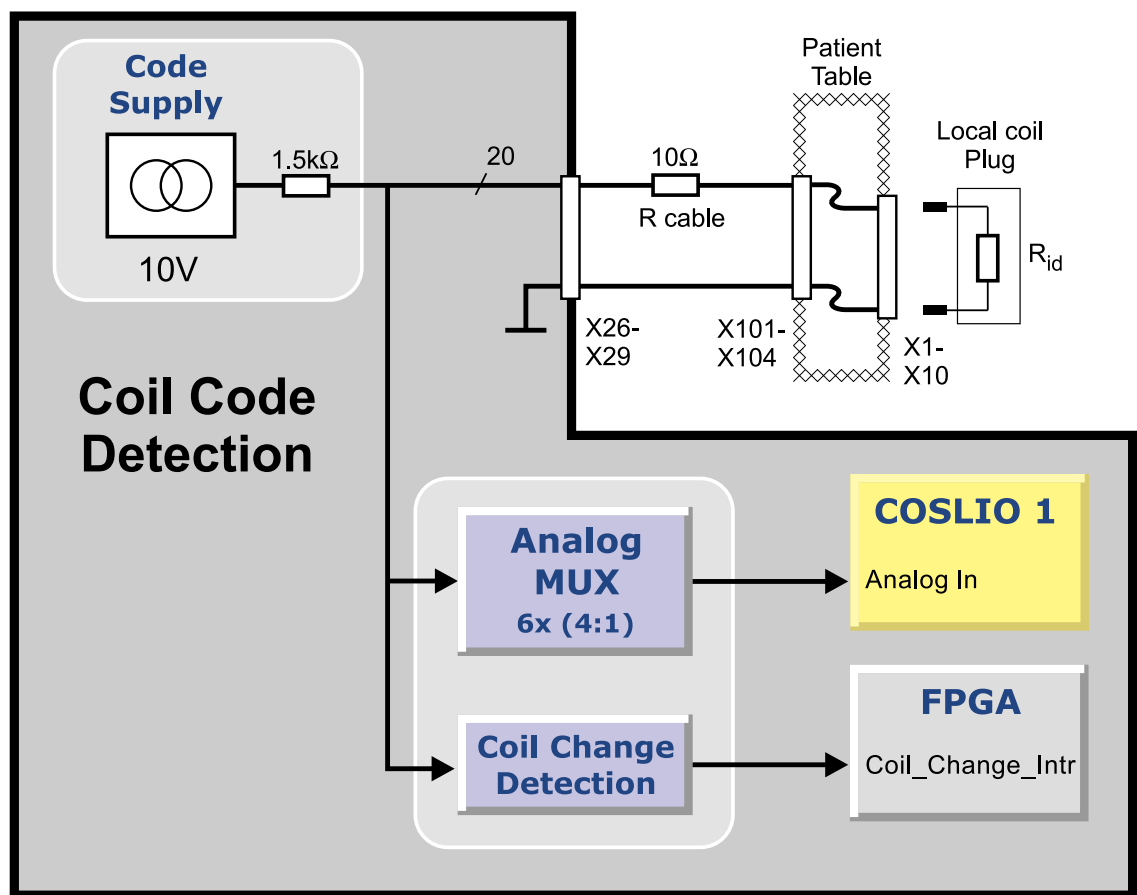
The code circuit is floating with respect to the rest of the coil electronics and thus uses a separate ground!

9.4.3 Coil Code change

A current source generates a constant current which is fed to the coil connector. The local coils are equipped with one or more resistors connected to the 8 code signal lines. Where a resistor is found, a voltage drop is generated which is read and recognized by the CAN module controller.

During the sequence, the coil code is continuously monitored. If it changes (e.g. the patient pulls a coil connector), an interrupt to the CAN module will be generated. This will cause the sequence to be stopped.

Fig. 37: RFIS Coil Code Detection Diagram



Tab. 27 Code resistor values

Coil code in Hex	Resistance in Ohm	Voltage at MUX	Notes
0	<63	0,314	Signifies a short = error!

Coil code in Hex	Resistance in Ohm	Voltage at MUX	Notes
1	132 - 162	0,938	
2	256 - 291	1,564	
3	402 - 444	2,19	
4	577 - 627	2,816	
5	790 - 853	3,442	
6	1056 - 1135	4,066	
7	1397 - 1500	4,692	
8	1849 - 1989	5,318	
9	2478 - 2680	5,944	
A	3412 - 3729	6,570	
B	4946 - 5511	7,196	
C	7930 - 9207	7,82	
D	16259 - 21493	8,446	
E	14643	9,072	Signifies no coils connected
F	37572	9,698	Signifies an open = error!

Tab. 28 Matrix Coil Codes

Coil	Connector	Code
Head Matrix Upper	X6	5D
Head Matrix Lower	X3	AD
Neck Matrix Upper	X2	B7
Neck Matrix Lower	X2	37
Spine Matrix	X7	B1
Spine Matrix	X9	B2
Body Matrix 0°	X1	53
Body Matrix 180°	X10	57
PAA Matrix	X4, X8	51

Tab. 29 Local Coil Codes

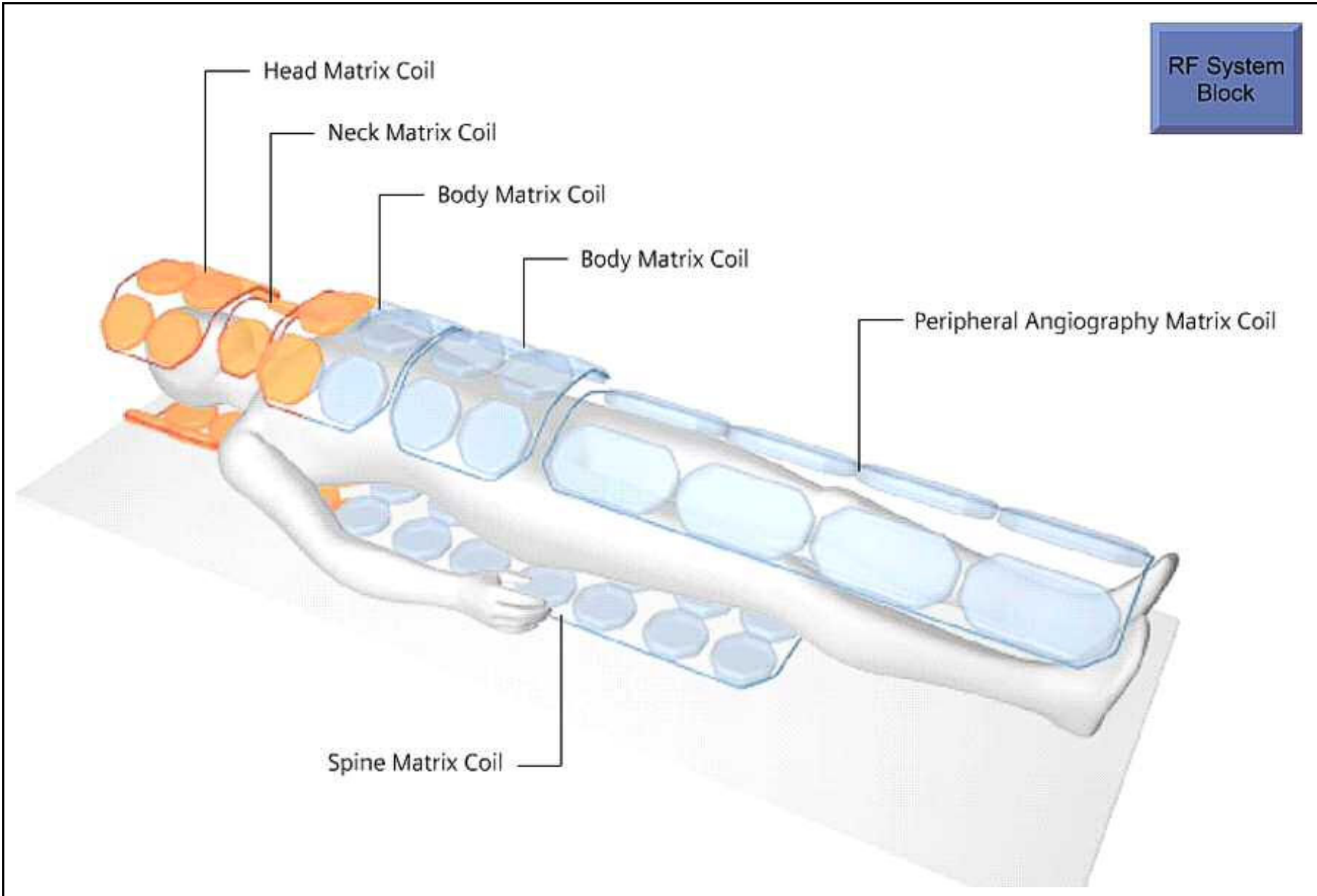
Short Name	Coil Name	Coil Code
	Reserved for Siemens Coil	11

Short Name	Coil Name	Coil Code
HLS	Heart Liver (Spec.)	12
USI	USAI Head Spec	14
EX	Extremity Coil	2b
DL	Double Loop Array Left	2c
DR	Double Loop Array Right	2d
CA	MAI Carotid Array	31
UE	MAI Upper Extremity Flex	32
LE	MAI Lower Extremity Flex	33
WT	MAI Wrist Array	34
WR	Wrist	44
SH	Shoulder Array	4a
HB	MRIDC High Res.Head/Brain	4b
Shim	Shim Device	55
BR	CP Breast Array	58
LF	CP Flex Large	89
FL7	Flex Loop 7	8b
FH1	FMRI Head Array	a4
FH2	FMRI Head Array	aa
LL	Flex Loop Large	bd
C1	Customer Coil 1 (RX - Coil)	c1
C2	Customer Coil 2 (Tx/Rx - Coil)	c2
SL	Flex Loop Small	d9
BC	Integrated Body resonator	ff
Ser._P	Service Plug: mode TX / RX	cc / cd
SF	CP Flex Small	98
EN	Endorectal	9d

10.1 General

The MAGNETOM Avanto is the first system to use the newly designed set of receive-only Matrix Coils which are part of the Tim system. Many of the Local Coils designed for use with the MAGNETOM Symphony system can also be used on the MAGNETOM Avanto. For more information, click on the graphics below.

Fig. 38: Matrix Coils



10.2 Matrix Coil Concept

The coil concept for the MAGNETOM Avanto consists of a set of receive-only coils which allow imaging of the whole body with parallel imaging capabilities in all directions. All coils can be applied simultaneously to the patient and subsets of elements of each coil can be selected as required for a particular examination region. The Matrix Coil concept is the extension of the IPA concept of the MAGNETOM Symphony and Sonata to parallel imaging and whole body imaging. This new concept is called Total imaging matrix, or Tim, and includes the following individual coils:

Tab. 30 Matrix Coils

Coil	Element Structure
Head Matrix	12 element coil, upper part removable
Neck Matrix	4 element coil, upper part removable
Spine Matrix	24 element coil
Body Matrix	6 element coil
PA Matrix	16 element coil combined internally to 8 CP elements (PA: Peripheral Angiography)

10.3 Technical Features

All Matrix coils are **pre-tuned** in the factory to a fixed load and no further tuning at the MR system is required. When the Body Coil is transmitting, all elements of the Matrix Coils are detuned (switched off resonance).

10.3.1 Detuning and Safety Concept

The coil safety mechanisms built into all Matrix and Local Coils are designed to protect both patient and coil: that is, The patient from uncomfortable RF burns and the coils from disintegration.

The safety mechanisms offer three safety levels: Each coil element is equipped with a detune circuit consisting of an inductor and a PIN diode which, when forward biased, creates a parallel resonance circuit in series with the coil element effectively opening the element and thus minimizing induced RF currents.

The second safety layer is a passive detune circuit which ensures proper detuning in case the PIN diode bias current should fail or the situation that the user forgets to connect a coil that is left on the patient table.

As a third safety layer, a fuse is placed in every coil element that will blow as soon as the induced current reaches a dangerous level should the passive detune circuit fail.

10.3.2 Coil Recognition

Connected coils are recognized the same way as in the Harmony/Symphony systems by means of analog coil codes realized with code resistors. This allows the Avanto coil control circuitry to recognize the Symphony Local Coils as well.

10.3.3 Shock Indicators

Coils are exposed to heavy mechanical stress through daily usage and thus susceptible to defects through wear and tear as well as possible misuse. Shock indicators are placed on each of the hard-cased coils, i.e. those coils with hard shells, to indicate whether a coil was exposed to excessive mechanical stress.

10.3.4 Amplifiers

Each receive coil element is equipped with a low noise preamplifier to maximize signal-to-noise ratio.

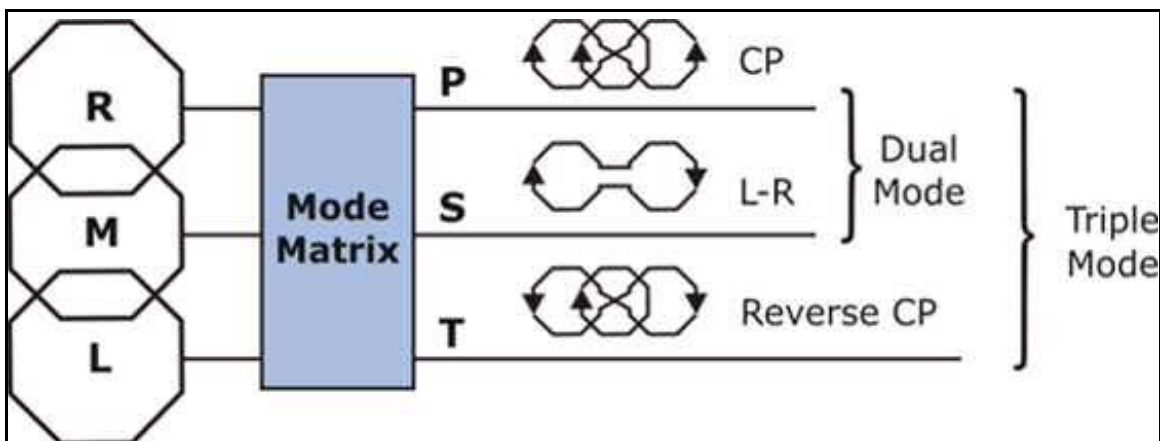
10.3.5 Sheath Currents

RF traps have been included in the coils as well as in the connecting cables to suppress cable shield currents arising from the electrical field of the body coil. These currents, if not suppressed, will in turn create an RF field which will interfere with the MR experiment.

10.4 Mode Matrix

A key feature for all Matrix Coils, except the PA Matrix Coil, is that they are not delivering signals from individual coil elements to the system, but rather combinations from two or three coil elements. To this end, the signals of these elements are fed into a so-called mode-matrix, where they are combined to two or three orthogonal (in the mathematical sense) signals. For example, three coil elements arranged in the left-right direction of the patient are considered. The corresponding three signals are called L, M and R for left, middle and right, respectively. The first output signal of the mode-matrix, or primary mode (P), corresponds to the CP signal of a loop and butterfly antenna design which is commonly used as a CP coil element. The second output signal or secondary mode (S), is a combination of the signals from the left and right coil elements. The third output signal, or tertiary mode (T), corresponds to the reverse CP signal, i.e. the signal of a CP element which is sensitive to the reverse CP components. Since the output signals of the mode-matrix are orthogonal, no information or energy is lost.

Fig. 39: Mode Matrix

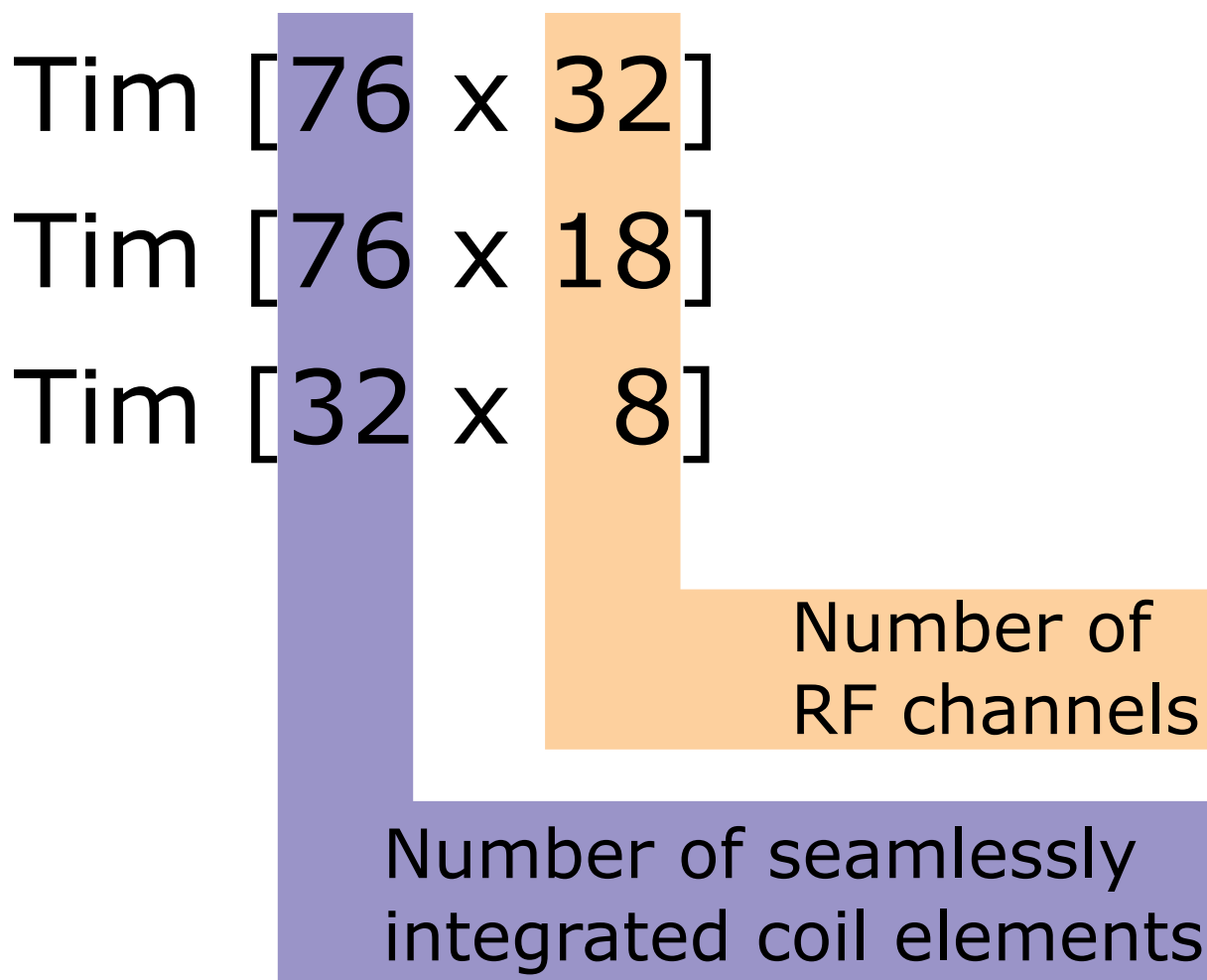


The signals of the original coil elements could be recovered if needed. The reason and advantage for the use of mode signals instead of the original coil element signals is the more efficient use of receiver channels: If no parallel imaging acquisition (iPAT) is used, the primary mode is sufficient and only one instead of three receiver channels have to be used in this case.

Higher modes can be selected in order to obtain enhanced parallel imaging capability or to gain higher signal-to-noise ratio in the periphery of the field-of-view.

10.5 The Matrix Notation

Fig. 40: Tim Nomenclature

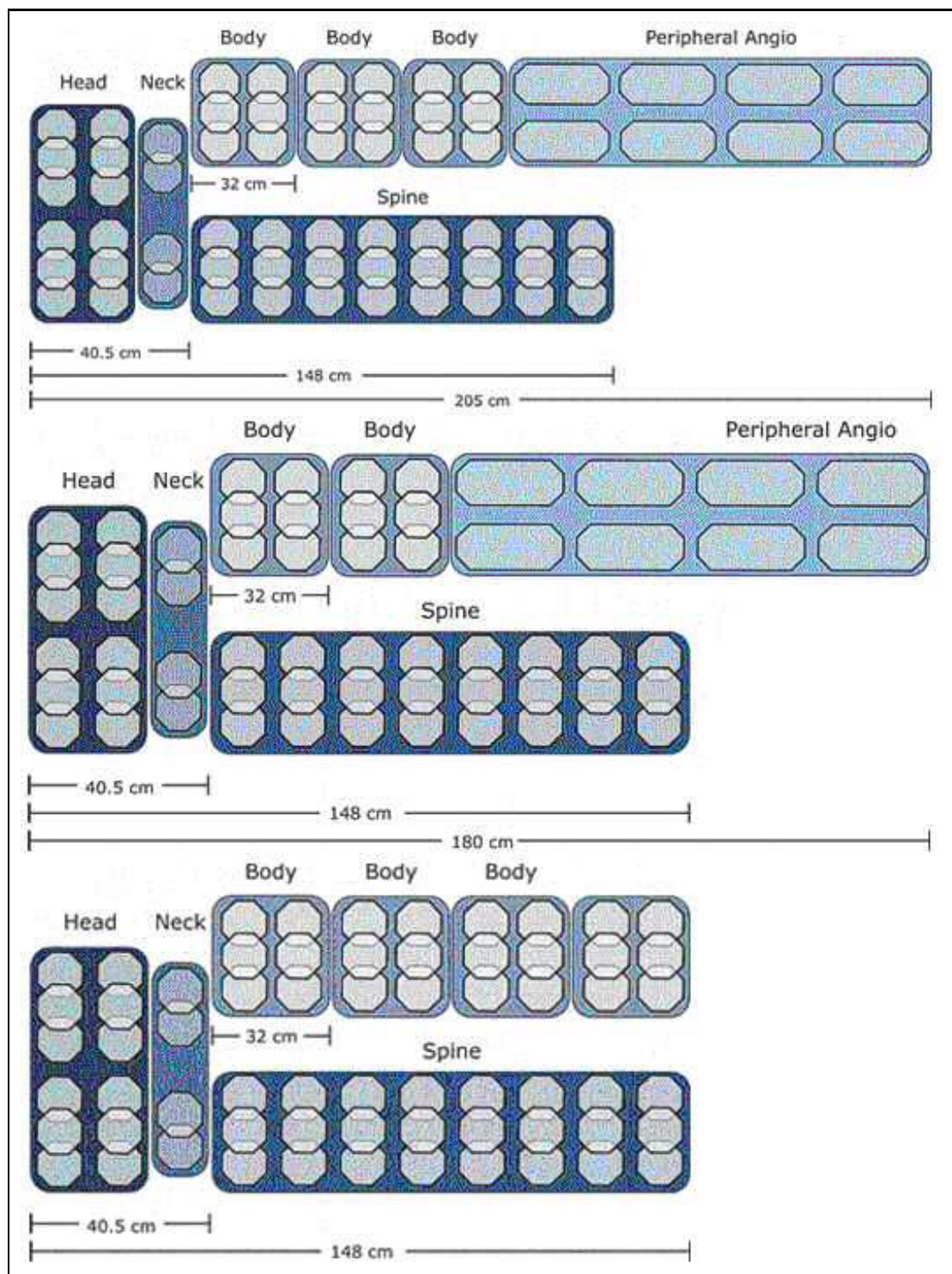


76 - There are 10 coil plugs with 64 pins ($4 \times 8 + 4 \times 6 + 2 \times 4$). The Spine Matrix coil has 24 elements, but has a built-in dynamic switch. As a result only 12 pins are required. Thus : $64 + 12 = 76$ coil elements which can be seamlessly integrated into one single examination.

32 - is the number of effective coil elements in the CP mode: Head Matrix (4), Neck Matrix (2), Spine Matrix (8), Body Matrix (3×6), PA Matrix (8) plus an additional 4-channel coil connected to one of the two Symphony-compatible coil connectors = 32 coil elements which can be seamlessly integrated into one single examination.

The last number is the number of independent RF receiver channels.

Fig. 41: Matrix Coil Overview



10.5.1 Flexibility

Another key feature of the Matrix Coil concept is that Matrix Coils can be combined to increase coverage and allow whole body exams. Up to four identical Body Matrix Coils can be connected in addition with other Matrix Coils. The total coverage without repositioning the patient is 205 cm. Besides the Matrix Coils, other coils, e.g. the flex coils can be connected simultaneously and integrated into an examination.

10.6 Head Matrix Coil

The Head Matrix Coil is a 12 element receive only coil. The 12 elements are arranged in two rings of 6 elements each around the head - one ring covers the crown of the head and the other the base of the head. For positioning the patient, the upper part of the Head Matrix is completely removable. Therefore, 6 of these elements (first half of each ring) are in the lower part, and 6 elements (second half of each ring) are in the removable upper part. Both parts are connected by a cable to the system. There are no electrical contacts between the upper part and the lower part of the Head Matrix.

Two fixation devices can be used to immobilize the head. When imaging patients who are claustrophobic, the lower part of the Head Matrix can be used without the upper part. A mirror can be attached to the Head Matrix so that the patient can look outside the bore. Viewing to either the front or rear side of the magnet is supported.

The mode matrix implementation for this coil is to connect 3 adjacent elements of the two 6-element rings to a mode matrix. When only the primary modes are selected for examination, the coil behaves like a 4 element array coil: 2 elements in the anterior and in the posterior part, and 2 elements in the brain and in the feet-side section.

Applications include among others:

- Head examination
- MR-Angiography
- Combined Head/Neck examination

Fig. 42: Head Matrix Coil

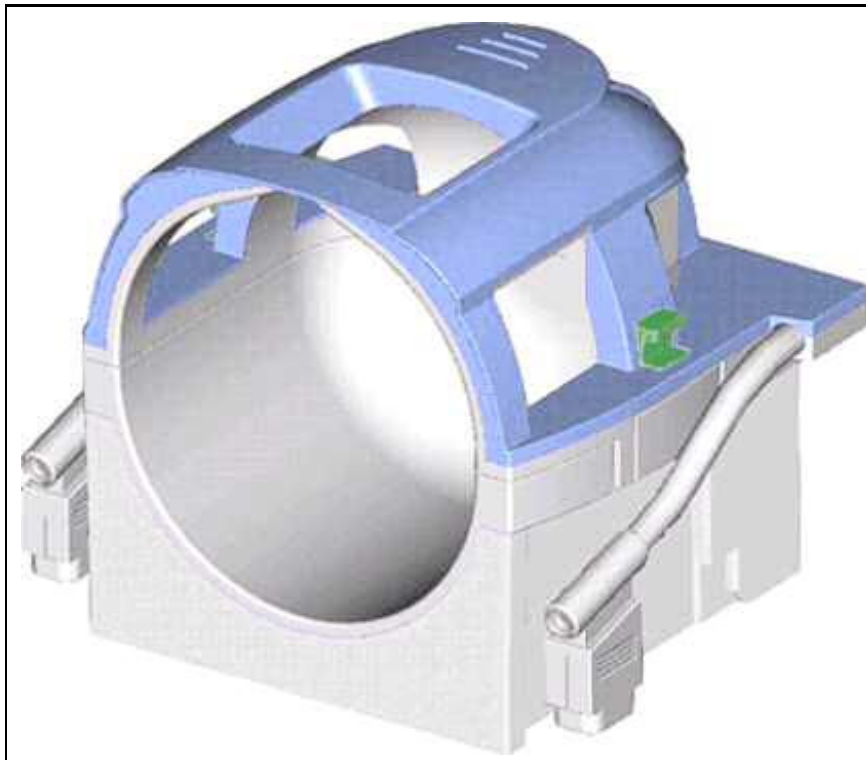
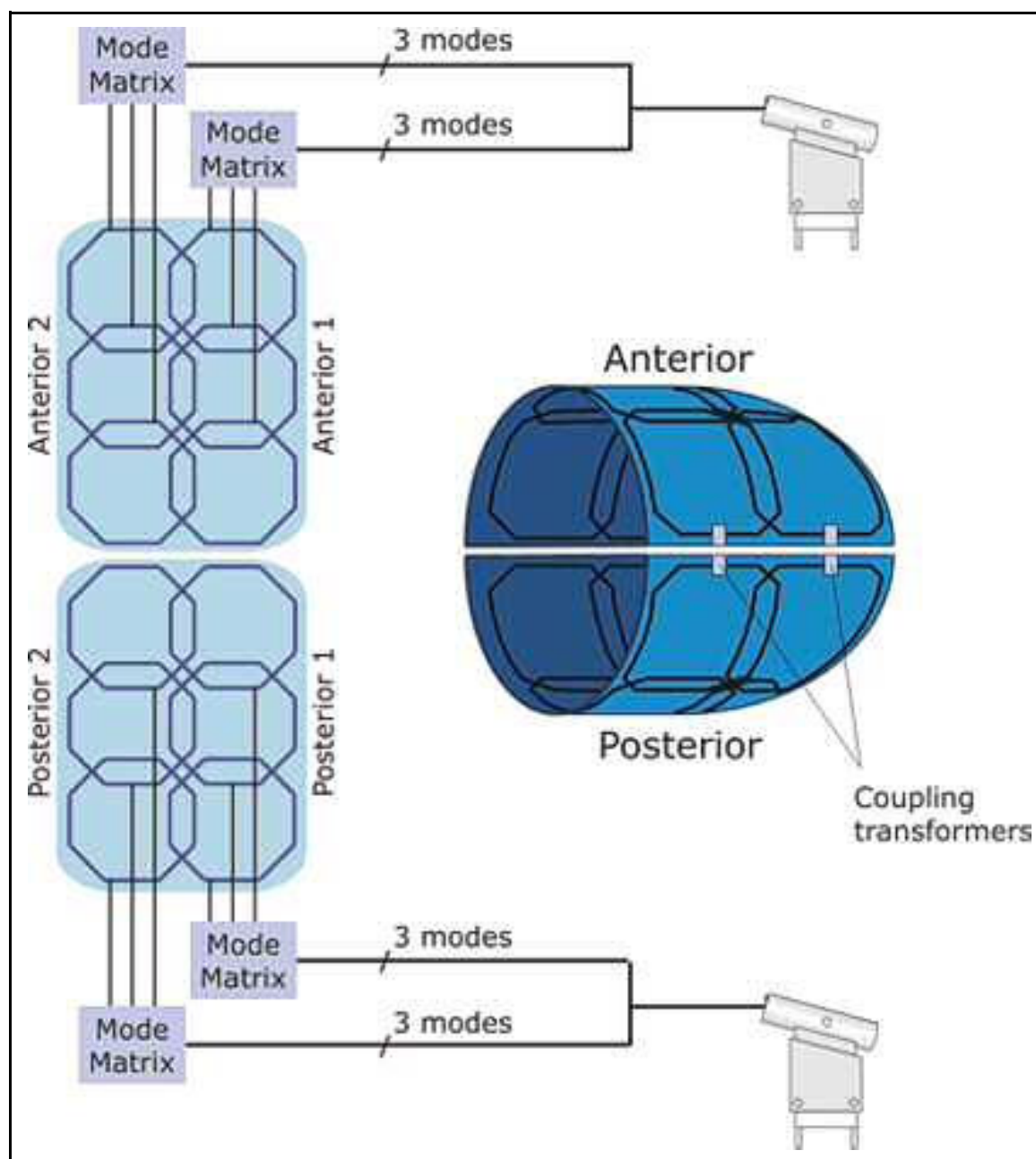


Fig. 43: Head Matrix Coil Construction



10.7 Neck Matrix Coil

The Neck Matrix coil is a 4 element receive-only coil. Two of these elements are in the lower part, and two are in the removable upper part. The upper part connects to the lower one.

When positioning the patient, the upper part is completely removable. The lower part of the coil can be used by itself. Mechanically, the Neck Matrix coils fits to the Head Matrix coil on one side and to the Spine Matrix coil on the other side.

The mode matrix implementation for this coil is the CP mode as the primary one, and the reverse-CP mode as the secondary one. When only the primary mode is selected for examination, the coil behaves like a 2-element array coil with one upper and one lower element.



Although the Neck coil uses the same coil connector as the Symphony Local Coils, the Neck Matrix coil CANNOT be used on the Symphony system!

Applications include among others:

- Neck
- Larynx/Esophagus
- Cervical Spine
- MR Angiography
- Mediastinum

Fig. 44: Neck Matrix Coil

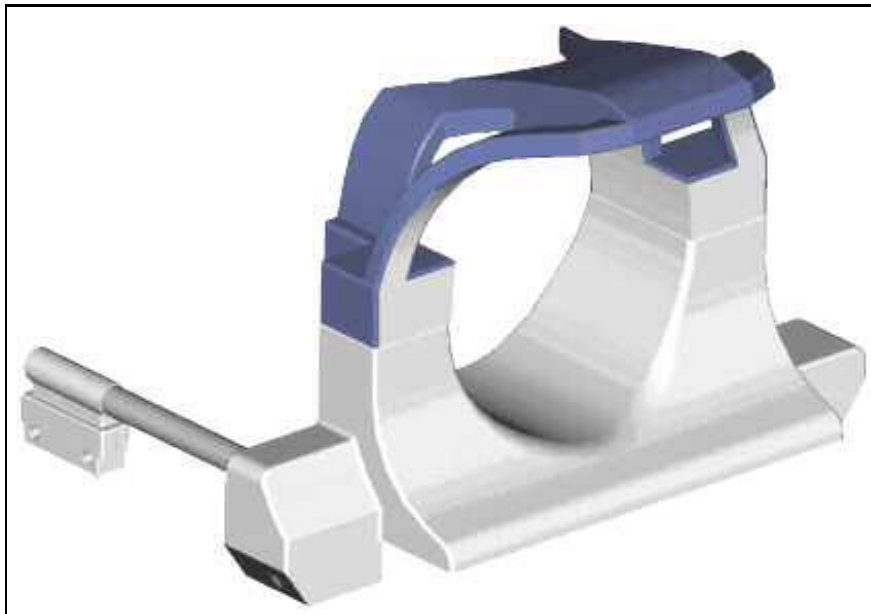
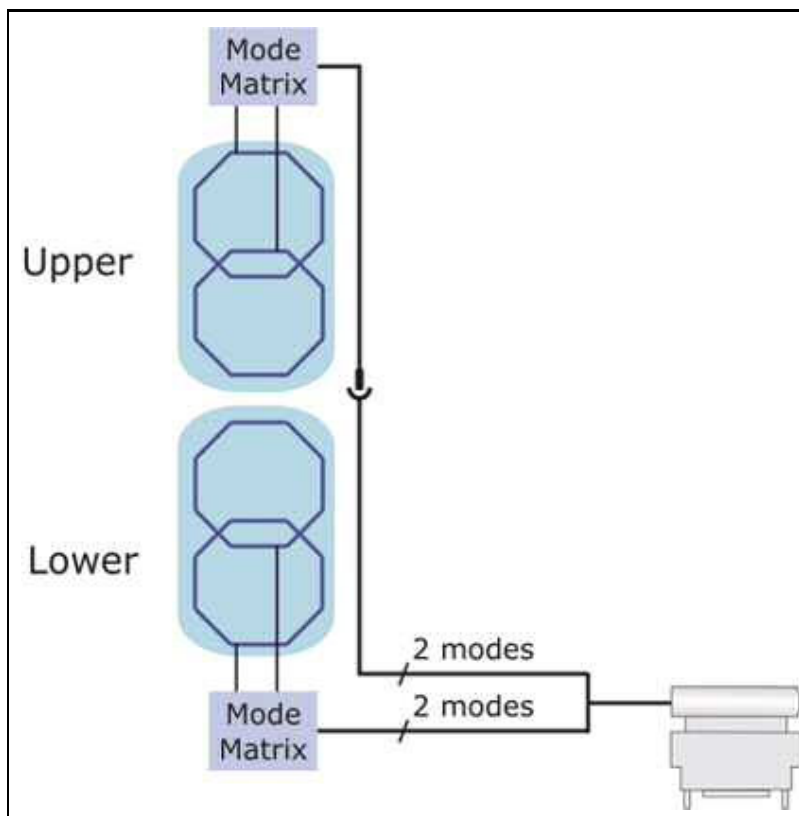


Fig. 45: Neck Matrix Coil Construction



10.8 Spine Matrix Coil

The Spine Matrix coil is a 24 element receive-only coil. The 24 elements are stacked in 8 clusters along the head-feed direction with each cluster consisting of 3 coil elements in left-right direction. Since not all of these 8 element clusters can be in the field-of-view at the same time, subsets of up to 4 adjacent groups out of the 8 groups, or up to 12 out of the 24 elements can be selected for examination.

The mode matrix implementation for this coil is to connect the 3 elements of one element group to a mode matrix. When only the primary mode is selected for examination, the coil behaves like an 8 element CP array coil from which up to four adjacent CP elements can be selected.

Fig. 46: Spine Matrix Coil

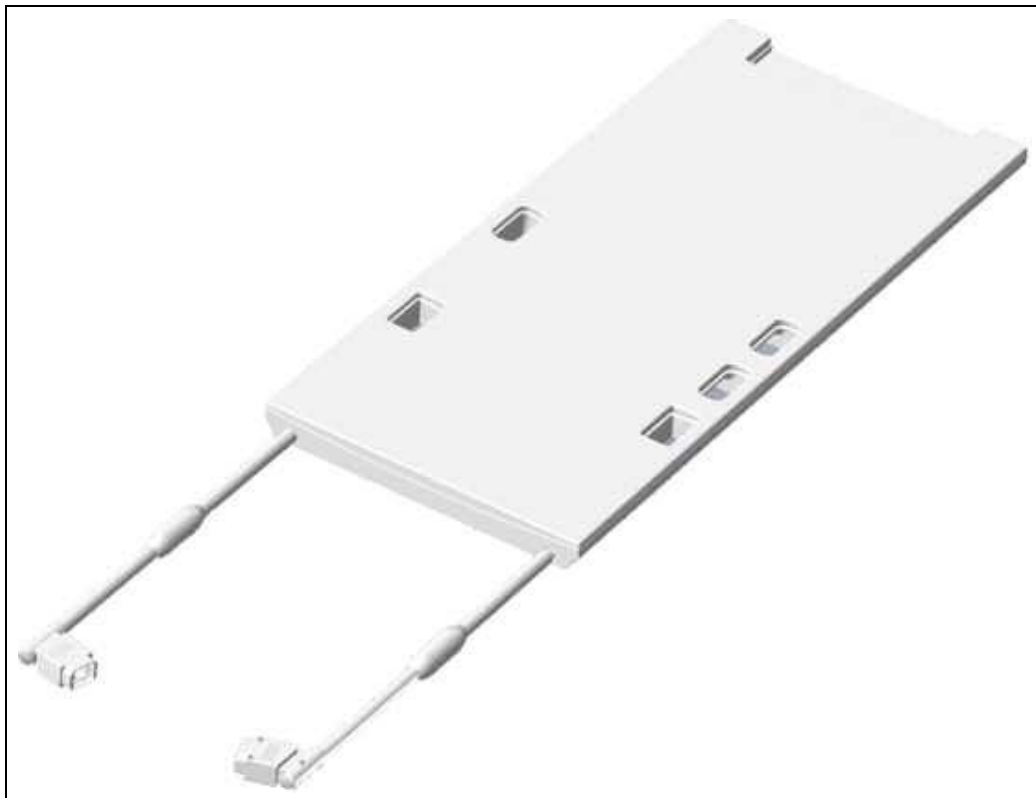
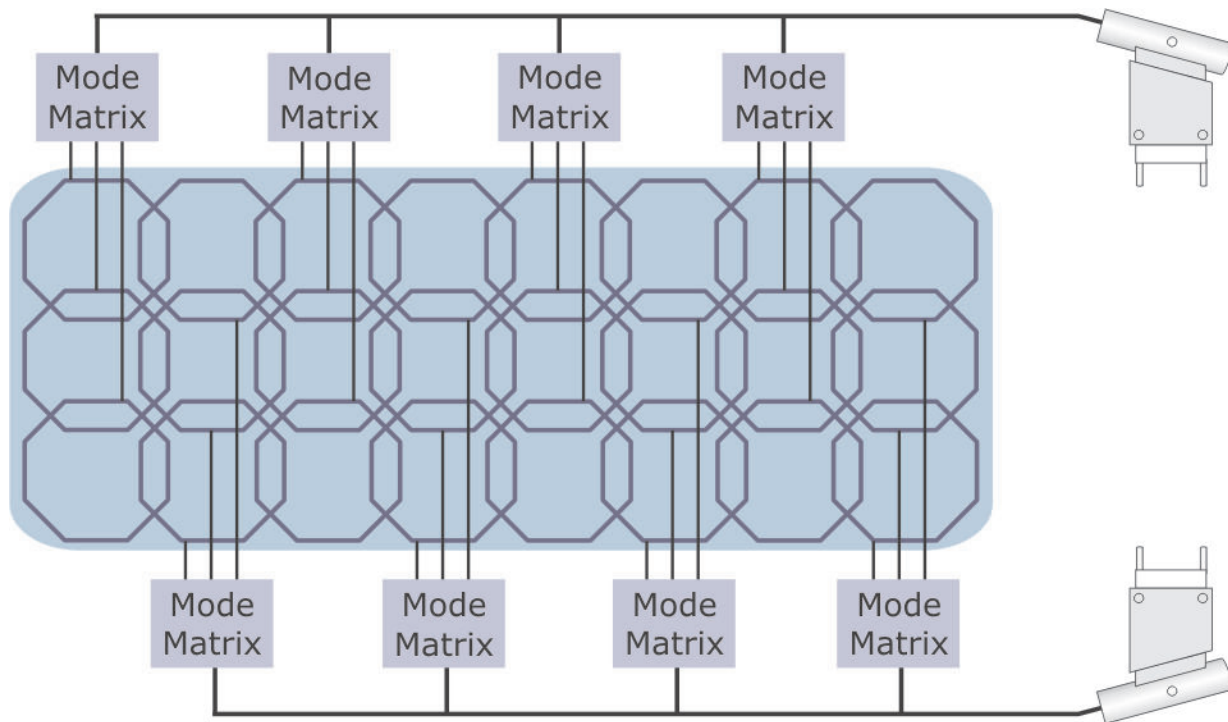


Fig. 47: Spine Matrix Coil Construction



Applications include among others:

- High resolution imaging of the whole spine
- Combination with various coils

The Body Matrix coil is similar to the 6-channel Body Array coil with several improvements.

- The connection to the system with one cable instead of two.
- A newly designed coil connector.
- The additional mode matrix.
- A slightly different coil size.
- A **direction detector** to detect the coil's orientation in the magnet. The last feature is needed because the coil can be rotated by 180° by the user with respect to the anterior-posterior direction. In this case the CP mode and the reverse-CP mode are exchanged.

Applications include:

- High resolution imaging of thorax (including heart), abdomen and pelvis

10.10 Peripheral Angio Coil

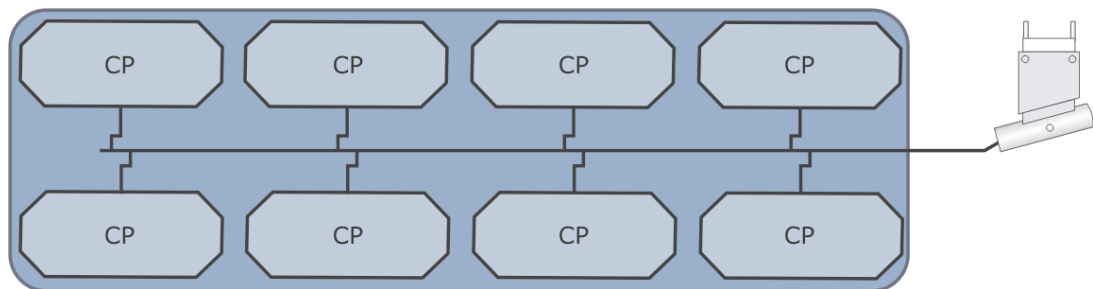
The coil is practically the same as the Harmony/Symphony coil. The connector and some internal adaptations for the Avanto coil interface were made.

Fig. 50: Peripheral Angio Coil



This coil can be used in both directions for feet-first and head-first examinations.

Fig. 51: Peripheral Angio Coil Construction



Applications include:

- High resolution imaging of thorax (including. heart), abdomen and pelvis
- MR Angiography
- Combination with various coils

11.1 General

The next couple of pages provide the coil connector pin-out diagrams and associated signal tables.

Fig. 52: Location of the Matrix and Local Coil connectors

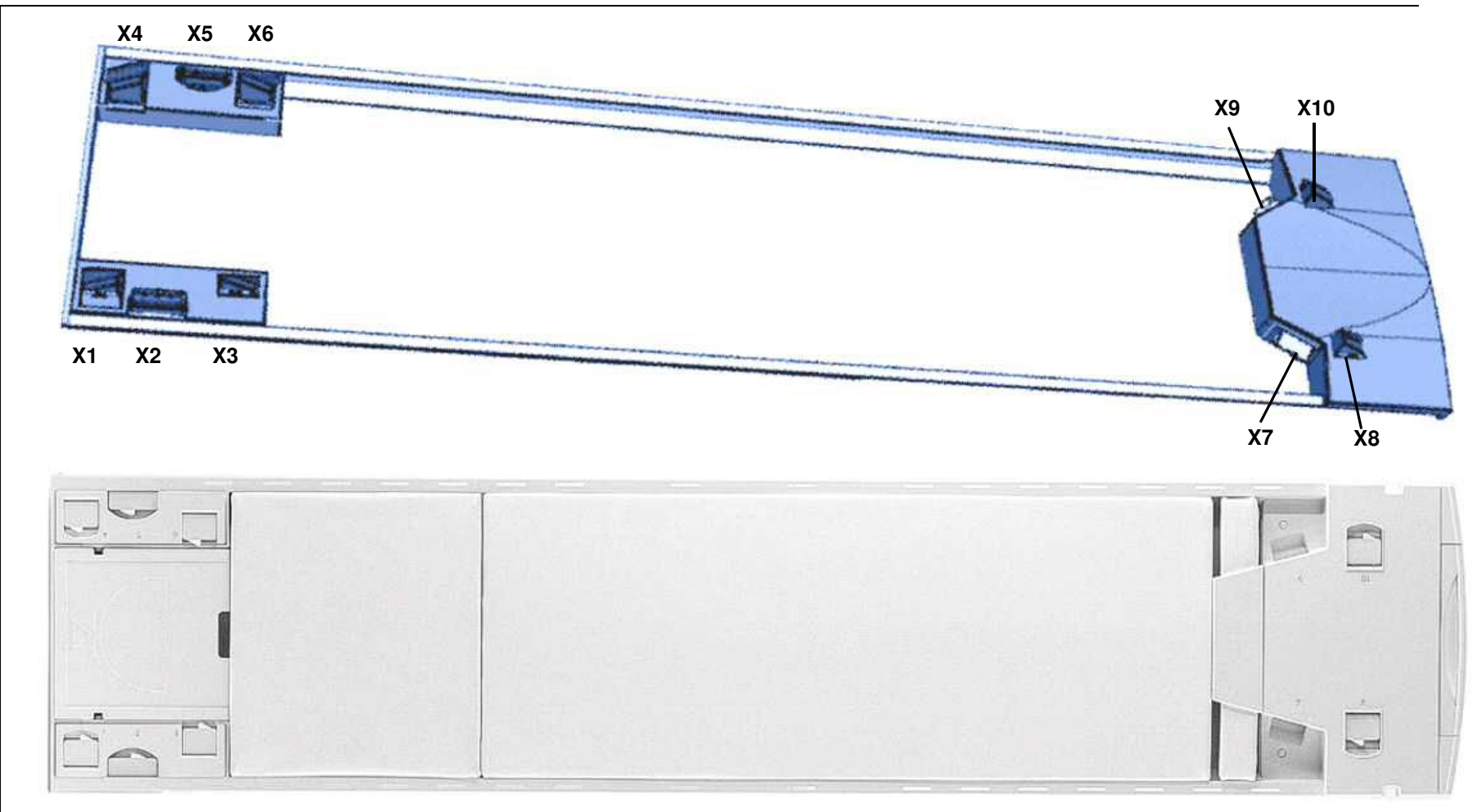
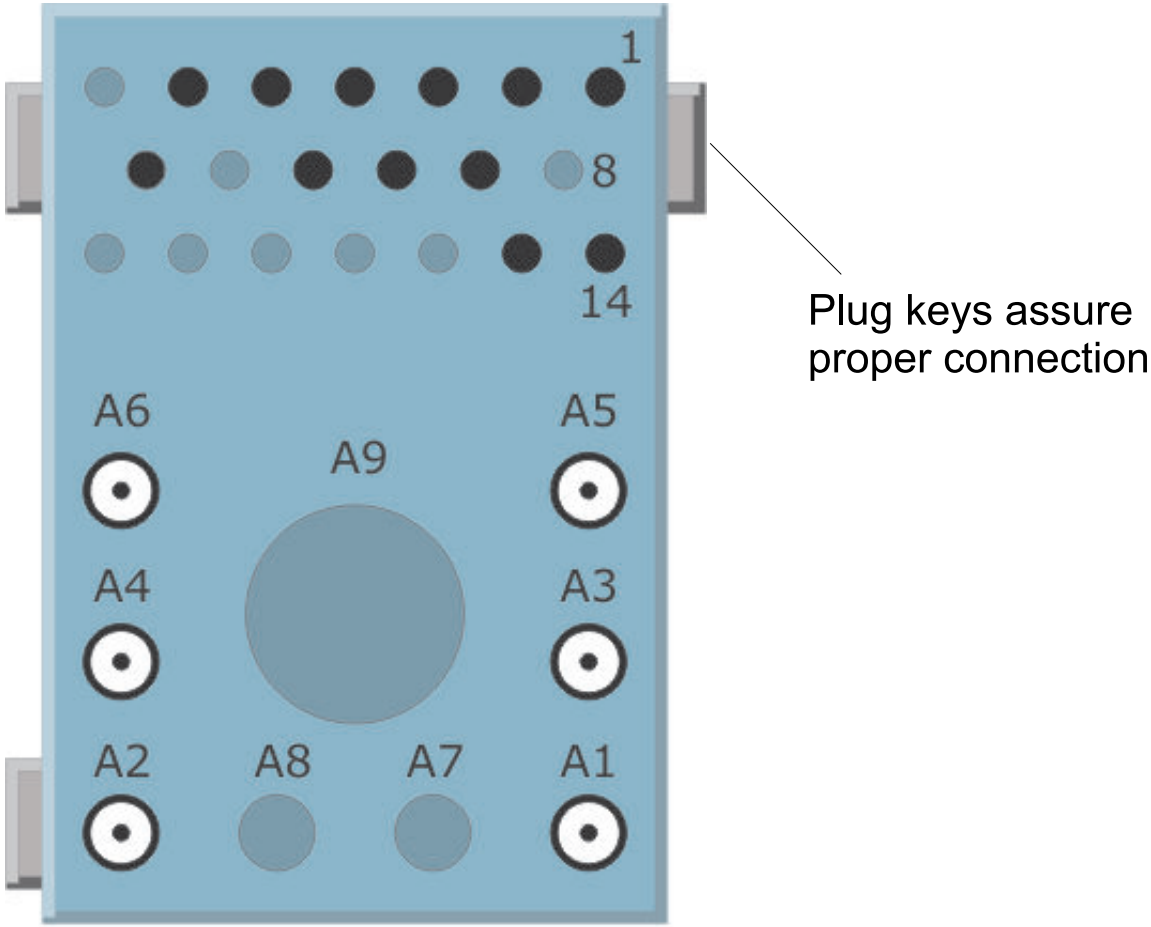


Fig. 53: Matrix Coil Connector - 6 Channel



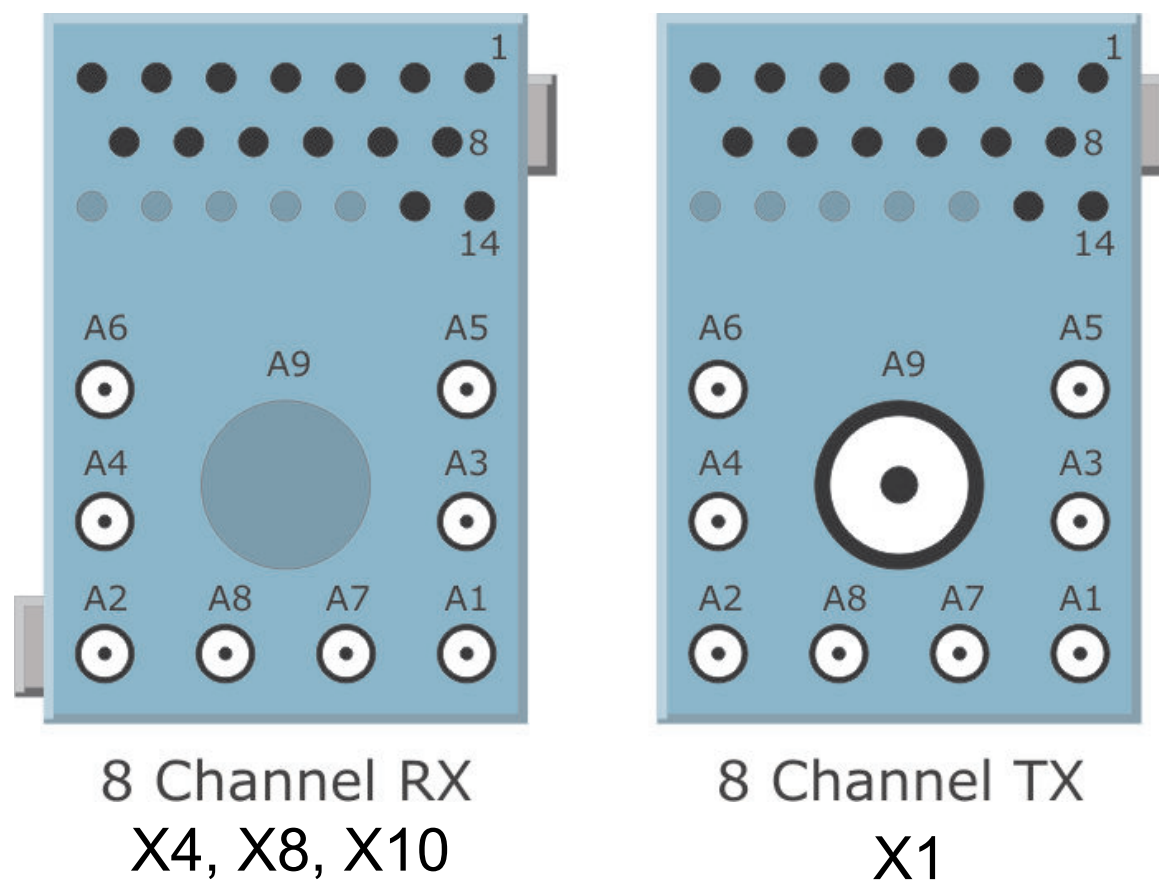
6 Channel RX
X3, X6, X7, X9

Tab. 31 Pin-out of 6 Channel Coil Connector

Signal Name	Pin	Description
PIN1	1	I=100mA (to 12V) Reverse V = -30V
PIN2	2	I=100mA (to 12V) Reverse V = -30V
PIN3	3	I=100mA (to 12V) Reverse V = -30V
PIN4	4	I=100mA (to 12V) Reverse V = -30V
PIN5	5	I=100mA (to 12V) Reverse V = -30V
PIN6	6	I=100mA (to 12V) Reverse V = -30V
n.a.	7	n.a.
n.a.	8	n.a.

Signal Name	Pin	Description
PIN GND 1	9	
PIN GND 2	10	
PIN GND 3	11	
n.a.	12	n.a.
Code1 / High Nibble	13	Code resistor *
Code2 / Low Nibble	14	Code resistor *
Code GND	15	
	16	
	17	
	18	
	19	
	20	
RX1	A1	Rx Coax Connector
RX2	A2	Rx Coax Connector
RX3	A3	Rx Coax Connector
RX4	A4	Rx Coax Connector
RX5	A5	Rx Coax Connector
RX6	A6	Rx Coax Connector
n.a.	A7	n.a.
n.a.	A8	n.a.
n.a.	A9	n.a.

Fig. 54: Matrix Coil Connector - 8 Channel

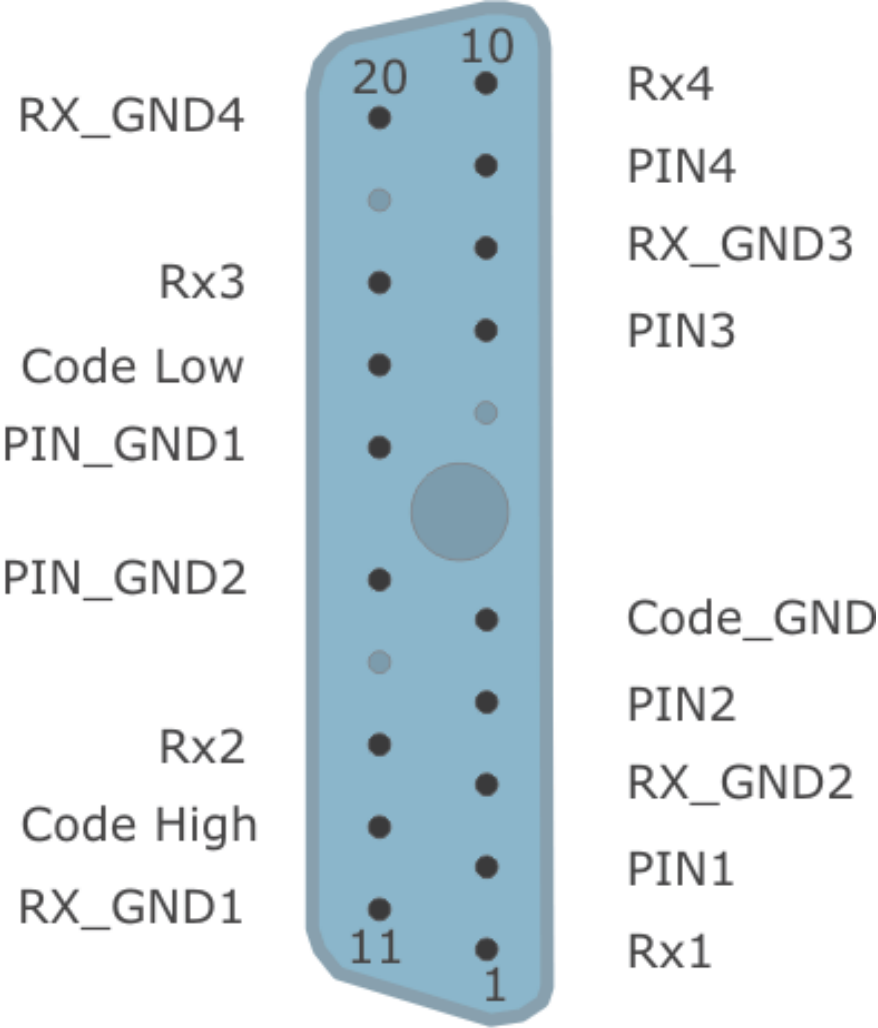


Tab. 32 Pin-out of 8 Channel Coil Connector

Signal Name	Pin	Description
PIN1	1	I=100mA (to 12V) Reverse V = -30V
PIN2	2	I=100mA (to 12V) Reverse V = -30V
PIN3	3	I=100mA (to 12V) Reverse V = -30V
PIN4	4	I=100mA (to 12V) Reverse V = -30V
PIN5	5	I=100mA (to 12V) Reverse V = -30V
PIN6	6	I=100mA (to 12V) Reverse V = -30V
PIN7	7	I=100mA (to 12V) Reverse V = -30V
PIN8	8	I=100mA (to 12V) Reverse V = -30V
PIN GND 1	9	
PIN GND 2	10	
PIN GND 3	11	
PIN GND 4	12	
Code1 / High Nibble	13	Code resistor *

Signal Name	Pin	Description
Code2 / Low Nibble	14	Code resistor *
Code GND	15	
	16	
	17	
	18	
	19	
	20	
RX1	A1	Rx Coax Connector
RX2	A2	Rx Coax Connector
RX3	A3	Rx Coax Connector
RX4	A4	Rx Coax Connector
RX5	A5	Rx Coax Connector
RX6	A6	Rx Coax Connector
RX7	A7	Rx Coax Connector
RX8	A8	Rx Coax Connector
TX	A9	Tx Coax Connector

Fig. 55: Matrix Coil Connector - 4 Channel



4 Channel RX

X2, X5

This coil connector is compatible to the Symphony Local Coils. There are two such connectors on the patient table, neither of which, however, has a transmit pin. For this reason, the transmit-receive local coils, such as the Tx/Rx CP Extremity coil, cannot be used on the Avanto.

Tab. 33 Pin-out of 8 Channel Coil Connector

Signal Name	Pin	Description
PIN1	2	I=100mA (to 12V) Reverse V = -30V
PIN2	4	I=100mA (to 12V) Reverse V = -30V
PIN3	7	I=100mA (to 12V) Reverse V = -30V

Signal Name	Pin	Description
PIN4	9	I=100mA (to 12V) Reverse V = -30V
PIN GND 1	16	
PIN GND 2	15	
Code1 / High Nibble	12	Code resistor *
Code2 / Low Nibble	17	Code resistor *
Code GND	5	
	19	
	14	
	6	
RX1	1	Rx Coax Connector
RX2	13	Rx Coax Connector
RX3	18	Rx Coax Connector
RX4	10	Rx Coax Connector
RX_GND1	11	Rx Coax Connector
RX_GND2	3	Rx Coax Connector
RX_GND3	8	Rx Coax Connector
RX_GND4	20	Rx Coax Connector

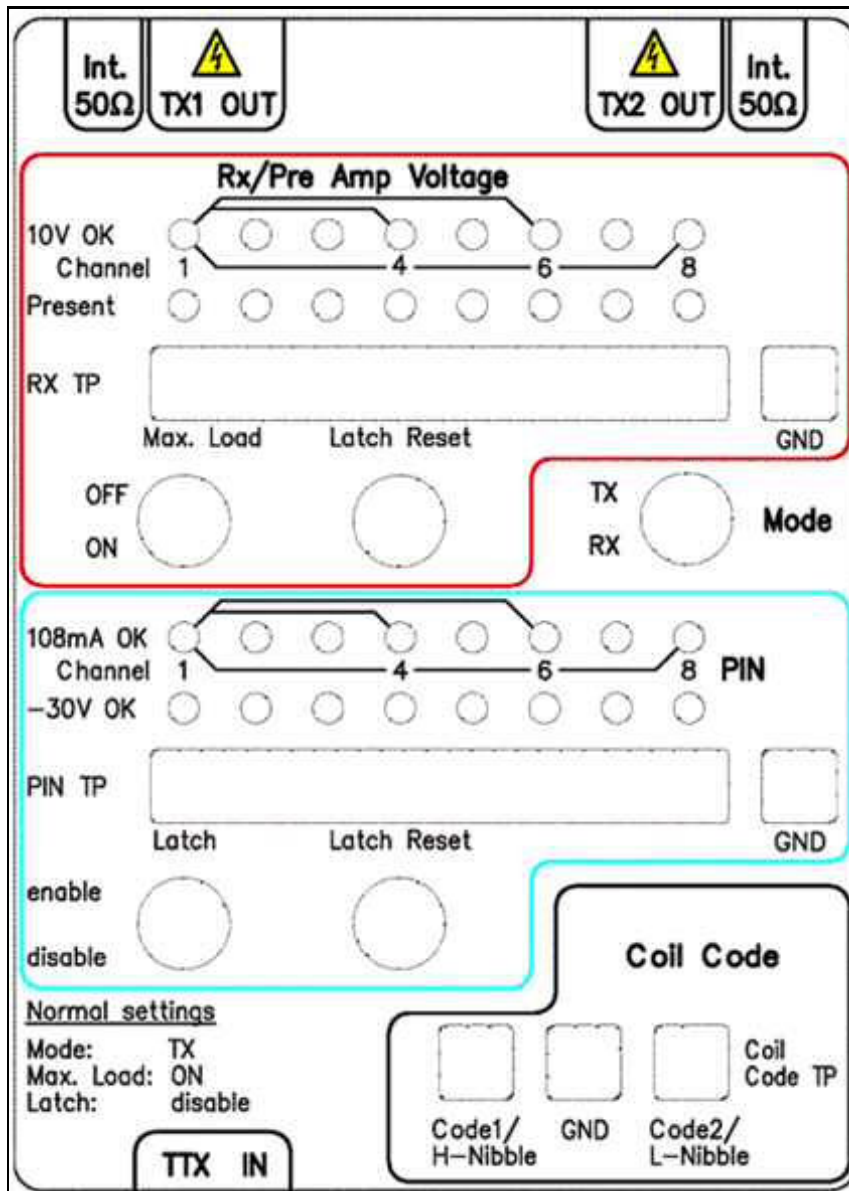


The pins are not in numerical order in this table!

12.1 General

12.1.1 Layout

Fig. 56:



12.1.2 Features

- Automatic evaluation of measured voltages and currents, such as 10 V preamplifier voltage and bias current and reverse bias voltage for the PIN diodes. Every channel is evaluated separately from the others.
- Besides automatic evaluation, manual measurements at external test points are still possible.

- Allows checking the 10 V preamplifiers under normal and extended load conditions (normal/maximum current).
- Helps to find wrong cabling between RF-electronics at the magnet (RCCS, RFIS) and the coil connectors at the patient table by use of semi-automatic/interactive TestTools.
- Enables to troubleshoot sporadic errors as well, such as broken cables in the patient table by use of error latches, detects also interrupts of ground wires, separately for each channel.
- Offers the possibility of connecting external 50-Ohm-terminators to the LC-TX-path to increase the maximum allowed power.
- Allows to measure the voltage used for coil code detection.
- Does not need any separate power supply, everything is supplied by the MR-system.

12.1.3 Components

- Service plug with two cables: one 6/8-channel plug and one 4-channel plug
- Auxiliary plug for TTX signal
- BNC cable for TTX signal
- Suitcase for transportation

12.2 Control Panel

12.2.1 Rx/Pre Amp Voltage

10V OK	Green LED's to indicate the PreAmp voltage of the corresponding RX-channel is included in the specifications (specification 9.4 V - 10.9 V, in service plug 8.7 V - 11.8 V).
Present	Yellow LED's to indicate that some PreAmp voltage of the corresponding RX-channel is present. No specification check, lower voltage results in less brightness. LEDs start to glow at approx. 3 V and are bright between 5 and 7 V.
RX TP, GND	Sockets/test points for voltmeter, values measured between 7.9 V and 11.0 V are OK. Due to the internal structure of the service plug, the measured voltage differs from the real voltage of the coil plug by approx. 0.75 V - 1 V. Unused channels (e.g. channel 7 and 8 with a 6-channel-plug) will also show a voltage: approx. 1.1 V/5.5 V without/with maximum load.
Channel 1 .. 8	Label to identify the correct channel.



The 10 V of RX-channel 1 supplies the Rx/PreAmp-section with power. When the channel has no voltage either, the complete Rx/PreAmp-section will be "dark". The 10 V of RX-channel 2 supplies the PIN-section with power. When RX channel 2 has no voltage, the complete PIN-section will also be "dark".

12.2.2 PIN

108mA OK	Yellow LED's to indicate the bias current of the corresponding PIN-channel is within specifications (specification 101 - 115 mA, in service plug 88 - 131.5 mA).
-30V OK	Green LED's to indicate the reverse bias voltage of the corresponding PIN-channel is within specifications (specification -31.3 V through -27.7 V, in service plug -33.9 V through -25.6 V).
PIN TP, GND	Sockets/test points for voltmeter current: Measurements values between 0.65 V and 0.87 V are OK, representing 95 to 120 mA. voltage: Measurements values between -31.3 V and -27.7 V are OK. Unused channels (e.g. channel 7 and 8 with a 6-channel-plug) will also show a voltage: approx. 2.3 V with reverse bias voltage and approx. 0.98 V with bias current.
Channel 1 .. 8	Label to identify the correct channel.

12.2.3 Mode

TX	Normal mode of the service plug, for both transmit and receive with the service plug. Results in coil code "CC".
RX	RX mode of the service plug, for transmit with Body Coil and receive with the Service Plug/Pick-Up coil. Results in coil code "CD".

12.2.4 Coil Code

Code 1 / High Nibble	Sense line for first part of the coil code (high nibble, with Service Plug in TX and RX-mode this is the left "C").
Code 2 / Low Nibble	Sense line for second part of the coil code (low nibble, with service plug in TX-mode, this is the right "C", and in RX-mode the "D").
Coil Code TP, GND	Sockets/test points for voltmeter. Approx. values for coil code "CC" is 7.8 V for both sockets.

12.3 Controls

12.3.1 Max. Load

Switches the load condition of the 10 V supplies in the RCCS between "normal" (10 - 20 mA) and "maximum" (approx. 70 mA). The measured values at the test points will change with the position of this switch by approx. 0.6 V and depends on the length of the cables.



The 10 V of RX-channel 2 supplies the PIN-section with power. Thus a normal load condition is not possible here. Additionally, the measured voltage via RX-TP will be always quite low.

The normal setting is "Max. Load ON".

12.3.2 Latch

The latches are intended for troubleshooting sporadic errors caused by breaks in the cables from the RCCS and RFIS to the coil connectors. Both breaks of the ground and the life wire can be detected.

This switch in the PIN-section allows to enable or disable the error latch. This is necessary for testing the RFIS via TestTools which would not be possible otherwise.

The normal setting is "disabled".

12.3.3 Latch Reset

Resets the error condition latches of the corresponding section (Rx/PreAmp or PIN).

In the Rx/PreAmp-section the latch cannot be enabled or disabled as this is not necessary. Thus after plug-in of the service plug, the latch there has to be resetted. In the PIN-section, this is only necessary in case the latch is enabled.

12.3.4 Latch

This switch in the PIN-section allows to enable or disable the error latch. This is necessary for testing the RFIS via TestTools which would not be possible otherwise.

The normal setting is "disabled".

12.3.5 Mode

Contrary to the old service plug from MAGNETOM Harmony/Symphony, the new service plug has two coil codes, "CC" and "CD".

Coil code "CC" behaves like the old service plug and simulates a TxRx-coil (transmit and receive both with the service plug), whereas coil code "CD" simulates a Rx-coil only (transmit with Body Coil and receive with the service plug). This mode is intended for use with an external pick-up-coil (e.g. simple loop-coil), e.g. for spike troubleshooting.

The normal setting is "TX-mode".



Although the new service plug can be used with MAGNETOM Harmony/Symphony as well, it is not a complete replacement for the old plug. There are some tests (mainly LCCS-tests) which cannot be done with the new plug. Thus for MAGNETOM Harmony/Symphony, the new plug is just an add-on and the old plug is still necessary there. From the new plug the automatic test features can be used and the detachable 50-Ohm-termination resistor.

12.4 Connections

12.4.1 TX1 Out, TX2 Out

Possibility of connecting an external 50-Ohm-power attenuator to the LC-TX-path to increase the maximum allowed power.

12.4.2 Int. 50 Ohm

Internal 50 Ohm termination resistors to allow also transmit of RF-power to the service plug (plug simulates a Tx-coil, same as with MAGNETOM Harmony/Symphony).

The internal resistor allows RF up to an average power of approx. 2 W for one sequence. If more power is necessary while troubleshooting, an external 50 Ohm termination has to be used.

One resistor for each connector is available. The 4-channel termination is not used with MAGNETOM Avanto, but was designed for MAGNETOM Harmony/Symphony. In case one of the resistors gets defective, it is possible to use a small BNC-cable to "activate" the other resistor as both are never used at the same system type.

12.4.3 TTX IN

For test signal TTX, necessary for the TestTools where the auxiliary plug is used (RCCS plug tests).

The input can also be used for pick-up coils without a built-in preamplifier.

12.5 Notes on use

12.5.1 General



CAUTION

Connecting both the 6/8-channel plug and the 4-channel plug at the same time may damage the service plug.

Irreversible damage to the service plug could result if the plugs are connected for a long time. A few seconds up to one minute is no problem.

- » Never connect both the 6/8-channel plug and the 4-channel plug at the same time.

- Never have external cables (e.g. for DVM, oscilloscope) connected when performing RF-measurements with the service plug. The results would be incorrect.
- The normal setting of the control is "TX mode", "Latch disabled" and "Max. Load ON" or "Max. Load OFF" (doesn't matter).

12.5.2 Rx/PreAmp-Voltage section

- After plug-in of the service plug, the latches have to be reset in case the display of the LED's is required. In case the latch of the PIN-section is disabled (normal setting), this is only necessary for the latch of the Rx/PreAmp-section.
- The 10 V of RX-channel 1 supplies the Rx/PreAmp-section with power. When the channel has no voltage, the complete section will also be "dark".
- Unused channels (e.g. channel 7 and 8 with a 6-channel-plug) will also show a voltage: approx. 1.1 V without load and 5.5 V with maximum load.
- The values measured at the RX test points will change with the position of the Max.-Load-switch by approx. 0.6 V and depend on the length of the cables.

12.5.3 PIN section

- The 10 V of RX-channel 2 supplies the PIN-section with power.
 - A normal load condition is not possible here.
 - The voltages measured via RX-TP will always be quite low.
 - When RX channel 2 has no voltage, the complete PIN-section will also be "dark".
- Unused channels (e.g. channel 7 and 8 with a 6-channel-plug) will also show a voltage: approx. 2.3 V with reverse bias voltage and approx. 0.98 V with bias current.
- PIN_GND_1 is used as a reference ground. This may result in small differences (some mV) when comparing with PIN_GND_2, PIN_GND_3 or PIN_GND_4.

- In bias current mode in case PIN_GND_1 is missing, channel one and two will be dark. With PIN_GND_2 it will be channel three and four, PIN_GND_3 channel five and six and with PIN_GND_4 channel seven and eight.

12.5.4 Pick-Up coil

- Pick-up coils can be connected to TTX IN or RX-TP and GND (every channel, preferred is channel 1).
- Pick-up coils connected to RX-TP and GND have to block the 10 V coming from the RCCS either by using an internal preamplifier or by a capacitor, otherwise a connection between RX-TP and GND may result in a shortcut of the 10 V.
- Pick-up coils connected to TTX IN do not need to have this, no 10 V are available here. The disadvantage is that an additional attenuation (6-10 dB) has to be taken into account due to the service plug's internal structure.

12.5.5 TestTools

- The test results with the RCCS plug tests differ from system to system, depending on the length of the RF-cables (additional attenuation). In addition, the results from the head and the feet end of the patient table differ by approx. 10%.
- In case you measure the PIN-diode voltages and currents using the RFIS test, it is no problem to keep the test running on the same coil connector as long as necessary.

- » The common print number (M6-015.850.01) from Function Description is divided into separate Component print numbers.

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